

simpsons-paradox RuleSpeak®

Digital mirror of the Simpson's Paradox domain. The entities are Studies, Treatments, Strata, and CaseCells (raw patient counts). SuccessRates, TreatmentSummaries, and TreatmentRankings are derived. The paradox falls out of the DAG as an emergent derived fact — it is never modeled directly.

Declarative business rules rendered from the rulebook — the single source of truth. Every statement below is the plain-language reading of a formula; nothing here is hand-authored prose.

Use the [⚙ options](#) in the top-right corner to turn each part of the documentation on or off — RuleSpeak®, the literal reading, the raw formula, the author's description, and the links to a rule's inputs and consumers. Your choices are remembered.

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Loop

A loop is identified by its name.

f **Name** Calculated 1 hop from ground truth string DR-1

IN RULESPEAK®

A loop's name is the same as its loop ID.

▾ **Inputs** 1 fields this rule reads

● **Loop ID** Loop

Unique identifier: loop-01 through loop-10.

▸ **Ground truth at the leaves** 1 the raw values this ultimately depends on

Display label.

Title Ground truth string

Ground truth. Written directly to the database (datatype string); not computed from anything else. Everything derived ultimately traces back to values like this one.

Short title for what this loop adds to the domain model.

Status Ground truth string

Ground truth. Written directly to the database (datatype string); not computed from anything else. Everything derived ultimately traces back to values like this one.

planned / in-progress / complete

New Concept Ground truth string

Ground truth. Written directly to the database (datatype string); not computed from anything else. Everything derived ultimately traces back to values like this one.

The new first-class domain concept introduced this loop.

Domain Question Ground truth string

Ground truth. Written directly to the database (datatype string); not computed from anything else. Everything derived ultimately traces back to values like this one.

The natural-language question this loop's concept answers FROM the domain data.



Mock Data Note

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

What the seed data is designed to show.



Next Suggestion

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Suggestion for the next loop, written after witnessing the DAG at this loop's end.



Tradition ID

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

FK → ResearchTraditions.TraditionId. The scholarly tradition whose conceptual vocabulary this loop primarily draws on — tracks which tradition has driven each stage of the instrument's development.



Tradition Name

Lookup

1 hop from ground truth

string

DR-2

IN RULESPEAK®

A loop's tradition name is the value of LOOKUP(TraditionId, ResearchTraditions[TraditionId], ResearchTraditions[Name]).

▼ Inputs 1 fields this rule reads

● Tradition ID

Loop

FK → ResearchTraditions.TraditionId. The scholarly tradition whose conceptual vocabulary this loop primarily draws on — tracks which tradition has driven each stage of the instrument's development.

► Ground truth at the leaves 1 the raw values this ultimately depends on

Lookup: ResearchTraditions.Name via TraditionId.



Tradition Core Concern

Lookup

1 hop from ground truth

string

DR-3

IN RULESPEAK®

A loop's [tradition core concern](#) is the value of `LOOKUP(TraditionId, ResearchTraditions[TraditionId], ResearchTraditions[CoreConcern])`.

▼ **Inputs** 1 fields this rule reads

● **Tradition ID**

Loop

FK → `ResearchTraditions.TraditionId`. The scholarly tradition whose conceptual vocabulary this loop primarily draws on — tracks which tradition has driven each stage of the instrument's development.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: `ResearchTraditions.CoreConcern` via `TraditionId` — which intellectual concern drove this loop.

● **Commit Hash** Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Full git SHA where this loop row first landed in `effortless-rulebook.json` — the replay anchor for this turn of the Effortless loop.

f **Commit Short** Calculated 1 hop from ground truth string DR-4

IN RULESPEAK®

A loop's [commit short](#) is computed as the first 7 character(s) of the [commit hash](#).

▼ **Inputs** 1 fields this rule reads

● **Commit Hash**

Loop

Full git SHA where this loop row first landed in `effortless-rulebook.json` — the replay anchor for this turn of the Effortless loop.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

First 7 characters of `CommitHash` for display and git checkout.

● **Commit Date** Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

ISO date (`YYYY-MM-DD`) of the commit where this loop landed.



Commit Message

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Git commit subject line for this loop's landing commit.



Git Tag

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Optional milestone tag pointing at this loop's landing commit (sp-loop-milestone-). Used for discovery replay without memorizing SHAs.*

Study

A study is identified by its name.



Name

Calculated

1 hop from ground truth

string

DR-5

IN RULESPEAK®

A study's name is the same as its study ID.

▼ **Inputs** 1 fields this rule reads

Study ID

Study

Unique slug for this study (e.g. kidney-1986, berkeley-1973).

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Display label, mirrors StudyId.



Title

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Human-readable title of the study.



Source

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Citation or description of the original data source.

Source URL Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

URL to the original source.

Publication Year Ground truth integer

Ground truth. Written directly to the database (datatype `integer`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Year the study was published. Used for temporal stratification and pre-registration dating.

Domain Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Research domain: medicine | epidemiology | social-science | education | legal | sports | economics | synthetic. Enables cross-domain SignalPurity stratification.

Is Synthetic Ground truth boolean

Ground truth. Written directly to the database (datatype `boolean`); not computed from anything else. Everything derived ultimately traces back to values like this one.

TRUE for constructed/counterfactual studies; FALSE for real published studies. Filters the corpus for empirical claims.

Σ Total Cases Aggregation 1 hop from ground truth integer DR-6

IN RULESPEAK®

A study's total cases is the total cases across the case cells related to the study.

▼ Inputs 3 fields this rule reads

● **Cases** Case Cell
Total number of patients in this cell. RAW INPUT
— never derived.

→ **Study** Case Cell
FK to the parent Study.

● Study ID

Study

Unique slug for this study (e.g. kidney-1986, berkeley-1973).

- **Ground truth at the leaves** 3 the raw values this ultimately depends on

Total number of cases (trials) across all CaseCells in this study.



Cell Count

Aggregation

1 hop from ground truth

integer

DR-7

IN RULESPEAK®

A study's cell count is the number of case cells related to the study.

- ▼ **Inputs** 2 fields this rule reads

→ Study

Case Cell

FK to the parent Study.

● Study ID

Study

Unique slug for this study (e.g. kidney-1986, berkeley-1973).

- **Ground truth at the leaves** 2 the raw values this ultimately depends on

Number of CaseCell rows for this study.



Tradition ID

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

FK → ResearchTraditions.TraditionId. The scholarly tradition whose framing best characterizes this study's paradox type. Synthetic studies inherit the tradition of their conceptual origin.



Primary Researcher ID

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

FK → Researchers.ResearcherId. The researcher most associated with first surfacing or canonically citing this study in the Simpson's Paradox literature.



Tradition Name

Lookup

1 hop from ground truth

string

DR-8

IN RULESPEAK®

A study's tradition name is the value of LOOKUP(TraditionId, ResearchTraditions[TraditionId], ResearchTraditions[Name]).

▼ **Inputs** 1 fields this rule reads

● **Tradition ID**

Study

FK → ResearchTraditions.TraditionId. The scholarly tradition whose framing best characterizes this study's paradox type. Synthetic studies inherit the tradition of their conceptual origin.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: ResearchTraditions.Name via TraditionId.



Tradition Core Conern

Lookup

1 hop from ground truth

string

DR-9

IN RULESPEAK®

A study's tradition core conern is the value of LOOKUP(TraditionId, ResearchTraditions[TraditionId], ResearchTraditions[CoreConcern]).

▼ **Inputs** 1 fields this rule reads

● **Tradition ID**

Study

FK → ResearchTraditions.TraditionId. The scholarly tradition whose framing best characterizes this study's paradox type. Synthetic studies inherit the tradition of their conceptual origin.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: ResearchTraditions.CoreConcern via TraditionId.



Primary Researcher Name

Lookup

1 hop from ground truth

string

DR-10

IN RULESPEAK®

A study's primary researcher name is the value of LOOKUP(PrimaryResearcherId, Researchers[ResearcherId], Researchers[Name]).

▼ **Inputs** 1 fields this rule reads

● **Primary Researcher ID**

Study

FK → Researchers.ResearcherId. The researcher most associated with first surfacing or canonically citing this study in the Simpson's Paradox literature.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: *Researchers.Name* via *PrimaryResearcherId*.

↗ **Primary Researcher Affiliation** Lookup 1 hop from ground truth string DR-11

IN RULESPEAK®

A study's **primary researcher affiliation** is the value of LOOKUP(PrimaryResearcherId, Researchers[ResearcherId], Researchers[Affiliation]).

▼ **Inputs** 1 fields this rule reads

- **Primary Researcher ID** Study
FK → Researchers.ResearcherId. The researcher most associated with first surfacing or canonically citing this study in the Simpson's Paradox literature.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: *Researchers.Affiliation* via *PrimaryResearcherId*.

↗ **Distortion Type** Lookup 1 hop from ground truth string DR-12

IN RULESPEAK®

A study's **distortion type** is the value of LOOKUP(StudyId, TreatmentRankings[Study], TreatmentRankings[DistortionType]).

▼ **Inputs** 1 fields this rule reads

- **Study ID** Study
Unique slug for this study (e.g. kidney-1986, berkeley-1973).

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: *TreatmentRankings.DistortionType* for this study — the geometric classification of its paradox.

↗ **Policy Implication** Lookup 1 hop from ground truth string DR-13

IN RULESPEAK®

A study's **policy implication** is the value of LOOKUP(StudyId, TreatmentRankings[Study], TreatmentRankings[CorrectedPolicyImplication]).

▼ **Inputs** 1 fields this rule reads

● Study ID

Study

Unique slug for this study (e.g. kidney-1986, berkeley-1973).

- **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: TreatmentRankings.CorrectedPolicyImplication for this study.



Allocation Distortion

Lookup

1 hop from ground truth

float

DR-14

IN RULESPEAK®

A study's [allocation distortion](#) is the value of LOOKUP(StudyId, TreatmentRankings[Study], TreatmentRankings[AllocationDistortion]).

- ▼ **Inputs** 1 fields this rule reads

● Study ID

Study

Unique slug for this study (e.g. kidney-1986, berkeley-1973).

- **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: TreatmentRankings.AllocationDistortion for this study — the scalar severity of allocation-induced distortion.



Signal Purity

Lookup

1 hop from ground truth

float

DR-15

IN RULESPEAK®

A study's [signal purity](#) is the value of LOOKUP(StudyId, TreatmentRankings[Study], TreatmentRankings[SignalPurity]).

- ▼ **Inputs** 1 fields this rule reads

● Study ID

Study

Unique slug for this study (e.g. kidney-1986, berkeley-1973).

- **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: TreatmentRankings.SignalPurity for this study — ratio of true signal to total gap width.



Is Sign Flip

Lookup

1 hop from ground truth

boolean

DR-16

IN RULESPEAK®

A study's [is sign flip](#) is the value of LOOKUP(StudyId, TreatmentRankings[Study], TreatmentRankings[IsSignFlip]).

▼ **Inputs** 1 fields this rule reads

● **Study ID**

Study

Unique slug for this study (e.g. kidney-1986, berkeley-1973).

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: TreatmentRankings.IsSignFlip for this study — true if the paradox satisfies the continuous sign-flip definition.

↗ **Corrected Winner** **Lookup** 1 hop from ground truth string DR-17

IN RULESPEAK®

A study's [corrected winner](#) is the value of LOOKUP(StudyId, TreatmentRankings[Study], TreatmentRankings[CorrectedWinner]).

▼ **Inputs** 1 fields this rule reads

● **Study ID**

Study

Unique slug for this study (e.g. kidney-1986, berkeley-1973).

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: TreatmentRankings.CorrectedWinner for this study — which treatment wins after allocation correction.

↗ **Confounding Variable** **Lookup** 1 hop from ground truth string DR-18

IN RULESPEAK®

A study's [confounding variable](#) is the value of LOOKUP(StudyId, StratumVariables[Study], StratumVariables[VariableName]).

▼ **Inputs** 1 fields this rule reads

● **Study ID**

Study

Unique slug for this study (e.g. kidney-1986, berkeley-1973).

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: `StratumVariables.VariableName` for this study — the named stratifying variable.

Causal Role Lookup 1 hop from ground truth string DR-19

IN RULESPEAK®

A study's [causal role](#) is the value of `LOOKUP(StudyId, StratumVariables[Study], StratumVariables[CausalRole])`.

▼ **Inputs** 1 fields this rule reads

● **Study ID** Study
Unique slug for this study (e.g. kidney-1986, berkeley-1973).

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: `StratumVariables.CausalRole` for this study — confounder, mediator, collider, proxy, or contested.

Ingestion Cell Parity Calculated 2 hops from ground truth boolean DR-20

IN RULESPEAK®

A study is flagged ingestion cell parity if the value of `AND(LOOKUP(StudyId, TreatmentRankings[Study], TreatmentRankings[StratumCount]) >= 2, CellCount = LOOKUP(StudyId, TreatmentRankings[Study], TreatmentRankings[StratumCount]) * 2)`.

▼ **Inputs** 2 fields this rule reads

● **Study ID** Study
Unique slug for this study (e.g. kidney-1986, berkeley-1973).

Σ **Cell Count** Study
Count of case cells whose study is study ID.

► **Ground truth at the leaves** 2 the raw values this ultimately depends on

TRUE when $CellCount = 2 \times StratumCount$.

Ingestion Compliance Calculated 3 hops from ground truth string DR-21

IN RULESPEAK®

A study's [ingestion compliance](#) is computed as the value of `IF(IsSynthetic, "exempt-synthetic", IF(AND(IngestionCellParity, LOOKUP(StudyId, StratumVariables[Study], StratumVariables[VariableName]) <> ""), "all", "partial"))`.

▼ **Inputs** 3 fields this rule reads

● **Is Synthetic** Study

 **Ingestion Cell Parity** Study

TRUE for constructed/counterfactual studies;
FALSE for real published studies. Filters the
corpus for empirical claims.

```
AND(LOOKUP({{StudyId}},  
TreatmentRankings[Study],  
TreatmentRankings[StratumCount]) >= 2,  
{{CellCount}} = LOOKUP({{StudyId}},  
TreatmentRankings[Study],  
TreatmentRankings[StratumCount]) * 2).
```

● Study ID

Study

Unique slug for this study (e.g. kidney-1986,
berkeley-1973).

► **Ground truth at the leaves** 3 the raw values this ultimately depends on

all | partial | exempt-synthetic.

● Is Control Study Ground truth boolean

Ground truth. Written directly to the database (datatype `boolean`); not computed from anything else. Everything derived ultimately traces back to values like this one.

TRUE if this study is a control (no Simpson's paradox expected/found); FALSE for the main Simpson's Paradox corpus.

● Control Domain Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Domain slug for control studies (e.g. medicine-rct, education-higher). NULL for non-control studies.

● Valid Time Start Ground truth integer

Ground truth. Written directly to the database (datatype `integer`); not computed from anything else. Everything derived ultimately traces back to values like this one.

First year of the real-world cohort period this study's data covers (valid-time, not publication year). Defaults to PublicationYear when the true cohort window is not independently known.

● Valid Time End Ground truth integer

Ground truth. Written directly to the database (datatype `integer`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Last year of the real-world cohort period this study's data covers (valid-time, not publication year). Defaults to PublicationYear when the true cohort window is not independently known.



Valid Time Midpoint

Calculated

1 hop from ground truth

float

DR-22

IN RULESPEAK®

The study's valid time midpoint is determined by the following priority:

1. the valid time start plus the valid time end divided by 2 if all of these hold:
 - the valid time start has a value
 - the valid time end has a value
2. the publication year in all other cases

▼ Inputs 3 fields this rule reads

● Valid Time Start

Study

First year of the real-world cohort period this study's data covers (valid-time, not publication year). Defaults to PublicationYear when the true cohort window is not independently known.

● Valid Time End

Study

Last year of the real-world cohort period this study's data covers (valid-time, not publication year). Defaults to PublicationYear when the true cohort window is not independently known.

● Publication Year

Study

Year the study was published. Used for temporal stratification and pre-registration dating.

► Ground truth at the leaves 3 the raw values this ultimately depends on

Midpoint year of the valid-time cohort window, used to bucket studies by when their underlying data actually occurred rather than when the paper was published.



Valid Time Decade

Calculated

2 hops from ground truth

integer

DR-23

IN RULESPEAK®

The study's valid time decade is determined by the following priority:

1. an empty string if the valid time midpoint is blank
2. the valid time midpoint divided by 10 rounded down to a whole number times 10 in all other cases

▼ Inputs 1 fields this rule reads

f Valid Time Midpoint

Study

If valid time start is not "" and valid time end is not "" then ((valid time start plus valid time end) divided by 2), otherwise publication year.

► Ground truth at the leaves 3 the raw values this ultimately depends on

ValidTimeMidpoint floored to the decade (e.g. 1973 -> 1970). The bucket key ConfounderDistortionTimeline aggregates over.

Treatment

A treatment is identified by its name and is related to a study.

f **Name** Calculated 1 hop from ground truth string DR-24

IN RULESPEAK®

A treatment's name is the same as its treatment ID.

▼ **Inputs** 1 fields this rule reads

● **Treatment ID** Treatment
Unique slug (e.g. kidney-1986-A, kidney-1986-B).

► **Ground truth at the leaves** 1 the raw values this ultimately depends on
Display label.

... **Study** Relationship string

A pointer to Study. It is not derived from anything else — it is the link the rest of the DAG hangs off.

FK to the parent Study.

● **Treatment Label** Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Short label used in CaseCells and comparisons (e.g. A, B).

● **Description** Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Plain-language description of what this treatment is.

Σ **Total Cases** Aggregation 1 hop from ground truth integer DR-25

A treatment's total cases is the total cases across the case cells related to the treatment.

▼ **Inputs** 5 fields this rule reads

● **Cases**

Case Cell

Total number of patients in this cell. RAW INPUT
— never derived.

... Study

Case Cell

FK to the parent Study.

... Study

Treatment

FK to the parent Study.

● **Treatment Label**

Case Cell

The treatment applied in this cell (matches
Treatments.TreatmentLabel).

● **Treatment Label**

Treatment

Short label used in CaseCells and comparisons
(e.g. A, B).

► **Ground truth at the leaves** 5 the raw values this ultimately depends on

Total cases for this treatment across all strata.

**Total Successes**

Aggregation

1 hop from ground truth

integer

DR-26

A treatment's total successes is the total successes across the case cells related to the treatment.

▼ **Inputs** 5 fields this rule reads

● **Successes**

Case Cell

Number of patients in this cell who had a
successful outcome. RAW INPUT — never
derived.

... Study

Case Cell

FK to the parent Study.

... Study

Treatment

FK to the parent Study.

● **Treatment Label**

Case Cell

The treatment applied in this cell (matches
Treatments.TreatmentLabel).

● **Treatment Label**

Treatment

Short label used in CaseCells and comparisons
(e.g. A, B).

► **Ground truth at the leaves** 5 the raw values this ultimately depends on

Total successes for this treatment across all strata.



Pooled Success Rate

Calculated

2 hops from ground truth

number

DR-27

IN RULESPEAK®

The treatment's pooled success rate is determined by the following priority:

1. an empty string if the total cases is 0
2. the total successes divided by the total cases in all other cases

▼ Inputs 2 fields this rule reads

Σ Total Cases

Treatment

Sum of cases across all case cells whose study is study and treatment label is treatment label.

Σ Total Successes

Treatment

Sum of successes across all case cells whose study is study and treatment label is treatment label.

► Ground truth at the leaves 6 the raw values this ultimately depends on

Pooled (across all strata) success rate for this treatment: $TotalSuccesses / TotalCases$.

Strata

A strata is identified by its name and is related to a study.



Name

Calculated

1 hop from ground truth

string

DR-28

IN RULESPEAK®

A strata's name is the same as its stratum ID.

▼ Inputs 1 fields this rule reads

● Stratum ID

Strata

Unique slug (e.g. kidney-1986-small, kidney-1986-large).

► Ground truth at the leaves 1 the raw values this ultimately depends on

Display label.



Study

Relationship

string

A pointer to Study. It is not derived from anything else — it is the link the rest of the DAG hangs off.

FK to the parent Study.

● Stratum Label Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Short label used in CaseCells (e.g. *small, large, dept-A*).

● Description Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Plain-language description of what defines this stratum.

Σ Total Cases Aggregation 1 hop from ground truth integer DR-29

IN RULESPEAK®

A strata's total cases is the total cases across the case cells related to the strata.

▼ **Inputs** 5 fields this rule reads

● **Cases** Case Cell
Total number of patients in this cell. RAW INPUT
— never derived.

→ **Study** Case Cell
FK to the parent Study.

→ **Study** Strata
FK to the parent Study.

● **Stratum Label** Case Cell
The stratum this cell belongs to (matches `Strata.StratumLabel`).

● **Stratum Label** Strata
Short label used in CaseCells (e.g. *small, large, dept-A*).

► **Ground truth at the leaves** 5 the raw values this ultimately depends on

Total cases in this stratum across all treatments.

Case Cell

A case cell is identified by its name and is related to a study.

f Name Calculated 1 hop from ground truth string DR-30

IN RULESPEAK®

A case cell's name is the same as its case cell ID.

▼ **Inputs** 1 fields this rule reads

● **Case Cell ID** Case Cell
Unique slug (e.g. kidney-1986-small-A).

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Display label.

Study Relationship string

A pointer to [Study](#). It is not derived from anything else — it is the link the rest of the DAG hangs off.

FK to the parent Study.

Stratum Label Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

The stratum this cell belongs to (matches `Strata.StratumLabel`).

Treatment Label Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

The treatment applied in this cell (matches `Treatments.TreatmentLabel`).

Successes Ground truth integer

Ground truth. Written directly to the database (datatype `integer`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Number of patients in this cell who had a successful outcome. RAW INPUT — never derived.

Cases Ground truth integer

Ground truth. Written directly to the database (datatype `integer`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Total number of patients in this cell. RAW INPUT — never derived.

f Cell Success Rate Calculated 1 hop from ground truth number DR-31

IN RULESPEAK®

The case cell's cell success rate is determined by the following priority:

1. **0** if the cases is 0
2. **the successes divided by the cases** in all other cases

▼ **Inputs** 2 fields this rule reads

● Cases

Case Cell

Total number of patients in this cell. RAW INPUT — never derived.

● Successes

Case Cell

Number of patients in this cell who had a successful outcome. RAW INPUT — never derived.

► **Ground truth at the leaves** 2 the raw values this ultimately depends on

Success rate within this cell: Successes / Cases. The first derived fact in the DAG. Fixed loop-90: this rulebook's IF(x=0,"",...) idiom transpiles to a bare "::numeric cast that errors at runtime for the one CaseCells row with Cases=0 (surfaced only once _erb_tr_metrics stopped masking it). Returns 0 for the degenerate zero-cases cell instead.

Σ Total Cases for Treatment Aggregation 1 hop from ground truth integer DR-32

IN RULESPEAK®

A case cell's total cases for treatment is the total cases across the case cells related to the case cell.

▼ **Inputs** 3 fields this rule reads

● Cases

Case Cell

Total number of patients in this cell. RAW INPUT — never derived.

→ Study

Case Cell

FK to the parent Study.

● Treatment Label

Case Cell

The treatment applied in this cell (matches Treatments.TreatmentLabel).

► **Ground truth at the leaves** 3 the raw values this ultimately depends on

Total cases across ALL strata for this treatment in this study — the denominator for `TreatmentExposureFraction`.

f Treatment Exposure Fraction Calculated 2 hops from ground truth number DR-33

IN RULESPEAK®

The case cell's treatment exposure fraction is determined by the following priority:

1. an empty string if the total cases for treatment is 0
2. the cases divided by the total cases for treatment in all other cases

▼ Inputs 2 fields this rule reads

Σ Total Cases for Treatment Case Cell

Sum of cases across all case cells whose study is study and treatment label is treatment label.

● Cases Case Cell

Total number of patients in this cell. RAW INPUT — never derived.

► Ground truth at the leaves 3 the raw values this ultimately depends on

Fraction of this treatment's total cases that fall in this stratum: $Cases / TotalCasesForTreatment$. High imbalance across strata is the mechanism of confounding — when treatments are allocated very differently across strata, the pooled rate conflates treatment effect with stratum difficulty.

f Is Valid Ingestion Cell Calculated 1 hop from ground truth boolean DR-34

IN RULESPEAK®

A case cell is considered a valid ingestion cell if all of the following:

- the cases is at least 0
- the successes is at least 0
- the successes is at most the cases

▼ Inputs 2 fields this rule reads

● Cases Case Cell

Total number of patients in this cell. RAW INPUT — never derived.

● Successes Case Cell

Number of patients in this cell who had a successful outcome. RAW INPUT — never derived.

► Ground truth at the leaves 2 the raw values this ultimately depends on

TRUE when $Cases \geq 0$, $Successes \geq 0$, and $Successes \leq Cases$ (loop-92: a stratum-treatment combination can legitimately have zero observed cases in real source data -- e.g. `rdatasets-mtcars-engine` has zero 8-cylinder straight-engine cars -- so $Cases=0$ is a valid ingestion cell, not a defect).

Stratum Summary

A stratum summary is identified by its [name](#) and is related to a [study](#).

f **Name** Calculated 1 hop from ground truth string DR-35

IN RULESPEAK®

A stratum summary's [name](#) is the same as its stratum summary ID.

▾ **Inputs** 1 fields this rule reads

● **Stratum Summary ID** Stratum Summary
Unique slug (e.g. kidney-1986-small-A).

▸ **Ground truth at the leaves** 1 the raw values this ultimately depends on

Display label.

... **Study** Relationship string

A pointer to [Study](#). It is not derived from anything else — it is the link the rest of the DAG hangs off.

FK to the parent Study.

● **Stratum Label** Ground truth string

Ground truth. Written directly to the database (datatype string); not computed from anything else. Everything derived ultimately traces back to values like this one.

Which stratum (matches Strata.StratumLabel).

● **Treatment Label** Ground truth string

Ground truth. Written directly to the database (datatype string); not computed from anything else. Everything derived ultimately traces back to values like this one.

Which treatment (matches Treatments.TreatmentLabel).

Σ **Stratum Successes** Aggregation 1 hop from ground truth integer DR-36

IN RULESPEAK®

A stratum summary's [stratum successes](#) is the total [successes](#) across the [case cells](#) related to the stratum summary.

▼ **Inputs** 7 fields this rule reads

● **Successes**

Case Cell

Number of patients in this cell who had a successful outcome. RAW INPUT — never derived.

→ **Study**

Case Cell

FK to the parent Study.

→ **Study**

Stratum Summary

FK to the parent Study.

● **Stratum Label**

Case Cell

The stratum this cell belongs to (matches Strata.StratumLabel).

● **Stratum Label**

Stratum Summary

Which stratum (matches Strata.StratumLabel).

● **Treatment Label**

Case Cell

The treatment applied in this cell (matches Treatments.TreatmentLabel).

● **Treatment Label**

Stratum Summary

Which treatment (matches Treatments.TreatmentLabel).

► **Ground truth at the leaves** 7 the raw values this ultimately depends on

Total successes in this (stratum, treatment) cell for this study.

Σ Stratum Cases Aggregation 1 hop from ground truth integer DR-37

IN RULESPEAK®

A stratum summary's stratum cases is the total cases across the case cells related to the stratum summary.

▼ **Inputs** 7 fields this rule reads

● **Cases**

Case Cell

Total number of patients in this cell. RAW INPUT — never derived.

→ **Study**

Case Cell

FK to the parent Study.

→ **Study**

Stratum Summary

FK to the parent Study.

● **Stratum Label**

Case Cell

The stratum this cell belongs to (matches Strata.StratumLabel).

● **Stratum Label**

Stratum Summary

Which stratum (matches Strata.StratumLabel).

● **Treatment Label**

Case Cell

The treatment applied in this cell (matches Treatments.TreatmentLabel).

● Treatment Label

Stratum Summary

Which treatment (matches
Treatments.TreatmentLabel).

► **Ground truth at the leaves** 7 the raw values this ultimately depends on

Total cases in this (stratum, treatment) cell for this study.



Stratum Success Rate

Calculated

2 hops from ground truth

number

DR-38

IN RULESPEAK®

The stratum summary's stratum success rate is determined by the following priority:

1. **an empty string** if the stratum cases is 0
2. **the stratum successes divided by the stratum cases** in all other cases

▼ **Inputs** 2 fields this rule reads

Σ Stratum Cases

Stratum Summary

Sum of cases across all case cells whose study is
study, stratum label is stratum label, and
treatment label is treatment label.

Σ Stratum Successes

Stratum Summary

Sum of successes across all case cells whose
study is study, stratum label is stratum label, and
treatment label is treatment label.

► **Ground truth at the leaves** 8 the raw values this ultimately depends on

Success rate within this stratum for this treatment: StratumSuccesses / StratumCases. The true per-stratum picture.



Stratum Successes a

Aggregation

1 hop from ground truth

integer

DR-39

IN RULESPEAK®

A stratum summary's stratum successes a is the total successes across the case cells related to the stratum summary.

▼ **Inputs** 6 fields this rule reads

● Successes

Case Cell

Number of patients in this cell who had a
successful outcome. RAW INPUT — never
derived.

↗ Study

Case Cell

FK to the parent Study.

↗ Study

Stratum Summary

FK to the parent Study.

● Stratum Label

Case Cell

The stratum this cell belongs to (matches
Strata.StratumLabel).

● **Stratum Label** Stratum Summary

Which stratum (matches Strata.StratumLabel).

● **Treatment Label** Case Cell

The treatment applied in this cell (matches Treatments.TreatmentLabel).

► **Ground truth at the leaves** 6 the raw values this ultimately depends on

| *Successes for treatment A in this (study, stratum) — used to determine the per-stratum winner.*

Σ Stratum Cases a Aggregation 1 hop from ground truth integer DR-40

IN RULESPEAK®

A stratum summary's stratum cases a is the total cases across the case cells related to the stratum summary.

▼ **Inputs** 6 fields this rule reads

● **Cases** Case Cell

Total number of patients in this cell. RAW INPUT — never derived.

→ **Study** Case Cell

FK to the parent Study.

→ **Study** Stratum Summary

FK to the parent Study.

● **Stratum Label** Case Cell

The stratum this cell belongs to (matches Strata.StratumLabel).

● **Stratum Label** Stratum Summary

Which stratum (matches Strata.StratumLabel).

● **Treatment Label** Case Cell

The treatment applied in this cell (matches Treatments.TreatmentLabel).

► **Ground truth at the leaves** 6 the raw values this ultimately depends on

| *Cases for treatment A in this (study, stratum).*

f Stratum Rate a Calculated 2 hops from ground truth number DR-41

IN RULESPEAK®

The stratum summary's stratum rate a is determined by the following priority:

1. **an empty string** if the stratum cases a is 0
2. **the stratum successes a divided by the stratum cases a** in all other cases

▼ **Inputs** 2 fields this rule reads

Σ Stratum Cases a Stratum Summary

Sum of cases across all case cells whose study is study, stratum label is stratum label, and

Σ Stratum Successes a Stratum Summary

Sum of successes across all case cells whose study is study, stratum label is stratum label, and

treatment label is "A".

treatment label is "A".

► **Ground truth at the leaves** 7 the raw values this ultimately depends on

Success rate for treatment A in this stratum.

Σ **Stratum Successes B** Aggregation 1 hop from ground truth integer DR-42

IN RULESPEAK®

A stratum summary's stratum successes b is the total successes across the case cells related to the stratum summary.

▼ **Inputs** 6 fields this rule reads

● **Successes**

Case Cell

Number of patients in this cell who had a successful outcome. RAW INPUT — never derived.

→ **Study**

Case Cell

FK to the parent Study.

→ **Study**

Stratum Summary

FK to the parent Study.

● **Stratum Label**

Case Cell

The stratum this cell belongs to (matches Strata.StratumLabel).

● **Stratum Label**

Stratum Summary

Which stratum (matches Strata.StratumLabel).

● **Treatment Label**

Case Cell

The treatment applied in this cell (matches Treatments.TreatmentLabel).

► **Ground truth at the leaves** 6 the raw values this ultimately depends on

Successes for treatment B in this (study, stratum).

Σ **Stratum Cases B** Aggregation 1 hop from ground truth integer DR-43

IN RULESPEAK®

A stratum summary's stratum cases b is the total cases across the case cells related to the stratum summary.

▼ **Inputs** 6 fields this rule reads

● **Cases**

Case Cell

Total number of patients in this cell. RAW INPUT — never derived.

→ **Study**

Case Cell

FK to the parent Study.

→ **Study**

Stratum Summary

● **Stratum Label**

Case Cell

FK to the parent Study.

The stratum this cell belongs to (matches Strata.StratumLabel).

● Stratum Label

Stratum Summary

Which stratum (matches Strata.StratumLabel).

● Treatment Label

Case Cell

The treatment applied in this cell (matches Treatments.TreatmentLabel).

► **Ground truth at the leaves** 6 the raw values this ultimately depends on

Cases for treatment B in this (study, stratum).



Stratum Rate B

Calculated

2 hops from ground truth

number

DR-44

IN RULESPEAK®

The stratum summary's stratum rate b is determined by the following priority:

1. **an empty string** if the stratum cases b is 0
2. **the stratum successes b divided by the stratum cases b** in all other cases

▼ **Inputs** 2 fields this rule reads

Σ Stratum Cases B

Stratum Summary

Sum of cases across all case cells whose study is study, stratum label is stratum label, and treatment label is "B".

Σ Stratum Successes B

Stratum Summary

Sum of successes across all case cells whose study is study, stratum label is stratum label, and treatment label is "B".

► **Ground truth at the leaves** 7 the raw values this ultimately depends on

Success rate for treatment B in this stratum.



Stratum Winner

Calculated

3 hops from ground truth

string

DR-45

IN RULESPEAK®

The stratum summary's stratum winner is determined by the following priority:

1. **"A"** if the stratum rate a is greater than the stratum rate b
2. **"B"** in all other cases

▼ **Inputs** 2 fields this rule reads

f Stratum Rate a

Stratum Summary

If stratum cases a equals 0 then "", otherwise (stratum successes a divided by stratum cases a).

f Stratum Rate B

Stratum Summary

If stratum cases b equals 0 then "", otherwise (stratum successes b divided by stratum cases b).

► **Ground truth at the leaves** 7 the raw values this ultimately depends on

Which treatment wins in this stratum: 'A' if $\text{StratumRateA} > \text{StratumRateB}$, else 'B'. Used by *TreatmentRankings* to detect paradox.

Σ Study Total Cases Aggregation 1 hop from ground truth integer DR-46

IN RULESPEAK®

A stratum summary's study total cases is the total cases across the case cells related to the stratum summary.

▼ **Inputs** 3 fields this rule reads

● **Cases** Case Cell
Total number of patients in this cell. RAW INPUT
— never derived.

↗ **Study** Case Cell
FK to the parent Study.

↗ **Study** Stratum Summary
FK to the parent Study.

► **Ground truth at the leaves** 3 the raw values this ultimately depends on

Total cases across ALL strata and ALL treatments in this study. The denominator for StratumFraction.

Σ Stratum Total Cases Aggregation 1 hop from ground truth integer DR-47

IN RULESPEAK®

A stratum summary's stratum total cases is the total cases across the case cells related to the stratum summary.

▼ **Inputs** 5 fields this rule reads

● **Cases** Case Cell
Total number of patients in this cell. RAW INPUT
— never derived.

↗ **Study** Case Cell
FK to the parent Study.

↗ **Study** Stratum Summary
FK to the parent Study.

● **Stratum Label** Case Cell
The stratum this cell belongs to (matches `Strata.StratumLabel`).

● **Stratum Label** Stratum Summary
Which stratum (matches `Strata.StratumLabel`).

► **Ground truth at the leaves** 5 the raw values this ultimately depends on

Total cases in this stratum across both treatments (A + B combined). Numerator for StratumFraction.

f Stratum Fraction Calculated 2 hops from ground truth number DR-48

IN RULESPEAK®

The stratum summary's stratum fraction is determined by the following priority:

1. **an empty string** if the study total cases is 0
2. **the stratum total cases divided by the study total cases** in all other cases

▼ Inputs 2 fields this rule reads

Σ **Study Total Cases** Stratum Summary
Sum of cases across all case cells whose study is study.

Σ **Stratum Total Cases** Stratum Summary
Sum of cases across all case cells whose study is study and stratum label is stratum label.

► Ground truth at the leaves 5 the raw values this ultimately depends on

Fraction of the study's total cases that fall in this stratum: $\text{StratumTotalCases} / \text{StudyTotalCases}$. The weight the confounder assigns to each stratum. A large StratumFraction on a low-success stratum pulls the pooled rate down; when that stratum is also over-allocated to one treatment, it creates the reversal.

f Weighted Stratum Rate Calculated 3 hops from ground truth number DR-49

IN RULESPEAK®

The stratum summary's weighted stratum rate is determined by the following priority:

1. **an empty string** if the stratum fraction is blank
2. **the stratum fraction times the stratum success rate** in all other cases

▼ Inputs 2 fields this rule reads

f **Stratum Fraction** Stratum Summary
If study total cases equals 0 then "", otherwise (stratum total cases divided by study total cases).

f **Stratum Success Rate** Stratum Summary
If stratum cases equals 0 then "", otherwise (stratum successes divided by stratum cases).

► Ground truth at the leaves 8 the raw values this ultimately depends on

$\text{StratumFraction} \times \text{StratumSuccessRate}$. The contribution of this (stratum, treatment) row to the study-level pooled rate. Summing $\text{WeightedStratumRate}$ across all strata for a given treatment reconstructs that treatment's pooled rate from stratum weights — witnessing the mechanism equation.

Σ Treatment a Cases Here Aggregation 1 hop from ground truth integer DR-50

IN RULESPEAK®

A stratum summary's treatment a cases here is the total cases across the case cells related to the stratum summary.

▼ **Inputs** 6 fields this rule reads

● **Cases**

Case Cell

Total number of patients in this cell. RAW INPUT
— never derived.

→ **Study**

Case Cell

FK to the parent Study.

→ **Study**

Stratum Summary

FK to the parent Study.

● **Stratum Label**

Case Cell

The stratum this cell belongs to (matches Strata.StratumLabel).

● **Stratum Label**

Stratum Summary

Which stratum (matches Strata.StratumLabel).

● **Treatment Label**

Case Cell

The treatment applied in this cell (matches Treatments.TreatmentLabel).

► **Ground truth at the leaves** 6 the raw values this ultimately depends on

Cases for treatment A in this stratum — used to compute TreatmentA's allocation fraction in this stratum.

Σ **Treatment B Cases Here** Aggregation 1 hop from ground truth integer DR-51

IN RULESPEAK®

A stratum summary's treatment b cases here is the total cases across the case cells related to the stratum summary.

▼ **Inputs** 6 fields this rule reads

● **Cases**

Case Cell

Total number of patients in this cell. RAW INPUT
— never derived.

→ **Study**

Case Cell

FK to the parent Study.

→ **Study**

Stratum Summary

FK to the parent Study.

● **Stratum Label**

Case Cell

The stratum this cell belongs to (matches Strata.StratumLabel).

● **Stratum Label**

Stratum Summary

Which stratum (matches Strata.StratumLabel).

● **Treatment Label**

Case Cell

The treatment applied in this cell (matches Treatments.TreatmentLabel).

► **Ground truth at the leaves** 6 the raw values this ultimately depends on

Cases for treatment B in this stratum — used to compute TreatmentB's allocation fraction in this stratum.

Σ **Treatment a Total Cases** Aggregation 1 hop from ground truth integer DR-52

IN RULESPEAK®

A stratum summary's treatment a total cases is the total cases across the stratum summary's case cells that have a treatment label of "A".

▼ **Inputs** 4 fields this rule reads

● **Cases**

Case Cell

Total number of patients in this cell. RAW INPUT
— never derived.

→ **Study**

Case Cell

FK to the parent Study.

→ **Study**

Stratum Summary

FK to the parent Study.

● **Treatment Label**

Case Cell

The treatment applied in this cell (matches
Treatments.TreatmentLabel).

► **Ground truth at the leaves** 4 the raw values this ultimately depends on

Total cases for treatment A across all strata in this study — denominator for AllocationFractionA.

Σ **Treatment B Total Cases**

Aggregation

1 hop from ground truth

integer

DR-53

IN RULESPEAK®

A stratum summary's treatment b total cases is the total cases across the stratum summary's case cells that have a treatment label of "B".

▼ **Inputs** 4 fields this rule reads

● **Cases**

Case Cell

Total number of patients in this cell. RAW INPUT
— never derived.

→ **Study**

Case Cell

FK to the parent Study.

→ **Study**

Stratum Summary

FK to the parent Study.

● **Treatment Label**

Case Cell

The treatment applied in this cell (matches
Treatments.TreatmentLabel).

► **Ground truth at the leaves** 4 the raw values this ultimately depends on

Total cases for treatment B across all strata in this study — denominator for AllocationFractionB.

f **Allocation Fraction a**

Calculated

2 hops from ground truth

number

DR-54

IN RULESPEAK®

The stratum summary's allocation fraction a is determined by the following priority:

1. **an empty string** if the treatment a total cases is 0
2. **the treatment a cases here divided by the treatment a total cases** in all other cases

▼ **Inputs** 2 fields this rule reads

Σ **Treatment a Total Cases** Stratum Summary

Sum of cases across all case cells whose study is study and treatment label is "A".

Σ **Treatment a Cases Here** Stratum Summary

Sum of cases across all case cells whose study is study, stratum label is stratum label, and treatment label is "A".

► **Ground truth at the leaves** 6 the raw values this ultimately depends on

Fraction of treatment A's total cases that land in this stratum: $\text{TreatmentACasesHere} / \text{TreatmentATotalCases}$. Equivalent to $\text{TreatmentExposureFraction}$ for A.

f

Allocation Fraction B

Calculated

2 hops from ground truth

number

DR-55

IN RULESPEAK®

The stratum summary's allocation fraction b is determined by the following priority:

1. **an empty string** if the treatment b total cases is 0
2. **the treatment b cases here divided by the treatment b total cases** in all other cases

▼ **Inputs** 2 fields this rule reads

Σ **Treatment B Total Cases** Stratum Summary

Sum of cases across all case cells whose study is study and treatment label is "B".

Σ **Treatment B Cases Here** Stratum Summary

Sum of cases across all case cells whose study is study, stratum label is stratum label, and treatment label is "B".

► **Ground truth at the leaves** 6 the raw values this ultimately depends on

Fraction of treatment B's total cases that land in this stratum: $\text{TreatmentBCasesHere} / \text{TreatmentBTotalCases}$.

f

Allocation Bias

Calculated

3 hops from ground truth

number

DR-56

IN RULESPEAK®

The stratum summary's allocation bias is determined by the following priority:

1. **an empty string** if the allocation fraction a is blank
2. **the allocation fraction a minus the allocation fraction b** in all other cases

▼ **Inputs** 2 fields this rule reads

f **Allocation Fraction a** Stratum Summary

If treatment a total cases equals 0 then "", otherwise (treatment a cases here divided by treatment a total cases).

f **Allocation Fraction B** Stratum Summary

If treatment b total cases equals 0 then "", otherwise (treatment b cases here divided by treatment b total cases).

- **Ground truth at the leaves** 6 the raw values this ultimately depends on

AllocationFractionA minus AllocationFractionB: how much more of treatment A (relative to B) is concentrated in this stratum. Positive = A is over-allocated here; negative = B is over-allocated here. When AllocationBias is large and negative in a low-success stratum, treatment A is under-represented in the hard cases — making A look worse in the pooled view than it is in any stratum. This is the confounder's fingerprint as a signed number.

f Stratum Gap Calculated 3 hops from ground truth number DR-57

IN RULESPEAK®

The stratum summary's stratum gap is determined by the following priority:

1. **an empty string** if the stratum rate a is blank
2. **the stratum rate a minus the stratum rate b** in all other cases

- ▼ **Inputs** 2 fields this rule reads

f Stratum Rate a Stratum Summary
If stratum cases a equals 0 then "", otherwise
(stratum successes a divided by stratum cases a).

f Stratum Rate B Stratum Summary
If stratum cases b equals 0 then "", otherwise
(stratum successes b divided by stratum cases b).

- **Ground truth at the leaves** 7 the raw values this ultimately depends on

Signed rate difference within this stratum: StratumRateA minus StratumRateB. Positive = A wins here; negative = B wins here; zero = tied. When every stratum has StratumGap > 0 but the pooled gap is negative, the contradiction IS Simpson's Paradox made arithmetically visible in one column. Multiply StratumGap × StratumFraction and sum across strata to get the equal-weight pooled gap (the signal allocation would produce without confounding).

f Weighted Stratum Gap Calculated 4 hops from ground truth number DR-58

IN RULESPEAK®

The stratum summary's weighted stratum gap is determined by the following priority:

1. **an empty string** if the stratum gap is blank
2. **the stratum gap times the stratum fraction** in all other cases

- ▼ **Inputs** 2 fields this rule reads

f Stratum Gap Stratum Summary
If stratum rate a is "" then "", otherwise (stratum
rate a minus stratum rate b).

f Stratum Fraction Stratum Summary
If study total cases equals 0 then "", otherwise
(stratum total cases divided by study total cases).

- **Ground truth at the leaves** 7 the raw values this ultimately depends on

$\text{StratumGap} \times \text{StratumFraction}$: the contribution of this stratum's treatment-rate difference to the equal-weight pooled gap. Summing $\text{WeightedStratumGap}$ across all *TreatmentA* rows for a study gives $\text{WeightedStratumGapSum}$ — the pooled gap the study would show under balanced allocation. This is the per-stratum building block of the sign-flip test.



Abs Allocation Bias

Calculated

4 hops from ground truth

number

DR-59

IN RULESPEAK®

The stratum summary's abs allocation bias is determined by the following priority:

1. **an empty string** if the allocation bias is blank
2. **the absolute value of the allocation bias** in all other cases

▼ Inputs 1 fields this rule reads

f Allocation Bias

Stratum Summary

If allocation fraction a is "" then "", otherwise
(allocation fraction a minus allocation fraction b).

► Ground truth at the leaves 6 the raw values this ultimately depends on

$\text{ABS}(\text{AllocationBias})$. MAXIFS-friendly unsigned magnitude so $\text{TreatmentRankings.MaxStratumImbalance}$ can compute live via MAXIFS instead of a hand-written cache.



Abs Stratum Gap

Calculated

4 hops from ground truth

number

DR-60

IN RULESPEAK®

The stratum summary's abs stratum gap is determined by the following priority:

1. **an empty string** if the stratum gap is blank
2. **the absolute value of the stratum gap** in all other cases

▼ Inputs 1 fields this rule reads

f Stratum Gap

Stratum Summary

If stratum rate a is "" then "", otherwise (stratum
rate a minus stratum rate b).

► Ground truth at the leaves 7 the raw values this ultimately depends on

$\text{ABS}(\text{StratumGap})$. MAXIFS-friendly unsigned magnitude so $\text{TreatmentRankings.MaxStratumGap}$ can compute live via MAXIFS instead of a hand-written cache.

Model Summary

A model summary is identified by its name.

f Name Calculated 1 hop from ground truth string DR-61

IN RULESPEAK®

A model summary's name is the same as its model summary ID.

▼ **Inputs** 1 fields this rule reads

● **Model Summary ID** Model Summary
Unique identifier for this model instance.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Display label.

Σ Reversal Count Aggregation 8 hops from ground truth integer DR-62

IN RULESPEAK®

A model summary's reversal count is the number of treatment rankings related to the model summary.

▼ **Inputs** 1 fields this rule reads

f Is Reversal Treatment Ranking
If is sign flip is "" then "", otherwise is sign flip.

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

Number of studies with IsReversal=TRUE (sign-flip / allocation-driven reversal).

Σ Non Reversal Count Aggregation 8 hops from ground truth integer DR-63

IN RULESPEAK®

A model summary's non reversal count is the number of treatment rankings related to the model summary.

▼ **Inputs** 1 fields this rule reads

f Is Reversal Treatment Ranking
If is sign flip is "" then "", otherwise is sign flip.

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

Number of studies with *IsReversal*=FALSE.

f Study Count Calculated 9 hops from ground truth integer DR-64

IN RULESPEAK®

A model summary's study count is computed as the reversal count plus the non reversal count.

▼ Inputs 2 fields this rule reads

Σ **Reversal Count** Model Summary
Count of treatment rankings whose is reversal is true().

Σ **Non Reversal Count** Model Summary
Count of treatment rankings whose is reversal is false().

► Ground truth at the leaves 11 the raw values this ultimately depends on

Total number of TreatmentRankings in this model: ReversalCount + NonReversalCount.

Σ Explained Count Aggregation 9 hops from ground truth integer DR-65

IN RULESPEAK®

A model summary's explained count is the number of treatment rankings related to the model summary.

▼ Inputs 1 fields this rule reads

f **Is Paradox Explained** Treatment Ranking
Is reversal and confounders in study is greater than 0.

► Ground truth at the leaves 15 the raw values this ultimately depends on

Number of studies with IsParadoxExplained=TRUE (reversal AND confirmed confounder).

Σ Total Paradox Strength Aggregation 7 hops from ground truth number DR-66

IN RULESPEAK®

A model summary's total paradox strength is the total paradox strength across the treatment rankings related to the model summary.

▼ Inputs 2 fields this rule reads

f **Paradox Strength** Treatment Ranking

● **Treatment a** Treatment Ranking
Label for the first treatment in the comparison (matches Treatments.TreatmentLabel).

If stratum count equals 0 then "", otherwise
(pooled gap times (strata won by loser divided by
stratum count)).

- **Ground truth at the leaves** 11 the raw values this ultimately depends on

Sum of ParadoxStrength across all studies. Used to compute average.

f Avg Paradox Strength Calculated 10 hops from ground truth number DR-67

IN RULESPEAK®

The model summary's avg paradox strength is determined by the following priority:

1. **an empty string** if the study count is 0
2. **the total paradox strength divided by the study count** in all other cases

- ▼ **Inputs** 2 fields this rule reads

f **Study Count** Model Summary
Reversal count plus non reversal count.

Σ **Total Paradox Strength** Model Summary
Sum of paradox strength across all treatment
rankings whose treatment a is "A".

- **Ground truth at the leaves** 11 the raw values this ultimately depends on

*Average ParadoxStrength across all studies: TotalParadoxStrength / StudyCount. A scalar summary of how
paradox-rich this dataset is.*

Σ Type a Count Aggregation 8 hops from ground truth integer DR-68

IN RULESPEAK®

A model summary's type a count is the number of treatment rankings related to the model
summary.

- ▼ **Inputs** 1 fields this rule reads

f **Distortion Type** Treatment Ranking
If allocation distortion is "" then "", otherwise if is
sign flip then if is stratum unanimous then "A",
otherwise "B", otherwise if not is sign flip,
allocation distortion is greater than 0.01, and the
absolute value of signed pooled gap is greater
than (the absolute value of corrected gap plus
0.001) then "C+", otherwise if not is sign flip,
allocation distortion is greater than 0.01, and the
absolute value of signed pooled gap is less than

(the absolute value of corrected gap minus 0.001)
then "C-", otherwise "D".

- **Ground truth at the leaves** 11 the raw values this ultimately depends on

Studies classified as Type-A: sign flip AND full reversal (ReversalIntensity=1). The canonical Simpson's Paradox — allocation distortion reverses the pooled winner.

Σ **Danger Tier Count** Aggregation 9 hops from ground truth integer DR-69

IN RULESPEAK®

A model summary's danger tier count is the number of treatment rankings related to the model summary.

- ▼ **Inputs** 1 fields this rule reads

f **Screening Tier** Treatment Ranking

If distortion type is "A" or distortion type is "B"
then "DANGER", otherwise if distortion type is
"C+" or distortion type is "C-" then "CAUTION",
otherwise if distortion type is "D" then "SAFE",
otherwise "".

- **Ground truth at the leaves** 11 the raw values this ultimately depends on

Studies where TreatmentRankings.ScreeningTier = DANGER — pooled sign is wrong (DistortionType A or B).

Σ **Caution Tier Count** Aggregation 9 hops from ground truth integer DR-70

IN RULESPEAK®

A model summary's caution tier count is the number of treatment rankings related to the model summary.

- ▼ **Inputs** 1 fields this rule reads

f **Screening Tier** Treatment Ranking

If distortion type is "A" or distortion type is "B"
then "DANGER", otherwise if distortion type is
"C+" or distortion type is "C-" then "CAUTION",
otherwise if distortion type is "D" then "SAFE",
otherwise "".

- **Ground truth at the leaves** 11 the raw values this ultimately depends on

Studies where `TreatmentRankings.ScreeningTier = CAUTION` — pooled sign holds but magnitude is distorted (`DistortionType C+` or `C-`).

Σ Safe Tier Count Aggregation 9 hops from ground truth integer DR-71

IN RULESPEAK®

A model summary's safe tier count is the number of treatment rankings related to the model summary.

▼ **Inputs** 1 fields this rule reads

f **Screening Tier** Treatment Ranking

If distortion type is "A" or distortion type is "B" then "DANGER", otherwise if distortion type is "C+" or distortion type is "C-" then "CAUTION", otherwise if distortion type is "D" then "SAFE", otherwise "".

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

Studies where `TreatmentRankings.ScreeningTier = SAFE` — pooling is trustworthy (`DistortionType D`).

Σ Type B Count Aggregation 8 hops from ground truth integer DR-72

IN RULESPEAK®

A model summary's type b count is the number of treatment rankings related to the model summary.

▼ **Inputs** 1 fields this rule reads

f **Distortion Type** Treatment Ranking

If allocation distortion is "" then "", otherwise if is sign flip then if is stratum unanimous then "A", otherwise "B", otherwise if not is sign flip, allocation distortion is greater than 0.01, and the absolute value of signed pooled gap is greater than (the absolute value of corrected gap plus 0.001) then "C+", otherwise if not is sign flip, allocation distortion is greater than 0.01, and the absolute value of signed pooled gap is less than (the absolute value of corrected gap minus 0.001) then "C-", otherwise "D".

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

Studies classified as Type-B: sign flip but partial reversal ($\text{ReversalIntensity} < 1$). Allocation distortion creates a sign flip but the pooled signal partially reflects per-stratum reality.

Σ Type D Count Aggregation 8 hops from ground truth integer DR-73

IN RULESPEAK®

A model summary's type d count is the number of treatment rankings related to the model summary.

▼ **Inputs** 1 fields this rule reads

f **Distortion Type** Treatment Ranking

If allocation distortion is "" then "", otherwise if is sign flip then if is stratum unanimous then "A", otherwise "B", otherwise if not is sign flip, allocation distortion is greater than 0.01, and the absolute value of signed pooled gap is greater than (the absolute value of corrected gap plus 0.001) then "C+", otherwise if not is sign flip, allocation distortion is greater than 0.01, and the absolute value of signed pooled gap is less than (the absolute value of corrected gap minus 0.001) then "C-", otherwise "D".

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

Studies classified as Type-D: no sign flip, negligible distortion. Pooled analysis is trustworthy — allocation is sufficiently balanced.

Σ Type C Plus Count Aggregation 8 hops from ground truth integer DR-74

IN RULESPEAK®

A model summary's type c plus count is the number of treatment rankings related to the model summary.

▼ **Inputs** 1 fields this rule reads

f **Distortion Type** Treatment Ranking

If allocation distortion is "" then "", otherwise if is sign flip then if is stratum unanimous then "A", otherwise "B", otherwise if not is sign flip, allocation distortion is greater than 0.01, and the absolute value of signed pooled gap is greater than (the absolute value of corrected gap plus 0.001) then "C+", otherwise if not is sign flip, allocation distortion is greater than 0.01, and the

absolute value of signed pooled gap is less than (the absolute value of corrected gap minus 0.001) then "C-", otherwise "D".

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

Studies classified as Type-C+: sign flip with amplification (ReversalIntensity>1). Pooled effect overstates the true per-stratum effect.

Σ **Type C Minus Count** Aggregation 8 hops from ground truth integer DR-75

IN RULESPEAK®

A model summary's type c minus count is the number of treatment rankings related to the model summary.

▼ **Inputs** 1 fields this rule reads

f **Distortion Type** Treatment Ranking

If allocation distortion is "" then "", otherwise if is sign flip then if is stratum unanimous then "A", otherwise "B", otherwise if not is sign flip, allocation distortion is greater than 0.01, and the absolute value of signed pooled gap is greater than (the absolute value of corrected gap plus 0.001) then "C+", otherwise if not is sign flip, allocation distortion is greater than 0.01, and the absolute value of signed pooled gap is less than (the absolute value of corrected gap minus 0.001) then "C-", otherwise "D".

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

Studies classified as Type-C-: sign flip with attenuation. Pooled effect understates or masks the true per-stratum effect.

f **Type a Fraction** Calculated 10 hops from ground truth number DR-76

IN RULESPEAK®

The model summary's type a fraction is determined by the following priority:

1. an empty string if the study count is 0
2. the type a count divided by the study count in all other cases

▼ **Inputs** 2 fields this rule reads

f **Study Count** Model Summary

Reversal count plus non reversal count.

Σ **Type a Count** Model Summary

Count of treatment rankings whose distortion type is "A".

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

Fraction of studies that are Type-A (full canonical reversals). $\text{TypeACount} / \text{StudyCount}$.

f Distortion Taxonomy Coverage Calculated 9 hops from ground truth string DR-77

IN RULESPEAK®

A model summary's distortion taxonomy coverage is computed as "A:", followed by the type a count, followed by " B:", followed by the type b count, followed by " C+:", followed by the c amplification count, followed by " C-:", followed by the c compression count, followed by " D:", followed by the type d count.

▼ **Inputs** 5 fields this rule reads

Σ **Type a Count** Model Summary
Count of treatment rankings whose distortion type is "A".

Σ **Type B Count** Model Summary
Count of treatment rankings whose distortion type is "B".

Σ **C Amplification Count** Model Summary
Count of treatment rankings whose distortion type is "C+".

Σ **C Compression Count** Model Summary
Count of treatment rankings whose distortion type is "C-".

Σ **Type D Count** Model Summary
Count of treatment rankings whose distortion type is "D".

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

Human-readable summary of the four-type distribution: e.g. 'A:2 B:1 C:1 D:0'. The model's self-portrait of its own distortion geometry.

f Distortion Only Count Calculated 9 hops from ground truth integer DR-78

IN RULESPEAK®

A model summary's distortion only count is computed as the c amplification count plus the c compression count.

▼ **Inputs** 2 fields this rule reads

Σ **C Amplification Count** Model Summary

Σ **C Compression Count** Model Summary

Count of treatment rankings whose distortion type is "C+".

Count of treatment rankings whose distortion type is "C-".

- **Ground truth at the leaves** 11 the raw values this ultimately depends on

Number of studies with real allocation distortion but no sign flip: TypeCCount. The pooled winner is correct but the effect size is biased. Distinct from ZeroStrengthCount, which conflates Type-C (real distortion) with Type-D (neutral).

Σ C Amplification Count

Aggregation

8 hops from ground truth

integer

DR-79

IN RULESPEAK®

A model summary's c amplification count is the number of treatment rankings related to the model summary.

- ▼ **Inputs** 1 fields this rule reads

f **Distortion Type** Treatment Ranking

If allocation distortion is "" then "", otherwise if is sign flip then if is stratum unanimous then "A", otherwise "B", otherwise if not is sign flip, allocation distortion is greater than 0.01, and the absolute value of signed pooled gap is greater than (the absolute value of corrected gap plus 0.001) then "C+", otherwise if not is sign flip, allocation distortion is greater than 0.01, and the absolute value of signed pooled gap is less than (the absolute value of corrected gap minus 0.001) then "C-", otherwise "D".

- **Ground truth at the leaves** 11 the raw values this ultimately depends on

Number of studies classified as type C+ (allocation-amplified): pooled winner is correct but pooled gap overstates true advantage. Causes researcher overconfidence — may accept higher cost/risk than true effect warrants.

Σ C Compression Count

Aggregation

8 hops from ground truth

integer

DR-80

IN RULESPEAK®

A model summary's c compression count is the number of treatment rankings related to the model summary.

- ▼ **Inputs** 1 fields this rule reads

f **Distortion Type** Treatment Ranking

If allocation distortion is "" then "", otherwise if is sign flip then if is stratum unanimous then "A", otherwise "B", otherwise if not is sign flip, allocation distortion is greater than 0.01, and the absolute value of signed pooled gap is greater than (the absolute value of corrected gap plus 0.001) then "C+", otherwise if not is sign flip, allocation distortion is greater than 0.01, and the absolute value of signed pooled gap is less than (the absolute value of corrected gap minus 0.001) then "C-", otherwise "D".

- **Ground truth at the leaves** 11 the raw values this ultimately depends on

Number of studies classified as type C- (allocation-compressed): pooled winner is correct but pooled gap understates true advantage. Causes researcher underconfidence — effective treatment may be prematurely abandoned.

Σ Avg Signal Purity Aggregation 8 hops from ground truth number DR-81

IN RULESPEAK®

A model summary's avg signal purity is an aggregated value computed across the model summary's related records.

- ▼ **Inputs** 2 fields this rule reads

f **Signal Purity** Treatment Ranking

If corrected gap is "" then "", otherwise if (the absolute value of corrected gap plus allocation distortion) equals 0 then 1, otherwise (the absolute value of corrected gap divided by (the absolute value of corrected gap plus allocation distortion)).

● **Treatment Ranking ID** Treatment Ranking

Unique slug (e.g. kidney-1986-A-vs-B).

- **Ground truth at the leaves** 12 the raw values this ultimately depends on

Average SignalPurity across all TreatmentRankings. Below 0.5 means allocation noise exceeds true signal on average — the corpus is predominantly misleading at the pooled level. Above 0.5 means the corpus is predominantly trustworthy. Fixed loop-90: COUNTIFS(...,"<>")/SUMIFS(...,"<>") mistranslate to a literal '<>' string-equality filter (a known transpiler bug), always returning 0/NULL; replaced with SUM/COUNT, both of which transpile correctly.

Σ Sweep Corrected Gap Max Aggregation 2 hops from ground truth number DR-82

IN RULESPEAK®

A model summary's [sweep corrected gap max](#) is the largest [sweep corrected gap](#) across the [allocation sweep](#) related to the model summary.

▼ Inputs 2 fields this rule reads

f **Sweep Corrected Gap** Allocation Sweep
LOOKUP({{StudyId}},
SweepStudyConfig[StudyId],
SweepStudyConfig[SweepCorrectedGap]).

● **Study ID** Allocation Sweep
The study being swept — all 238 studies in the corpus.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Maximum SweepCorrectedGap across all AllocationSweep rows. Should equal SweepCorrectedGapMin — the invariant.

Σ **Sweep Corrected Gap Min** Aggregation 2 hops from ground truth number DR-83

IN RULESPEAK®

A model summary's [sweep corrected gap min](#) is the smallest [sweep corrected gap](#) across the [allocation sweep](#) related to the model summary.

▼ Inputs 2 fields this rule reads

f **Sweep Corrected Gap** Allocation Sweep
LOOKUP({{StudyId}},
SweepStudyConfig[StudyId],
SweepStudyConfig[SweepCorrectedGap]).

● **Study ID** Allocation Sweep
The study being swept — all 238 studies in the corpus.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Minimum SweepCorrectedGap across all AllocationSweep rows. Should equal SweepCorrectedGapMax — the invariant.

f **Sweep Corrected Gap Range** Calculated 3 hops from ground truth number DR-84

IN RULESPEAK®

The model summary's [sweep corrected gap range](#) is determined by the following priority:

1. **an empty string** if the [sweep corrected gap max](#) is blank
2. **the [sweep corrected gap max](#) minus the [sweep corrected gap min](#)** in all other cases

▼ Inputs 2 fields this rule reads

Σ **Sweep Corrected Gap Max** Model Summary
Largest sweep corrected gap across all allocation sweep whose study ID is "kidney-1986".

Σ **Sweep Corrected Gap Min** Model Summary
Smallest sweep corrected gap across all allocation sweep whose study ID is "kidney-

1986".

- **Ground truth at the leaves** 1 the raw values this ultimately depends on

SweepCorrectedGapMax – SweepCorrectedGapMin. The allocation-invariance witness: this should be zero (up to floating-point rounding). If nonzero, the corrected gap is not allocation-free.



Sweep Pooled Gap Range

Aggregation

6 hops from ground truth

number

DR-85

IN RULESPEAK®

A model summary's sweep pooled gap range is an aggregated value computed across the model summary's related records.

- ▼ **Inputs** 2 fields this rule reads

f Sweep Pooled Gap

Allocation Sweep

If sweep pooled rate a is "" then "", otherwise (sweep pooled rate a minus sweep pooled rate b).

● Study ID

Allocation Sweep

The study being swept — all 238 studies in the corpus.

- **Ground truth at the leaves** 2 the raw values this ultimately depends on

MAX(SweepPooledGap) – MIN(SweepPooledGap) across all AllocationSweep rows. Expected to be large (≈ 0.21) — the pooled gap wanders as allocation shifts.



Real Study Count

Aggregation

1 hop from ground truth

integer

DR-86

IN RULESPEAK®

A model summary's real study count is the number of studies related to the model summary.

- ▼ **Inputs** 1 fields this rule reads

● Is Synthetic

Study

TRUE for constructed/counterfactual studies;
FALSE for real published studies. Filters the corpus for empirical claims.

- **Ground truth at the leaves** 1 the raw values this ultimately depends on

Number of real (non-synthetic) published studies in the corpus. Retyped loop-90: was mis-declared raw with a formula, which is why the transpiler emitted no native function and 02b had to hand-write calc_model_summary_real_study_count.



Avg Signal Purity Reversal

Calculated

8 hops from ground truth

number

DR-87

IN RULESPEAK®

A model summary's avg signal purity reversal is the average signal purity across the treatment rankings related to the model summary.

▼ **Inputs** 2 fields this rule reads

f **Signal Purity** Treatment Ranking

If corrected gap is "" then "", otherwise if (the absolute value of corrected gap plus allocation distortion) equals 0 then 1, otherwise (the absolute value of corrected gap divided by (the absolute value of corrected gap plus allocation distortion)).

f **Allocation Direction** Treatment Ranking

If corrected gap is "" then "", otherwise if is sign flip then "reversal", otherwise if the absolute value of signed pooled gap is greater than (the absolute value of corrected gap plus 0.001) then "amplification", otherwise if the absolute value of signed pooled gap is less than (the absolute value of corrected gap minus 0.001) then "compression", otherwise "neutral".

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

Average SignalPurity across reversal studies (type A and B). Theorem predicts this must be < 0.5. Retyped loop-90 (see RealStudyCount); rewritten from SUMIFS/COUNTIFS division to AVERAGEIFS because the SUMIFS/COUNTIFS division form transpiled with a mangled cross-table alias.

f **Avg Signal Purity Non Reversal** Calculated 8 hops from ground truth number DR-88

IN RULESPEAK®

A model summary's avg signal purity non reversal is the average signal purity across the treatment rankings related to the model summary.

▼ **Inputs** 2 fields this rule reads

f **Signal Purity** Treatment Ranking

If corrected gap is "" then "", otherwise if (the absolute value of corrected gap plus allocation distortion) equals 0 then 1, otherwise (the absolute value of corrected gap divided by (the absolute value of corrected gap plus allocation distortion)).

f **Allocation Direction** Treatment Ranking

If corrected gap is "" then "", otherwise if is sign flip then "reversal", otherwise if the absolute value of signed pooled gap is greater than (the absolute value of corrected gap plus 0.001) then "amplification", otherwise if the absolute value of signed pooled gap is less than (the absolute value of corrected gap minus 0.001) then "compression", otherwise "neutral".

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

Average SignalPurity across non-reversal studies (type C+, C-, D). Theorem predicts this must be >= 0.5 except edge cases. Retyped loop-90 (see RealStudyCount); rewritten from SUMIFS/COUNTIFS division to AVERAGEIFS (see AvgSignalPurityReversal).

f **Signal Purity Gap** Calculated 9 hops from ground truth number DR-89

IN RULESPEAK®

A model summary's signal purity gap is computed as the avg signal purity non reversal minus the avg signal purity reversal.

▼ **Inputs** 2 fields this rule reads

f Avg Signal Purity Non Reversal

Model Summary

Average signal purity across all treatment rankings whose allocation direction is "<>reversal".

f Avg Signal Purity Reversal Model Summary

Average signal purity across all treatment rankings whose allocation direction is "reversal".

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

Separation between reversal and non-reversal average SignalPurity. The larger this gap, the more discriminating the threshold at 0.5. Retyped loop-90 (see RealStudyCount).

Σ Medicine Study Count

Aggregation

1 hop from ground truth

integer

DR-90

IN RULESPEAK®

A model summary's medicine study count is the number of the model summary's studies that are not synthetic.

▼ **Inputs** 2 fields this rule reads

● Domain

Study

Research domain: medicine | epidemiology | social-science | education | legal | sports | economics | synthetic. Enables cross-domain SignalPurity stratification.

● Is Synthetic

Study

TRUE for constructed/counterfactual studies; FALSE for real published studies. Filters the corpus for empirical claims.

► **Ground truth at the leaves** 2 the raw values this ultimately depends on

Real published studies in the medicine domain. Retyped loop-90 (see RealStudyCount).

Σ Epidemiology Study Count

Aggregation

1 hop from ground truth

integer

DR-91

IN RULESPEAK®

A model summary's epidemiology study count is the number of the model summary's studies that are not synthetic.

▼ **Inputs** 2 fields this rule reads

● Domain

Study

● Is Synthetic

Study

Research domain: medicine | epidemiology | social-science | education | legal | sports | economics | synthetic. Enables cross-domain SignalPurity stratification.

TRUE for constructed/counterfactual studies; FALSE for real published studies. Filters the corpus for empirical claims.

► **Ground truth at the leaves** 2 the raw values this ultimately depends on

Real published studies in the epidemiology domain. Retyped loop-90 (see RealStudyCount).

f

Other Domain Study Count

Calculated

2 hops from ground truth

integer

DR-92

IN RULESPEAK®

A model summary's other domain study count is computed as the real study count minus the medicine study count minus the epidemiology study count.

▼ **Inputs** 3 fields this rule reads

Σ Real Study Count

Model Summary

Count of studies whose is synthetic is false().

Σ Medicine Study Count

Model Summary

Count of studies whose domain is "medicine" and is synthetic is false().

Σ Epidemiology Study Count

Model Summary

Count of studies whose domain is "epidemiology" and is synthetic is false().

► **Ground truth at the leaves** 2 the raw values this ultimately depends on

Real published studies outside medicine and epidemiology (sports, legal, education, social-science). Retyped loop-90 (see RealStudyCount).

f

Domain Diversity Note

Calculated

9 hops from ground truth

string

DR-93

IN RULESPEAK® ⚠ mechanical

A model summary's domain diversity note is computed as "real=", followed by the real study count, followed by " studies; wave2=", followed by the expansion wave2 study count, followed by "; eduLatent=", followed by the education latent fraction times 100 rounded to 0 decimal place(s), followed by "%; sportsLatent=", followed by the sports latent fraction times 100 rounded to 0 decimal place(s), followed by "%; econFlipRate=", followed by the economics sign flip rate times 100 rounded to 0 decimal place(s), followed by "%".

▼ **Inputs** 5 fields this rule reads

Σ Real Study Count

Model Summary

Count of studies whose is synthetic is false().

f Expansion Wave2 Study Count

Model Summary

Count of candidate study catalog whose expansion wave is "expansion-wave-2" and is imported is true().

f **Education Latent Fraction** Model Summary

If count of treatment rankings whose study domain is "education" and distortion type is "D" equals 0 then "", otherwise (count of treatment rankings whose study domain is "education", distortion type is "D", and is latent only flip is true() divided by count of treatment rankings whose study domain is "education" and distortion type is "D").

f **Sports Latent Fraction** Model Summary

If count of treatment rankings whose study domain is "sports" and distortion type is "D" equals 0 then "", otherwise (count of treatment rankings whose study domain is "sports", distortion type is "D", and is latent only flip is true() divided by count of treatment rankings whose study domain is "sports" and distortion type is "D").

f **Economics Sign Flip Rate** Model Summary

If count of treatment rankings whose study domain is "economics" equals 0 then "", otherwise (count of treatment rankings whose study domain is "economics" and is sign flip is true() divided by count of treatment rankings whose study domain is "economics").

► **Ground truth at the leaves** 14 the raw values this ultimately depends on

*Human-readable domain breakdown. Witnesses corpus diversity — a prerequisite for cross-domain SignalPurity claims. Retyped loop-90 (see RealStudyCount); TEXT(...,"0%") is not a supported transpiler function, replaced with ROUND(x*100,0) + literal '%'.*

Synthetic Phase Count Ground truth integer

Ground truth. Written directly to the database (datatype `integer`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Number of parameter grid points in SyntheticPhase. Refreshed from vw_phase_diagram_summary by 05b-customize-data.sql.

Phase Diagram Complete Ground truth boolean

Ground truth. Written directly to the database (datatype `boolean`); not computed from anything else. Everything derived ultimately traces back to values like this one.

TRUE when all five DistortionType cells are populated in the synthetic phase grid. Refreshed from vw_phase_diagram_summary by 05b.

Phase Taxonomy Coverage Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Type counts across the synthetic phase grid. Refreshed from `vw_phase_diagram_summary` by 05b.

Σ Ingestion Protocol Item Count Aggregation 1 hop from ground truth integer DR-94

IN RULESPEAK®

A model summary's [ingestion protocol item count](#) is the number of [ingestion protocol](#) related to the model summary.

▼ **Inputs** 1 fields this rule reads

● **Protocol ID** Ingestion Protocol
Unique slug for this contract item.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Number of rows in `IngestionProtocol` — the documented contract items. Fixed loop-90 (see `IngestionSummary.ProtocolItemCount` for why `COUNTIFS(...,"<>")` is unsafe).

↗ Corpus Passes Ingestion Contract Lookup boolean DR-95

IN RULESPEAK®

A model summary's [corpus passes ingestion contract](#) is the value of `LOOKUP("ingestion-v1", IngestionSummary[IngestionSummaryId], IngestionSummary[IngestionContractPasses])`.

Lookup: `IngestionSummary.IngestionContractPasses`.

↗ Ingestion Witness Note Lookup string DR-96

IN RULESPEAK®

A model summary's [ingestion witness note](#) is the value of `LOOKUP("ingestion-v1", IngestionSummary[IngestionSummaryId], IngestionSummary[IngestionWitnessNote])`.

Human-readable ingestion contract witness from `IngestionSummary`.

↗ Catalog Entry Count Lookup integer DR-97

IN RULESPEAK®

A model summary's [catalog entry count](#) is the value of `LOOKUP("catalog-v1", CorpusCatalogSummary[CatalogSummaryId], CorpusCatalogSummary[TotalCatalogEntries])`.

Total catalog rows.

Pending Import Count Lookup integer DR-98

IN RULESPEAK®

A model summary's [pending import count](#) is the value of LOOKUP("catalog-v1", CorpusCatalogSummary[CatalogSummaryId], CorpusCatalogSummary[CandidateCount]).

Candidate studies queued.

Ready to Encode Count Lookup integer DR-99

IN RULESPEAK®

A model summary's [ready to encode count](#) is the value of LOOKUP("catalog-v1", CorpusCatalogSummary[CatalogSummaryId], CorpusCatalogSummary[ReadyToEncodeCount]).

Encode-ready candidates.

Import Session Ready Lookup boolean DR-100

IN RULESPEAK®

A model summary's [import session ready](#) is the value of LOOKUP("catalog-v1", CorpusCatalogSummary[CatalogSummaryId], CorpusCatalogSummary[ImportSessionReady]).

Bulk import session readiness flag.

Catalog Witness Note Lookup string DR-101

IN RULESPEAK®

A model summary's [catalog witness note](#) is the value of LOOKUP("catalog-v1", CorpusCatalogSummary[CatalogSummaryId], CorpusCatalogSummary[CatalogWitnessNote]).

Import backlog witness string.

Latent Type D Count Calculated 9 hops from ground truth number DR-102

IN RULESPEAK®

A model summary's [latent type d count](#) is the number of [treatment rankings](#) related to the model summary.

▼ **Inputs** 1 fields this rule reads

Latent Flip Potential Treatment Ranking

If distortion type is "" then "", otherwise distortion type is "D" and pooled gap crosses zero equals yes.

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

| *Type-D studies with LatentFlipPotential=TRUE.*

f **Stable Type D Count** Calculated 9 hops from ground truth number DR-103

IN RULESPEAK®

A model summary's stable type d count is the number of the model summary's treatment rankings that are not latent flip potential.

▼ **Inputs** 2 fields this rule reads

f **Distortion Type** Treatment Ranking

If allocation distortion is "" then "", otherwise if is sign flip then if is stratum unanimous then "A", otherwise "B", otherwise if not is sign flip, allocation distortion is greater than 0.01, and the absolute value of signed pooled gap is greater than (the absolute value of corrected gap plus 0.001) then "C+", otherwise if not is sign flip, allocation distortion is greater than 0.01, and the absolute value of signed pooled gap is less than (the absolute value of corrected gap minus 0.001) then "C-", otherwise "D".

f **Latent Flip Potential** Treatment Ranking

If distortion type is "" then "", otherwise distortion type is "D" and pooled gap crosses zero equals yes.

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

| *Type-D studies without latent flip potential (allocation-stable at observed and counterfactual sweep).*

f **Latent Type D Fraction** Calculated 10 hops from ground truth number DR-104

IN RULESPEAK®

The model summary's latent type d fraction is determined by the following priority:

1. an empty string if the type d count is 0
2. the latent type d count divided by the type d count in all other cases

▼ **Inputs** 2 fields this rule reads

Σ **Type D Count** Model Summary

Count of treatment rankings whose distortion type is "D".

f **Latent Type D Count** Model Summary

Count of treatment rankings whose latent flip potential is true().

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

| *Fraction of Type-D studies with latent flip potential.*



Cross Zero Count

Calculated

2 hops from ground truth

number

DR-105

IN RULESPEAK®

A model summary's cross zero count is the number of treatment rankings related to the model summary.

▼ Inputs 1 fields this rule reads

f **Pooled Gap Crosses Zero** Treatment Ranking

```
LOOKUP({{Study}},  
SweepStudySummary[SweepStudyId],  
SweepStudySummary[PooledGapCrossesZero]).
```

► Ground truth at the leaves 1 the raw values this ultimately depends on

Studies whose pooled gap crosses zero under allocation sweep.



Sign Flip Signal Purity Max

Calculated

8 hops from ground truth

number

DR-106

IN RULESPEAK®

A model summary's sign flip signal purity max is the largest signal purity across the treatment rankings related to the model summary.

▼ Inputs 2 fields this rule reads

f **Signal Purity** Treatment Ranking

If corrected gap is "" then "", otherwise if (the absolute value of corrected gap plus allocation distortion) equals 0 then 1, otherwise (the absolute value of corrected gap divided by (the absolute value of corrected gap plus allocation distortion)).

f **Is Sign Flip** Treatment Ranking

If weighted stratum gap sum is "" then "", otherwise if weighted stratum gap sum is greater than 0 then signed pooled gap is less than 0, otherwise signed pooled gap is greater than 0.

► Ground truth at the leaves 11 the raw values this ultimately depends on

Maximum SignalPurity among sign-flip studies. Must stay below 0.5 for the signal-purity theorem.



Economics Sign Flip Count

Calculated

7 hops from ground truth

number

DR-107

IN RULESPEAK®

A model summary's economics sign flip count is the number of the model summary's treatment rankings that are sign flips.

▼ Inputs 2 fields this rule reads

f Study Domain

Treatment Ranking

LOOKUP({{Study}}, Studies[StudyId],
Studies[Domain]).

f Is Sign Flip

Treatment Ranking

If weighted stratum gap sum is "" then "",
otherwise if weighted stratum gap sum is greater
than 0 then signed pooled gap is less than 0,
otherwise signed pooled gap is greater than 0.

- **Ground truth at the leaves** 11 the raw values this ultimately depends on

Sign-flip count in economics domain.

Σ

Sum Pooled Gap Latent D

Aggregation

9 hops from ground truth

number

DR-108

IN RULESPEAK®

A model summary's sum pooled gap latent d is the total pooled gap across the treatment rankings related to the model summary.

- ▼ **Inputs** 2 fields this rule reads

f Pooled Gap

Treatment Ranking

If pooled rate a is "" then "", otherwise the
absolute value of (pooled rate a minus pooled rate
b).

f Latent Flip Potential

Treatment Ranking

If distortion type is "" then "", otherwise distortion
type is "D" and pooled gap crosses zero equals
yes.

- **Ground truth at the leaves** 11 the raw values this ultimately depends on

SUM(PooledGap) among Type-D studies with latent flip potential — a bare, unwrapped SUMIFS split out as its own field because nesting a cross-table SUMIFS/AVERAGEIFS as an operand inside another IF/division expression transpiles with a mangled WHERE-clause column alias (e.g. rankings_latent_flip_potential) that errors at runtime; a bare top-level SUMIFS transpiles correctly (loop-90).

f

Avg Pooled Gap Latent D

Calculated

10 hops from ground truth

number

DR-109

IN RULESPEAK®

The model summary's avg pooled gap latent d is determined by the following priority:

1. 0 if the latent type d count is 0
2. the sum pooled gap latent d divided by the latent type d count in all other cases

- ▼ **Inputs** 2 fields this rule reads

f Latent Type D Count

Model Summary

Count of treatment rankings whose latent flip
potential is true().

Σ Sum Pooled Gap Latent D

Model Summary

Sum of pooled gap across all treatment rankings
whose latent flip potential is true().

- **Ground truth at the leaves** 11 the raw values this ultimately depends on

Average PooledGap among Type-D studies with latent flip potential. Fixed loop-90 (see SumPooledGapLatentD for why the sum is split into its own field).

Σ Sum Pooled Gap Stable D Aggregation 9 hops from ground truth number DR-110

IN RULESPEAK®

A model summary's sum pooled gap stable d is the total pooled gap across the model summary's treatment rankings that are not latent flip potential.

▼ **Inputs** 3 fields this rule reads

f **Pooled Gap** Treatment Ranking

If pooled rate a is "" then "", otherwise the absolute value of (pooled rate a minus pooled rate b).

f **Distortion Type** Treatment Ranking

If allocation distortion is "" then "", otherwise if is sign flip then if is stratum unanimous then "A", otherwise "B", otherwise if not is sign flip, allocation distortion is greater than 0.01, and the absolute value of signed pooled gap is greater than (the absolute value of corrected gap plus 0.001) then "C+", otherwise if not is sign flip, allocation distortion is greater than 0.01, and the absolute value of signed pooled gap is less than (the absolute value of corrected gap minus 0.001) then "C-", otherwise "D".

f **Latent Flip Potential** Treatment Ranking

If distortion type is "" then "", otherwise distortion type is "D" and pooled gap crosses zero equals yes.

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

SUM(PooledGap) among allocation-stable Type-D studies — split out as a bare field for the same reason as SumPooledGapLatentD.

f **Avg Pooled Gap Stable D** Calculated 10 hops from ground truth number DR-111

IN RULESPEAK®

The model summary's avg pooled gap stable d is determined by the following priority:

1. **0** if the stable type d count is 0
2. the sum pooled gap stable d divided by the stable type d count in all other cases

▼ **Inputs** 2 fields this rule reads

f **Stable Type D Count** Model Summary

Σ **Sum Pooled Gap Stable D** Model Summary

Count of treatment rankings whose distortion type is "D" and latent flip potential is false().

Sum of pooled gap across all treatment rankings whose distortion type is "D" and latent flip potential is false().

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

Average PooledGap among allocation-stable Type-D studies. Fixed loop-90 (see SumPooledGapLatentD).

f Epidemiology Avg Distortion Calculated 7 hops from ground truth number DR-112

IN RULESPEAK®

A model summary's epidemiology avg distortion is the average allocation distortion across the treatment rankings related to the model summary.

▼ **Inputs** 2 fields this rule reads

f **Allocation Distortion** Treatment Ranking
If weighted stratum gap sum is "" then "",
otherwise the absolute value of (weighted stratum
gap sum minus signed pooled gap).

f **Study Domain** Treatment Ranking
LOOKUP({{Study}}, Studies[StudyId],
Studies[Domain]).

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

Mean AllocationDistortion in epidemiology studies.

f Education Avg Distortion Calculated 7 hops from ground truth number DR-113

IN RULESPEAK®

A model summary's education avg distortion is the average allocation distortion across the treatment rankings related to the model summary.

▼ **Inputs** 2 fields this rule reads

f **Allocation Distortion** Treatment Ranking
If weighted stratum gap sum is "" then "",
otherwise the absolute value of (weighted stratum
gap sum minus signed pooled gap).

f **Study Domain** Treatment Ranking
LOOKUP({{Study}}, Studies[StudyId],
Studies[Domain]).

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

Mean AllocationDistortion in education studies.

f Confounder Sign Flip Count Calculated 7 hops from ground truth number DR-114

IN RULESPEAK®

A model summary's [confounder sign flip count](#) is the number of the model summary's [treatment rankings](#) that [are sign flips](#).

▼ **Inputs** 2 fields this rule reads

↗ **Stratum Causal Role** Treatment Ranking
LOOKUP({{Study}}, StratumVariables[Study],
StratumVariables[CausalRole]).

f **Is Sign Flip** Treatment Ranking
If weighted stratum gap sum is "" then "",
otherwise if weighted stratum gap sum is greater
than 0 then signed pooled gap is less than 0,
otherwise signed pooled gap is greater than 0.

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

| *Sign-flip count among confounder-labeled studies.*

f **Confounder Latent Only Count** Calculated 8 hops from ground truth number DR-115

IN RULESPEAK®

A model summary's [confounder latent only count](#) is the number of the model summary's [treatment rankings](#) that [are latent only flips](#).

▼ **Inputs** 2 fields this rule reads

↗ **Stratum Causal Role** Treatment Ranking
LOOKUP({{Study}}, StratumVariables[Study],
StratumVariables[CausalRole]).

f **Is Latent Only Flip** Treatment Ranking
If pooled gap crosses zero is "" then "", otherwise
pooled gap crosses zero equals true() and is sign
flip equals false().

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

| *Latent-only flip count among confounder-labeled studies.*

f **Collider Selection Count** Calculated 2 hops from ground truth number DR-116

IN RULESPEAK®

A model summary's [collider selection count](#) is computed as the number of [treatment rankings](#) related to the model summary plus the number of [treatment rankings](#) related to the model summary.

▼ **Inputs** 1 fields this rule reads

↗ **Stratum Causal Role** Treatment Ranking
LOOKUP({{Study}}, StratumVariables[Study],
StratumVariables[CausalRole]).

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

| *Studies tagged collider or selection on StratumVariables.*

f Collider Selection Manifest Count Calculated 7 hops from ground truth number

DR-117

IN RULESPEAK®

A model summary's collider selection manifest count is computed as the number of the model summary's treatment rankings that are sign flips plus the number of the model summary's treatment rankings that are sign flips.

▼ **Inputs** 2 fields this rule reads

↗ **Stratum Causal Role** Treatment Ranking

LOOKUP({{Study}}, StratumVariables[Study],
StratumVariables[CausalRole]).

f **Is Sign Flip** Treatment Ranking

If weighted stratum gap sum is "" then "",
otherwise if weighted stratum gap sum is greater
than 0 then signed pooled gap is less than 0,
otherwise signed pooled gap is greater than 0.

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

| *Manifest sign-flips among collider/selection studies.*

f Collider Selection Latent Only Count Calculated 8 hops from ground truth number

DR-118

IN RULESPEAK®

A model summary's collider selection latent only count is computed as the number of the model summary's treatment rankings that are latent only flips plus the number of the model summary's treatment rankings that are latent only flips.

▼ **Inputs** 2 fields this rule reads

↗ **Stratum Causal Role** Treatment Ranking

LOOKUP({{Study}}, StratumVariables[Study],
StratumVariables[CausalRole]).

f **Is Latent Only Flip** Treatment Ranking

If pooled gap crosses zero is "" then "", otherwise
pooled gap crosses zero equals true() and is sign
flip equals false().

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

| *Latent-only flips among collider/selection studies.*

Σ Explained Confounder Count Aggregation 9 hops from ground truth integer DR-119

IN RULESPEAK®

A model summary's explained confounder count is the number of treatment rankings related to the model summary.

▼ **Inputs** 1 fields this rule reads

f **Is Paradox Explained** Treatment Ranking
Is reversal and confounders in study is greater than 0.

► **Ground truth at the leaves** 15 the raw values this ultimately depends on

Count of TreatmentRankings rows where IsParadoxExplained=TRUE (should equal ConfounderSignFlipCount).

Σ Contested or Mediator Explained Count Aggregation 9 hops from ground truth integer

DR-120

IN RULESPEAK®

A model summary's contested or mediator explained count is the number of the model summary's treatment rankings that are paradox-explained and are not adjustment appropriate.

▼ **Inputs** 3 fields this rule reads

f **Is Sign Flip** Treatment Ranking
If weighted stratum gap sum is "" then "",
otherwise if weighted stratum gap sum is greater than 0 then signed pooled gap is less than 0,
otherwise signed pooled gap is greater than 0.

f **Is Paradox Explained** Treatment Ranking
Is reversal and confounders in study is greater than 0.

f **Adjustment Appropriate** Treatment Ranking
Mediator risk count equals 0 and causal claim status is "established".

► **Ground truth at the leaves** 16 the raw values this ultimately depends on

Count of contested or mediator sign-flip rows where IsParadoxExplained=TRUE (should be zero).

f Discovery Witness Note Calculated 11 hops from ground truth string DR-121

IN RULESPEAK®

A model summary's discovery witness note is computed as "sweep: latentD=", followed by the latent type d fraction, followed by " purityMax=", followed by the sign flip signal purity max,

followed by " catalogExact=", followed by the [type prediction match rate](#), followed by " identities=", followed by the [confounder identity count](#).

▼ **Inputs** 4 fields this rule reads

f **Latent Type D Fraction** Model Summary

If type d count equals 0 then "", otherwise (latent type d count divided by type d count).

f **Sign Flip Signal Purity Max** Model Summary

Largest signal purity across all treatment rankings whose is sign flip is true().

↗ **Type Prediction Match Rate** Model Summary

LOOKUP("catalog-v1",
CorpusCatalogSummary[CatalogSummaryId],
CorpusCatalogSummary[TypePredictionMatchRate]).

Σ **Confounder Identity Count** Model Summary

Count(confounder identity ID).

► **Ground truth at the leaves** 12 the raw values this ultimately depends on

Research sweep synthesis rollout (loops 61–66).

↗ **Type Prediction Match Count** Lookup integer DR-122

IN RULESPEAK®

A model summary's [type prediction match count](#) is the value of LOOKUP("catalog-v1",
CorpusCatalogSummary[CatalogSummaryId],
CorpusCatalogSummary[TypePredictionMatchCount]).

Lookup: CorpusCatalogSummary.TypePredictionMatchCount.

↗ **Type Prediction Mismatch Count** Lookup integer DR-123

IN RULESPEAK®

A model summary's [type prediction mismatch count](#) is the value of LOOKUP("catalog-v1",
CorpusCatalogSummary[CatalogSummaryId],
CorpusCatalogSummary[TypePredictionMismatchCount]).

Lookup: CorpusCatalogSummary.TypePredictionMismatchCount.

↗ **Type Prediction Match Rate** Lookup number DR-124

IN RULESPEAK®

A model summary's [type prediction match rate](#) is the value of LOOKUP("catalog-v1",
CorpusCatalogSummary[CatalogSummaryId],
CorpusCatalogSummary[TypePredictionMatchRate]).

Lookup: *CorpusCatalogSummary.TypePredictionMatchRate*.



Sign Flip Prediction Match Rate

Lookup

number

DR-125

IN RULESPEAK®

A model summary's [sign flip prediction match rate](#) is the value of LOOKUP("catalog-v1", CorpusCatalogSummary[CatalogSummaryId], CorpusCatalogSummary[SignFlipPredictionMatchRate]).

Lookup: *CorpusCatalogSummary.SignFlipPredictionMatchRate*.



Catalog Prediction Witness Note

Lookup

string

DR-126

IN RULESPEAK®

A model summary's [catalog prediction witness note](#) is the value of LOOKUP("catalog-v1", CorpusCatalogSummary[CatalogSummaryId], CorpusCatalogSummary[CatalogPredictionWitnessNote]).

Lookup: *CorpusCatalogSummary.CatalogPredictionWitnessNote*.



High Imbalance Sign Flip Threshold

Ground truth

number

Ground truth. Written directly to the database (datatype `number`); not computed from anything else. Everything derived ultimately traces back to values like this one.

max_stratum_imbalance cutoff for geometry-controlled domain screening (loop-66). Set from corpus median by customize SQL.



Economics High Imbalance Sign Flip Count

Ground truth

number

Ground truth. Written directly to the database (datatype `number`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Sign-flip count among economics studies with max_stratum_imbalance >= threshold.



Epidemiology High Imbalance Sign Flip Rate

Ground truth

number

Ground truth. Written directly to the database (datatype `number`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Sign-flip rate among high-imbalance epidemiology studies.



Legal High Imbalance Sign Flip Rate

Ground truth

number

Ground truth. Written directly to the database (datatype `number`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Sign-flip rate among high-imbalance legal studies.

○ Sports High Imbalance Sign Flip Rate Ground truth number

Ground truth. Written directly to the database (datatype `number`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Sign-flip rate among high-imbalance sports studies.

f C Plus Avg Distortion Calculated 8 hops from ground truth number DR-127

IN RULESPEAK®

A model summary's c plus avg distortion is the average allocation distortion across the treatment rankings related to the model summary.

▼ Inputs 2 fields this rule reads

f Allocation Distortion Treatment Ranking

If weighted stratum gap sum is "" then "", otherwise the absolute value of (weighted stratum gap sum minus signed pooled gap).

f Distortion Type Treatment Ranking

If allocation distortion is "" then "", otherwise if is sign flip then if is stratum unanimous then "A", otherwise "B", otherwise if not is sign flip, allocation distortion is greater than 0.01, and the absolute value of signed pooled gap is greater than (the absolute value of corrected gap plus 0.001) then "C+", otherwise if not is sign flip, allocation distortion is greater than 0.01, and the absolute value of signed pooled gap is less than (the absolute value of corrected gap minus 0.001) then "C-", otherwise "D".

► Ground truth at the leaves 11 the raw values this ultimately depends on

Mean AllocationDistortion among Type-C+ studies.

f C Minus Avg Distortion Calculated 8 hops from ground truth number DR-128

IN RULESPEAK®

A model summary's c minus avg distortion is the average allocation distortion across the treatment rankings related to the model summary.

▼ Inputs 2 fields this rule reads

f Allocation Distortion Treatment Ranking

f Distortion Type Treatment Ranking

If weighted stratum gap sum is "" then "", otherwise the absolute value of (weighted stratum gap sum minus signed pooled gap).

If allocation distortion is "" then "", otherwise if is sign flip then if is stratum unanimous then "A", otherwise "B", otherwise if not is sign flip, allocation distortion is greater than 0.01, and the absolute value of signed pooled gap is greater than (the absolute value of corrected gap plus 0.001) then "C+", otherwise if not is sign flip, allocation distortion is greater than 0.01, and the absolute value of signed pooled gap is less than (the absolute value of corrected gap minus 0.001) then "C-", otherwise "D".

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

Mean AllocationDistortion among Type-C- studies.

f **Type D Avg Distortion** Calculated 8 hops from ground truth number DR-129

IN RULESPEAK®

A model summary's type d avg distortion is the average allocation distortion across the treatment rankings related to the model summary.

▼ **Inputs** 2 fields this rule reads

f **Allocation Distortion** Treatment Ranking
If weighted stratum gap sum is "" then "", otherwise the absolute value of (weighted stratum gap sum minus signed pooled gap).

f **Distortion Type** Treatment Ranking
If allocation distortion is "" then "", otherwise if is sign flip then if is stratum unanimous then "A", otherwise "B", otherwise if not is sign flip, allocation distortion is greater than 0.01, and the absolute value of signed pooled gap is greater than (the absolute value of corrected gap plus 0.001) then "C+", otherwise if not is sign flip, allocation distortion is greater than 0.01, and the absolute value of signed pooled gap is less than (the absolute value of corrected gap minus 0.001) then "C-", otherwise "D".

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

Mean AllocationDistortion among Type-D studies.

f **Sweep Fragile Count** Calculated 9 hops from ground truth integer DR-130

IN RULESPEAK®

A model summary's sweep fragile count is the number of treatment rankings related to the model summary.

▼ **Inputs** 1 fields this rule reads

f Is Sweep Fragile Treatment Ranking
Distortion type is "D", signal purity equals 1, allocation distortion equals 0, sweep pooled gap range is greater than 0.3, and allocation fragility is at least 10.

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

Count of TreatmentRankings with IsSweepFragile=TRUE.

f Expansion Wave1 Economics Expected a Count Calculated 2 hops from ground truth

integer DR-131

IN RULESPEAK®

A model summary's expansion wave1 economics expected a count is the number of the model summary's candidate study catalog that are imported; have a domain of "economics"; and have a expected distortion type of "A".

▼ **Inputs** 4 fields this rule reads

● **Expansion Wave** Candidate Study Catalog
Catalog wave tag; loop-67 expansion rows = expansion-wave-1.

● **Domain** Candidate Study Catalog
Domain tag: medicine, epidemiology, legal, sports, education, economics, social-science.

f Is Imported Candidate Study Catalog
Ingestion status is "imported".

● **Expected Distortion Type** Candidate Study Catalog
Hypothesized DistortionType after classification: A, B, C+, C-, D, or unknown.

► **Ground truth at the leaves** 4 the raw values this ultimately depends on

Imported expansion-wave-1 economics catalog rows tagged Expected A.

f Expansion Wave1 Economics Expected AD Count Calculated 2 hops from ground truth

integer DR-132

IN RULESPEAK®

A model summary's expansion wave1 economics expected AD count is the number of the model summary's candidate study catalog that are imported; have a domain of "economics"; have a expected distortion type of "A"; and have a observed distortion type of "D".

▼ **Inputs** 5 fields this rule reads

● **Expansion Wave** Candidate Study Catalog

Catalog wave tag; loop-67 expansion rows = expansion-wave-1.

f **Is Imported** Candidate Study Catalog

Ingestion status is "imported".

↗ **Observed Distortion Type** Candidate Study Catalog

LOOKUP({{LinkedStudyId}},
TreatmentRankings[Study],
TreatmentRankings[DistortionType]).

● **Domain** Candidate Study Catalog

Domain tag: medicine, epidemiology, legal, sports, education, economics, social-science.

● **Expected Distortion Type** Candidate Study Catalog

Hypothesized DistortionType after classification: A, B, C+, C-, D, or unknown.

► **Ground truth at the leaves** 5 the raw values this ultimately depends on

Expansion-wave-1 economics Expected-A imports observed as Type D.

f **Economics Expected a Mismatch Rate** Calculated 3 hops from ground truth number

DR-133

IN RULESPEAK®

The model summary's economics expected a mismatch rate is determined by the following priority:

1. **an empty string** if the expansion wave1 economics expected a count is 0
2. **the expansion wave1 economics expected AD count divided by the expansion wave1 economics expected a count** in all other cases

▼ **Inputs** 2 fields this rule reads

f **Expansion Wave1 Economics Expected a Count** Model Summary

Count of candidate study catalog whose expansion wave is "expansion-wave-1", domain is "economics", is imported is true(), and expected distortion type is "A".

f **Expansion Wave1 Economics Expected AD Count** Model Summary

Count of candidate study catalog whose expansion wave is "expansion-wave-1", domain is "economics", is imported is true(), expected distortion type is "A", and observed distortion type is "D".

► **Ground truth at the leaves** 5 the raw values this ultimately depends on

Fraction of expansion-wave-1 economics Expected-A imports observed as Type D.



Unanimous Sign Flip Count

Calculated

7 hops from ground truth

integer

DR-134

IN RULESPEAK®

A model summary's unanimous sign flip count is the number of the model summary's treatment rankings that are stratum-unanimous.

▼ Inputs 2 fields this rule reads

f Is Sign Flip

Treatment Ranking

If weighted stratum gap sum is "" then "",
otherwise if weighted stratum gap sum is greater
than 0 then signed pooled gap is less than 0,
otherwise signed pooled gap is greater than 0.

f Is Stratum Unanimous

Treatment Ranking

Strata won by a equals stratum count or strata
won by b equals stratum count.

► Ground truth at the leaves 11 the raw values this ultimately depends on

Sign-flip studies with unanimous stratum direction (IsSignFlip AND IsStratumUnanimous).



Max Study Sweep Corrected Gap Range

Aggregation

1 hop from ground truth

number

DR-135

IN RULESPEAK®

A model summary's max study sweep corrected gap range is the largest sweep corrected gap range across the sweep study summary related to the model summary.

▼ Inputs 1 fields this rule reads

● Sweep Corrected Gap Range

Sweep Study
Summary

Principled materialization
(refresh_sweep_study_summary()). Range of
SweepCorrectedGap across the sweep =
SweepCorrectedGapMax -
SweepCorrectedGapMin. Should be effectively
zero (< 0.0001) for all studies if CorrectedGap is
truly allocation-free. See CorrectedGapConstant
for why this is materialized rather than calculated.

► Ground truth at the leaves 1 the raw values this ultimately depends on

MAX(SweepStudySummary.SweepCorrectedGapRange) — must stay below 0.0001 for CorrectedGap invariance theorem.



Corrected Gap Invariant Fail Count

Aggregation

1 hop from ground truth

integer

IN RULESPEAK®

A model summary's corrected gap invariant fail count is the number of sweep study summary related to the model summary.

▼ **Inputs** 1 fields this rule reads

● **Sweep Corrected Gap Range** Sweep Study Summary

Principled materialization
(refresh_sweep_study_summary()). Range of SweepCorrectedGap across the sweep = SweepCorrectedGapMax - SweepCorrectedGapMin. Should be effectively zero (< 0.0001) for all studies if CorrectedGap is truly allocation-free. See CorrectedGapConstant for why this is materialized rather than calculated.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Count of SweepStudySummary rows where SweepCorrectedGapRange ≥ 0.0001 .



False Positive Explained Count

Aggregation

9 hops from ground truth

integer

DR-137

IN RULESPEAK®

A model summary's false positive explained count is the number of the model summary's false positive explaineds.

▼ **Inputs** 3 fields this rule reads

f **Is Paradox Explained** Treatment Ranking

Is reversal and confounders in study is greater than 0.

f **Is Sign Flip** Treatment Ranking

If weighted stratum gap sum is "" then "", otherwise if weighted stratum gap sum is greater than 0 then signed pooled gap is less than 0, otherwise signed pooled gap is greater than 0.

↗ **Stratum Causal Role** Treatment Ranking

LOOKUP({{Study}}, StratumVariables[Study], StratumVariables[CausalRole]).

► **Ground truth at the leaves** 15 the raw values this ultimately depends on

IsParadoxExplained=TRUE rows that are not confounder manifest sign-flips — must be 0 for explained $\Leftrightarrow$ confounder theorem.



Unexplained Confounder Sign Flip Count

Aggregation

9 hops from ground truth

integer

DR-138

IN RULESPEAK®

A model summary's unexplained confounder sign flip count is the number of the model summary's treatment rankings that are not paradox-explained and have a stratum causal role of "confounder".

▼ **Inputs** 3 fields this rule reads

f Is Sign Flip

Treatment Ranking

If weighted stratum gap sum is "" then "",
otherwise if weighted stratum gap sum is greater
than 0 then signed pooled gap is less than 0,
otherwise signed pooled gap is greater than 0.

Stratum Causal Role

Treatment Ranking

LOOKUP({{Study}}, StratumVariables[Study],
StratumVariables[CausalRole]).

f Is Paradox Explained

Treatment Ranking

Is reversal and confounders in study is greater
than 0.

► **Ground truth at the leaves** 15 the raw values this ultimately depends on

Confounder sign-flips with IsParadoxExplained=FALSE — must be 0 for explained↔confounder theorem.



Theorem Count

Calculated

1 hop from ground truth

integer

DR-139

IN RULESPEAK®

A model summary's theorem count is the number of the model summary's conclusions that have a status of "witnessed".

▼ **Inputs** 2 fields this rule reads

● Category

Conclusion

domain | instrument | methodology | open-
question | corpus-theorem

● Status

Conclusion

witnessed | planned | open

► **Ground truth at the leaves** 2 the raw values this ultimately depends on

Count of witnessed Conclusions with Category=corpus-theorem (portfolio rollup for loop-76; category renamed from theorem in loop-91).



Education Latent Fraction

Calculated

8 hops from ground truth

number

DR-140

IN RULESPEAK®

The model summary's [education latent fraction](#) is determined by the following priority:

1. **an empty string** if the model summary has no [treatment rankings](#) that have a [distortion type](#) of "D"
2. **the number of the model summary's [treatment rankings](#) that are latent only flips and have a [distortion type](#) of "D" divided by the number of the model summary's [treatment rankings](#) that have a [distortion type](#) of "D"** in all other cases

▼ Inputs 3 fields this rule reads

f **Study Domain** Treatment Ranking
LOOKUP({{Study}}, Studies[StudyId],
Studies[Domain]).

f **Distortion Type** Treatment Ranking
If allocation distortion is "" then "", otherwise if is
sign flip then if is stratum unanimous then "A",
otherwise "B", otherwise if not is sign flip,
allocation distortion is greater than 0.01, and the
absolute value of signed pooled gap is greater
than (the absolute value of corrected gap plus
0.001) then "C+", otherwise if not is sign flip,
allocation distortion is greater than 0.01, and the
absolute value of signed pooled gap is less than
(the absolute value of corrected gap minus 0.001)
then "C-", otherwise "D".

f **Is Latent Only Flip** Treatment Ranking
If pooled gap crosses zero is "" then "", otherwise
pooled gap crosses zero equals true() and is sign
flip equals false().

► Ground truth at the leaves 11 the raw values this ultimately depends on

Fraction of education Type-D studies with LatentFlipPotential=TRUE.

f **Sports Latent Fraction** Calculated 8 hops from ground truth number DR-141

IN RULESPEAK®

The model summary's [sports latent fraction](#) is determined by the following priority:

1. **an empty string** if the model summary has no [treatment rankings](#) that have a [distortion type](#) of "D"
2. **the number of the model summary's [treatment rankings](#) that are latent only flips and have a [distortion type](#) of "D" divided by the number of the model summary's [treatment rankings](#) that have a [distortion type](#) of "D"** in all other cases

▼ Inputs 3 fields this rule reads

f **Study Domain** Treatment Ranking

f **Distortion Type** Treatment Ranking

LOOKUP({{Study}}, Studies[StudyId],
Studies[Domain]).

If allocation distortion is "" then "", otherwise if is
sign flip then if is stratum unanimous then "A",
otherwise "B", otherwise if not is sign flip,
allocation distortion is greater than 0.01, and the
absolute value of signed pooled gap is greater
than (the absolute value of corrected gap plus
0.001) then "C+", otherwise if not is sign flip,
allocation distortion is greater than 0.01, and the
absolute value of signed pooled gap is less than
(the absolute value of corrected gap minus 0.001)
then "C-", otherwise "D".

f Is Latent Only Flip

Treatment Ranking

If pooled gap crosses zero is "" then "", otherwise
pooled gap crosses zero equals true() and is sign
flip equals false().

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

Fraction of sports Type-D studies with LatentFlipPotential=TRUE.



Economics Sign Flip Rate

Calculated

7 hops from ground truth

number

DR-142

IN RULESPEAK®

The model summary's economics sign flip rate is determined by the following priority:

1. **an empty string** if the model summary has no treatment rankings
2. **the number of the model summary's treatment rankings that are sign flips divided by the number of treatment rankings related to the model summary** in all other cases

▼ **Inputs** 2 fields this rule reads

f Study Domain

Treatment Ranking

LOOKUP({{Study}}, Studies[StudyId],
Studies[Domain]).

f Is Sign Flip

Treatment Ranking

If weighted stratum gap sum is "" then "",
otherwise if weighted stratum gap sum is greater
than 0 then signed pooled gap is less than 0,
otherwise signed pooled gap is greater than 0.

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

Manifest sign-flip rate among economics studies.



Education Type D Count

Calculated

8 hops from ground truth

integer

DR-143

IN RULESPEAK®

A model summary's education type d count is the number of the model summary's treatment rankings that have a distortion type of "D".

▼ **Inputs** 2 fields this rule reads

f **Study Domain** Treatment Ranking
LOOKUP({{Study}}, Studies[StudyId],
Studies[Domain]).

f **Distortion Type** Treatment Ranking
If allocation distortion is "" then "", otherwise if is
sign flip then if is stratum unanimous then "A",
otherwise "B", otherwise if not is sign flip,
allocation distortion is greater than 0.01, and the
absolute value of signed pooled gap is greater
than (the absolute value of corrected gap plus
0.001) then "C+", otherwise if not is sign flip,
allocation distortion is greater than 0.01, and the
absolute value of signed pooled gap is less than
(the absolute value of corrected gap minus 0.001)
then "C-", otherwise "D".

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

Type-D studies in education domain.

f **Education Latent D Count** Calculated 9 hops from ground truth integer DR-144

IN RULESPEAK®

A model summary's education latent d count is the number of the model summary's treatment rankings that are latent flip potential and have a distortion type of "D".

▼ **Inputs** 3 fields this rule reads

f **Study Domain** Treatment Ranking
LOOKUP({{Study}}, Studies[StudyId],
Studies[Domain]).

f **Distortion Type** Treatment Ranking
If allocation distortion is "" then "", otherwise if is
sign flip then if is stratum unanimous then "A",
otherwise "B", otherwise if not is sign flip,
allocation distortion is greater than 0.01, and the
absolute value of signed pooled gap is greater
than (the absolute value of corrected gap plus
0.001) then "C+", otherwise if not is sign flip,
allocation distortion is greater than 0.01, and the
absolute value of signed pooled gap is less than
(the absolute value of corrected gap minus 0.001)
then "C-", otherwise "D".

f **Latent Flip Potential** Treatment Ranking
If distortion type is "" then "", otherwise distortion
type is "D" and pooled gap crosses zero equals

yes.

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

Education Type-D studies with LatentFlipPotential.

f Sports Type D Count Calculated 8 hops from ground truth integer DR-145

IN RULESPEAK®

A model summary's sports type d count is the number of the model summary's treatment rankings that have a distortion type of "D".

▼ **Inputs** 2 fields this rule reads

f **Study Domain** Treatment Ranking
LOOKUP({{Study}}, Studies[StudyId],
Studies[Domain]).

f **Distortion Type** Treatment Ranking
If allocation distortion is "" then "", otherwise if is sign flip then if is stratum unanimous then "A", otherwise "B", otherwise if not is sign flip, allocation distortion is greater than 0.01, and the absolute value of signed pooled gap is greater than (the absolute value of corrected gap plus 0.001) then "C+", otherwise if not is sign flip, allocation distortion is greater than 0.01, and the absolute value of signed pooled gap is less than (the absolute value of corrected gap minus 0.001) then "C-", otherwise "D".

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

Type-D studies in sports domain.

f Sports Latent D Count Calculated 9 hops from ground truth integer DR-146

IN RULESPEAK®

A model summary's sports latent d count is the number of the model summary's treatment rankings that are latent flip potential and have a distortion type of "D".

▼ **Inputs** 3 fields this rule reads

f **Study Domain** Treatment Ranking
LOOKUP({{Study}}, Studies[StudyId],
Studies[Domain]).

f **Distortion Type** Treatment Ranking
If allocation distortion is "" then "", otherwise if is sign flip then if is stratum unanimous then "A", otherwise "B", otherwise if not is sign flip, allocation distortion is greater than 0.01, and the absolute value of signed pooled gap is greater than (the absolute value of corrected gap plus

0.001) then "C+", otherwise if not is sign flip, allocation distortion is greater than 0.01, and the absolute value of signed pooled gap is less than (the absolute value of corrected gap minus 0.001) then "C-", otherwise "D".

f Latent Flip Potential

Treatment Ranking

If distortion type is "" then "", otherwise distortion type is "D" and pooled gap crosses zero equals yes.

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

Sports Type-D studies with LatentFlipPotential.

f Economics Study Count

Calculated

1 hop from ground truth

integer

DR-147

IN RULESPEAK®

A model summary's economics study count is the number of the model summary's studies that are not synthetic.

▼ **Inputs** 2 fields this rule reads

● Domain

Study

Research domain: medicine | epidemiology | social-science | education | legal | sports | economics | synthetic. Enables cross-domain SignalPurity stratification.

● Is Synthetic

Study

TRUE for constructed/counterfactual studies; FALSE for real published studies. Filters the corpus for empirical claims.

► **Ground truth at the leaves** 2 the raw values this ultimately depends on

Real economics studies in corpus.

f Expansion Wave2 Study Count

Calculated

2 hops from ground truth

integer

DR-148

IN RULESPEAK®

A model summary's expansion wave2 study count is the number of the model summary's candidate study catalog that are imported.

▼ **Inputs** 2 fields this rule reads

● Expansion Wave

Candidate Study Catalog

Catalog wave tag; loop-67 expansion rows = expansion-wave-1.

f Is Imported

Candidate Study Catalog

Ingestion status is "imported".

► **Ground truth at the leaves** 2 the raw values this ultimately depends on

Studies imported in expansion wave 2.

Σ **Confounder Identity Count** Aggregation 1 hop from ground truth integer DR-149

IN RULESPEAK®

A model summary's confounder identity count is the number of confounder identities related to the model summary.

▼ **Inputs** 1 fields this rule reads

- **Confounder Identity ID** Confounder Identity
Unique slug (e.g. id-age-composition).

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Canonical ConfounderIdentity archetypes in ontology (loop-80).

Σ **Mapped Stratum Variable Count** Aggregation 1 hop from ground truth integer DR-150

IN RULESPEAK®

A model summary's mapped stratum variable count is the number of stratum variable identity maps related to the model summary.

▼ **Inputs** 1 fields this rule reads

- **Map ID** Stratum Variable Identity Map
Unique slug.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Stratum variables with an identity map row.

f **Unmapped Stratum Variable Count** Calculated 2 hops from ground truth integer

DR-151

IN RULESPEAK®

A model summary's unmapped stratum variable count is computed as the number of stratum variables related to the model summary minus the mapped stratum variable count.

▼ **Inputs** 2 fields this rule reads

● Stratum Variable ID

Stratum Variable

Unique slug (e.g. kidney-1986-stone-size).

Σ Mapped Stratum Variable Count

Model
Summary

Count(map ID).

► **Ground truth at the leaves** 2 the raw values this ultimately depends on

| *Stratum variables lacking identity map (should be 0).*



Age Identity Manifest Flip Rate

Calculated

number

DR-152

IN RULESPEAK®

A model summary's [age identity manifest flip rate](#) is computed as the value of LOOKUP("id-age-composition", IdentityClusterSummaries[ConfounderIdentity], IdentityClusterSummaries[ManifestFlipRate]).

| *Manifest flip rate for id-age-composition cluster (loop-81).*



Age Identity Study Count

Calculated

integer

DR-153

IN RULESPEAK®

A model summary's [age identity study count](#) is computed as the value of LOOKUP("id-age-composition", IdentityClusterSummaries[ConfounderIdentity], IdentityClusterSummaries[StudyCount]).

| *Real-study count in id-age-composition cluster (loop-81).*



Age Identity Latent Fraction Among Type D

Calculated

number

DR-154

IN RULESPEAK®

A model summary's [age identity latent fraction among type d](#) is computed as the value of LOOKUP("id-age-composition", IdentityClusterSummaries[ConfounderIdentity], IdentityClusterSummaries[LatentFractionAmongTypeD]).

| *Type-D latent fraction for id-age-composition cluster (loop-81).*



Severity Identity Latent Fraction Among Type D

Calculated

number

DR-155

IN RULESPEAK®

A model summary's [severity identity latent fraction among type d](#) is computed as the value of LOOKUP("id-disease-severity", IdentityClusterSummaries[ConfounderIdentity], IdentityClusterSummaries[LatentFractionAmongTypeD]).

| *Type-D latent fraction for id-disease-severity cluster (loop-81).*

f Identity Map Coverage Rate Calculated 2 hops from ground truth number DR-156

IN RULESPEAK®

The model summary's identity map coverage rate is determined by the following priority:

1. **an empty string** if the real study count is 0
2. **the mapped stratum variable count divided by the real study count** in all other cases

▼ **Inputs** 2 fields this rule reads

Σ **Real Study Count** Model Summary
Count of studies whose is synthetic is false().

Σ **Mapped Stratum Variable Count** Model Summary
Count(map ID).

► **Ground truth at the leaves** 2 the raw values this ultimately depends on

Fraction of real-study StratumVariables with a ConfounderIdentity map (loop-81 customize SQL).

o Identity Cluster Witness Note Ground truth string

Ground truth. Written directly to the database (datatype string); not computed from anything else. Everything derived ultimately traces back to values like this one.

Loop-80 identity layer witness rollup (customize SQL).

f Severity Medicine Manifest Flip Rate Calculated number DR-157

IN RULESPEAK®

A model summary's severity medicine manifest flip rate is computed as the value of LOOKUP("cell-id-disease-severity-x-medicine", IdentityDomainCells[CellId], IdentityDomainCells[ManifestFlipRate]).

Severity identity manifest flip rate in medicine domain.

f Severity Epi Manifest Flip Count Calculated integer DR-158

IN RULESPEAK®

A model summary's severity epi manifest flip count is computed as the value of LOOKUP("cell-id-disease-severity-x-epidemiology", IdentityDomainCells[CellId], IdentityDomainCells[ManifestFlipCount]).

Severity identity manifest flip count in epidemiology domain.

Σ Identity Domain Cell Count Aggregation 1 hop from ground truth integer DR-159

IN RULESPEAK®

A model summary's [identity domain cell count](#) is the number of [identity domain cells](#) related to the model summary.

▼ **Inputs** 1 fields this rule reads

- **Cell ID** Identity Domain Cell
Unique slug: identity-id + '-x-' + domain slug.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

| *Number of identity×domain cells.*

f **Selection Frailty Manifest Flip Count** Calculated integer DR-160

IN RULESPEAK®

A model summary's [selection frailty manifest flip count](#) is computed as the value of LOOKUP("cluster-id-selection-frailty", IdentityClusterSummaries[IdentityClusterId], IdentityClusterSummaries[ManifestFlipCount]).

| *Manifest flip count for selection-frailty identity.*

f **Selection Frailty Study Count** Calculated integer DR-161

IN RULESPEAK®

A model summary's [selection frailty study count](#) is computed as the value of LOOKUP("cluster-id-selection-frailty", IdentityClusterSummaries[IdentityClusterId], IdentityClusterSummaries[StudyCount]).

| *Study count for selection-frailty identity cluster.*

f **Collider Identity Manifest Flip Rate** Calculated number DR-162

IN RULESPEAK®

A model summary's [collider identity manifest flip rate](#) is computed as the value of LOOKUP("cluster-id-collider-proxy", IdentityClusterSummaries[IdentityClusterId], IdentityClusterSummaries[ManifestFlipRate]).

| *Manifest flip rate for collider-proxy identity.*

f **Geographic Type D Fraction** Calculated number DR-163

IN RULESPEAK®

A model summary's [geographic type d fraction](#) is computed as the value of `IF(LOOKUP("cluster-id-geographic-composition", IdentityClusterSummaries[IdentityClusterId], IdentityClusterSummaries[StudyCount])=0, "", LOOKUP("cluster-id-geographic-composition", IdentityClusterSummaries[IdentityClusterId], IdentityClusterSummaries[TypeDCount]) / LOOKUP("cluster-id-geographic-composition", IdentityClusterSummaries[IdentityClusterId], IdentityClusterSummaries[StudyCount]))`.

Type-D fraction for geographic-composition identity.



Severity Drift Direction

Calculated

string

DR-164

IN RULESPEAK®

A model summary's [severity drift direction](#) is computed as the value of `LOOKUP("id-disease-severity", ConfounderDistortionTimeline[ConfounderIdentity], ConfounderDistortionTimeline[DriftDirection])`.

loop-92: ConfounderDistortionTimeline.DriftDirection for id-disease-severity (severity-case-mix archetype), valid-time Type-D drift. Overridden in 02b-customize-functions.sql: native transpiler mistranslates LOOKUP(constant, Table[NonPKColumn], Table[TargetColumn]) into a lookup keyed on the table's PK instead of the specified match column.



Geographic Drift Direction

Calculated

string

DR-165

IN RULESPEAK®

A model summary's [geographic drift direction](#) is computed as the value of `LOOKUP("id-geographic-composition", ConfounderDistortionTimeline[ConfounderIdentity], ConfounderDistortionTimeline[DriftDirection])`.

loop-92: ConfounderDistortionTimeline.DriftDirection for id-geographic-composition (geography archetype), valid-time Type-D drift. Overridden in 02b-customize-functions.sql: native transpiler mistranslates LOOKUP(constant, Table[NonPKColumn], Table[TargetColumn]) into a lookup keyed on the table's PK instead of the specified match column.



Situational Drift Direction

Calculated

string

DR-166

IN RULESPEAK®

A model summary's [situational drift direction](#) is computed as the value of `LOOKUP("id-situational-context", ConfounderDistortionTimeline[ConfounderIdentity], ConfounderDistortionTimeline[DriftDirection])`.

loop-92: ConfounderDistortionTimeline.DriftDirection for id-situational-context (sports-situation archetype), valid-time Type-D drift. Overridden in 02b-customize-functions.sql: native transpiler mistranslates LOOKUP(constant, Table[NonPKColumn], Table[TargetColumn]) into a lookup keyed on the table's PK instead of the specified match column.

f Age Drift Direction Calculated string DR-167

IN RULESPEAK®

A model summary's [age drift direction](#) is computed as the value of LOOKUP("id-age-composition", ConfounderDistortionTimeline[ConfounderIdentity], ConfounderDistortionTimeline[DriftDirection]).

loop-92: ConfounderDistortionTimeline.DriftDirection for id-age-composition (demographic-composition archetype), valid-time Type-D drift. Overridden in 02b-customize-functions.sql: native transpiler mistranslates LOOKUP(constant, Table[NonPKColumn], Table[TargetColumn]) into a lookup keyed on the table's PK instead of the specified match column.

f Institutional Drift Direction Calculated string DR-168

IN RULESPEAK®

A model summary's [institutional drift direction](#) is computed as the value of LOOKUP("id-institutional-unit", ConfounderDistortionTimeline[ConfounderIdentity], ConfounderDistortionTimeline[DriftDirection]).

loop-92: ConfounderDistortionTimeline.DriftDirection for id-institutional-unit (organizational-unit archetype), valid-time Type-D drift. Overridden in 02b-customize-functions.sql: native transpiler mistranslates LOOKUP(constant, Table[NonPKColumn], Table[TargetColumn]) into a lookup keyed on the table's PK instead of the specified match column.

f Socioeconomic Drift Direction Calculated string DR-169

IN RULESPEAK®

A model summary's [socioeconomic drift direction](#) is computed as the value of LOOKUP("id-socioeconomic-status", ConfounderDistortionTimeline[ConfounderIdentity], ConfounderDistortionTimeline[DriftDirection]).

loop-92: ConfounderDistortionTimeline.DriftDirection for id-socioeconomic-status (ses-income archetype), valid-time Type-D drift. Overridden in 02b-customize-functions.sql: native transpiler mistranslates LOOKUP(constant, Table[NonPKColumn], Table[TargetColumn]) into a lookup keyed on the table's PK instead of the specified match column.

f Domain Flip Gap Survives Geometry Control Calculated 1 hop from ground truth

boolean DR-170

IN RULESPEAK®

A model summary is flagged domain flip gap survives geometry control if all of the following:

- the [economics high imbalance sign flip count](#) is 0
- the [epidemiology high imbalance sign flip rate](#) is greater than 0.15

▼ **Inputs** 2 fields this rule reads

● **Economics High Imbalance Sign Flip Count** Model Summary

Sign-flip count among economics studies with `max_stratum_imbalance >= threshold`.

● **Epidemiology High Imbalance Sign Flip Rate** Model Summary

Sign-flip rate among high-imbalance epidemiology studies.

► **Ground truth at the leaves** 2 the raw values this ultimately depends on

TRUE when the economics-vs-epidemiology sign-flip-rate gap for H-domain-flip-geometry-controlled survives after restricting to high-imbalance studies (loop-90 native promotion of the former 02b hand function).

f **Corpus Pattern Superseded Fail Count** Calculated 2 hops from ground truth integer

DR-171

IN RULESPEAK®

A model summary's corpus pattern superseded fail count is computed as the number of the model summary's discovery findings that are not confirmed plus the number of the model summary's discovery findings that are not confirmed plus the number of the model summary's discovery findings that are not confirmed plus the number of the model summary's discovery findings that are not confirmed plus the number of the model summary's discovery findings that are not confirmed plus the number of the model summary's discovery findings that are not confirmed.

▼ **Inputs** 2 fields this rule reads

● **Hypothesis ID** Discovery Finding

FK → DiscoveryHypotheses.HypothesisId.

f **Is Confirmed** Discovery Finding

```
IF({{HypothesisId}} = "H-latent-d",
  LOOKUP("simpsons-paradox-v1",
    ModelSummary[ModelSummaryId],
    ModelSummary[LatentTypeDFraction]) > 0.5,
  IF({{HypothesisId}} = "H-purity",
    LOOKUP("simpsons-paradox-v1",
      ModelSummary[ModelSummaryId],
      ModelSummary[SignFlipSignalPurityMax]) < 0.5,
    IF({{HypothesisId}} = "H-small-effect",
      LOOKUP("simpsons-paradox-v1",
        ModelSummary[ModelSummaryId],
        ModelSummary[AvgPooledGapStableD]) >
      LOOKUP("simpsons-paradox-v1",
        ModelSummary[ModelSummaryId],
        ModelSummary[AvgPooledGapLatentD]),
      IF({{HypothesisId}} = "H-econ-zero",
        LOOKUP("simpsons-paradox-v1",
```

```
ModelSummary[ModelSummaryId],
ModelSummary[EconomicsSignFlipCount]) = 0,
IF({{HypothesisId}} = "H-domain-dist",
LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[EpidemiologyAvgDistortion]) >
LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[EducationAvgDistortion])),
IF({{HypothesisId}} = "H-causal-manifest",
AND(LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[ConfounderSignFlipCount]) >
LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[ColliderSelectionManifestCount])),
LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[ConfounderSignFlipCount]) >=
10), IF({{HypothesisId}} = "H-causal-latent",
AND(LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[ColliderSelectionManifestCount])
= 0, LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[ColliderSelectionLatentOnlyCount])
>= 5), IF({{HypothesisId}} = "H-explained-
confounder", AND(LOOKUP("simpsons-paradox-
v1", ModelSummary[ModelSummaryId],
ModelSummary[ExplainedConfounderCount]) =
LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[ConfounderSignFlipCount]),
LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[ExplainedConfounderCount]) >=
10), IF({{HypothesisId}} = "H-unexplained-
nonconfounder", LOOKUP("simpsons-paradox-
v1", ModelSummary[ModelSummaryId],
ModelSummary[ContestedOrMediatorExplainedCount])
= 0, IF({{HypothesisId}} = "H-catalog-exact-
match", LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[TypePredictionMatchRate]) < 0.5,
IF({{HypothesisId}} = "H-catalog-flip-prediction",
LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[SignFlipPredictionMatchRate]) <
0.5, IF({{HypothesisId}} = "H-domain-flip-
```

geometry-controlled", AND(LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[EconomicsHighImbalanceSignFlipCount]) = 0, LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[EpidemiologyHighImbalanceSignFlipRate]) > 0.15), IF({{HypothesisId}} = "H-collider-no-manifest-v2", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[ColliderSelectionManifestCount]) = 0, IF({{HypothesisId}} = "H-cplus-magnitude", AND(LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[CPlusAvgDistortion]) > LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[CMinusAvgDistortion])), LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[CPlusAvgDistortion]) > LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[TypeDAvgDistortion])), IF({{HypothesisId}} = "H-econ-encoding-selection", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[EconomicsExpectedAMismatchRate]) > 0.5, IF({{HypothesisId}} = "H-domain-profiles-stable", AND(LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[EducationLatentFraction]) > 0.5, LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[SportsLatentFraction]) > 0.5, LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[EconomicsSignFlipRate]) < 0.05), IF({{HypothesisId}} = "H-corrected-gap-invariant", AND(LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[MaxStudySweepCorrectedGapRange]) < 0.0001, LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[CorrectedGapInvariantFailCount]) = 0), IF({{HypothesisId}} = "H-explained-bidirectional", AND(LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[ExplainedConfounderCount]) = LOOKUP("simpsons-paradox-v1",


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ModelSummary[ModelSummaryId],
ModelSummary[ConfounderSignFlipCount]),
LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[FalsePositiveExplainedCount]) =
0, LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[UnexplainedConfounderSignFlipCount])
= 0), IF({{HypothesisId}} = "H-collider-no-
manifest-theorem", AND(LOOKUP("simpsons-
paradox-v1", ModelSummary[ModelSummaryId],
ModelSummary[ColliderSelectionManifestCount])
= 0, LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[ColliderSelectionCount]) >= 10),
IF({{HypothesisId}} = "H-theorem-portfolio",
LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[TheoremCount]) >= 4,
IF({{HypothesisId}} = "H-age-identity-flip-rate",
AND(LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[AgeIdentityManifestFlipRate]) >=
0.40, LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[AgeIdentityStudyCount]) >= 10),
IF({{HypothesisId}} = "H-identity-map-coverage",
LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[IdentityMapCoverageRate]) >=
0.95, IF({{HypothesisId}} = "H-severity-medicine-
manifest", LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[SeverityMedicineManifestFlipRate])
>= 0.45, IF({{HypothesisId}} = "H-severity-epi-
latent-only", LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[SeverityEpiManifestFlipCount]) =
0, IF({{HypothesisId}} = "H-mechanism-other-
drain", LOOKUP("cluster-id-mechanism-other",
IdentityClusterSummaries[IdentityClusterId],
IdentityClusterSummaries[StudyCount]) <= 5,
IF({{HypothesisId}} = "H-selection-frailty-zero-
manifest", AND(LOOKUP("cluster-id-selection-
frailty",
IdentityClusterSummaries[IdentityClusterId],
IdentityClusterSummaries[ManifestFlipCount]) =
0, LOOKUP("cluster-id-selection-frailty",
IdentityClusterSummaries[IdentityClusterId],

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IdentityClusterSummaries[StudyCount]) >= 3),
IF({{HypothesisId}} = "H-collider-identity-low-
manifest", AND(LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[ColliderIdentityManifestFlipRate])
<= 0.15, LOOKUP("cluster-id-collider-proxy",
IdentityClusterSummaries[IdentityClusterId],
IdentityClusterSummaries[StudyCount]) >= 5),
IF({{HypothesisId}} = "H-geographic-stable-type-
d", AND(LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[GeographicTypeDFraction]) >=
0.25, LOOKUP("cluster-id-geographic-
composition",
IdentityClusterSummaries[IdentityClusterId],
IdentityClusterSummaries[StudyCount]) >= 3),
IF({{HypothesisId}} = "H-unexplained-flips-only-
nonconfounders", LOOKUP("simpsons-paradox-
v1", ModelSummary[ModelSummaryId],
ModelSummary[UnexplainedConfounderSignFlipCount])
= 0, IF({{HypothesisId}} = "H-confounder-drift-
type-d", AND(OR(LOOKUP("simpsons-paradox-
v1", ModelSummary[ModelSummaryId],
ModelSummary[SeverityDriftDirection]) = "rising",
LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[SeverityDriftDirection]) = "single-
decade"), OR(LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[GeographicDriftDirection]) =
"rising", LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[GeographicDriftDirection]) =
"single-decade"), OR(LOOKUP("simpsons-
paradox-v1", ModelSummary[ModelSummaryId],
ModelSummary[SituationalDriftDirection]) =
"rising", LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[SituationalDriftDirection]) =
"single-decade"), LOOKUP("simpsons-paradox-
v1", ModelSummary[ModelSummaryId],
ModelSummary[AgeDriftDirection]) <> "rising",
LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[InstitutionalDriftDirection]) <>
"rising", LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[SocioeconomicDriftDirection]) <>
"rising"), FALSE()))))))))))))))))))))))))))).

```

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Count of DiscoveryFindings rows for the 6 superseded-pattern hypotheses (H-econ-zero, H-domain-dist, H-catalog-flip-prediction, H-domain-flip-geometry-controlled, H-econ-encoding-selection, H-domain-profiles-stable) where IsConfirmed=FALSE. Six separate COUNTIFS summed with '+' because this rulebook's formula vocabulary has no IN-list/pipe-delimited OR criterion for COUNTIFS (loop-90 native promotion of the former 02b hand function).

f Expansion Wave3 Discovery Note Calculated 8 hops from ground truth string DR-172

IN RULESPEAK®

A model summary's expansion wave3 discovery note is computed as "superseded=", followed by the corpus pattern superseded fail count, followed by "; econFlips=", followed by the economics sign flip count, followed by "; flipPred=", followed by the sign flip prediction match rate, followed by "; catalogExact=", followed by the type prediction match rate, followed by "; theorems=", followed by the theorem count.

▼ **Inputs** 5 fields this rule reads

f Corpus Pattern Superseded Fail Count Model Summary

((((count of discovery findings whose hypothesis ID is "H-econ-zero" and is confirmed is false() plus count of discovery findings whose hypothesis ID is "H-domain-dist" and is confirmed is false()) plus count of discovery findings whose hypothesis ID is "H-catalog-flip-prediction" and is confirmed is false()) plus count of discovery findings whose hypothesis ID is "H-domain-flip-geometry-controlled" and is confirmed is false()) plus count of discovery findings whose hypothesis ID is "H-econ-encoding-selection" and is confirmed is false()) plus count of discovery findings whose hypothesis ID is "H-domain-profiles-stable" and is confirmed is false()).

f Economics Sign Flip Count Model Summary

Count of treatment rankings whose study domain is "economics" and is sign flip is true().

↗ Sign Flip Prediction Match Rate Model Summary

LOOKUP("catalog-v1", CorpusCatalogSummary[CatalogSummaryId], CorpusCatalogSummary[SignFlipPredictionMatchRate]

↗ Type Prediction Match Rate Model Summary

LOOKUP("catalog-v1", CorpusCatalogSummary[CatalogSummaryId], CorpusCatalogSummary[TypePredictionMatchRate]).

f Theorem Count Model Summary

Count of conclusions whose category is "corpus-theorem" and status is "witnessed".

- **Ground truth at the leaves** 14 the raw values this ultimately depends on

Expansion-wave-3 supersession-audit synthesis rollup (loop-78, promoted to native loop-90).

Stratum Variable

A stratum variable is identified by its name and is related to a study.

f **Name** Calculated 1 hop from ground truth string DR-173

IN RULESPEAK®

A stratum variable's name is the same as its stratum variable ID.

- ▼ **Inputs** 1 fields this rule reads

- **Stratum Variable ID** Stratum Variable
Unique slug (e.g. kidney-1986-stone-size).

- **Ground truth at the leaves** 1 the raw values this ultimately depends on

Display label.

... **Study** Relationship string

A pointer to Study. It is not derived from anything else — it is the link the rest of the DAG hangs off.

FK to the parent Study.

● **Variable Name** Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

The name of the stratifying variable (e.g. stone_size, department, age_group).

● **Causal Role** Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

The variable's causal role: 'confounder' (affects both assignment and outcome), 'mediator' (downstream of treatment), 'contested' (causal role disputed), 'unknown'.

Affects Treatment Assignment Ground truth boolean

Ground truth. Written directly to the database (datatype `boolean`); not computed from anything else. Everything derived ultimately traces back to values like this one.

TRUE if this variable causally influences which treatment a patient receives. Required for confounding.

Affects Outcome Ground truth boolean

Ground truth. Written directly to the database (datatype `boolean`); not computed from anything else. Everything derived ultimately traces back to values like this one.

TRUE if this variable causally influences the outcome independently of treatment. Required for confounding.

Is Confounder Calculated 1 hop from ground truth boolean DR-174

IN RULESPEAK®

A stratum variable is considered a confounder if all of the following:

- the [affects treatment assignment](#) flag is set
- the [affects outcome](#) flag is set
- the [causal role](#) is "confounder"

▼ **Inputs** 3 fields this rule reads

Affects Treatment Assignment Stratum Variable

TRUE if this variable causally influences which treatment a patient receives. Required for confounding.

Affects Outcome Stratum Variable

TRUE if this variable causally influences the outcome independently of treatment. Required for confounding.

Causal Role Stratum Variable

The variable's causal role: 'confounder' (affects both assignment and outcome), 'mediator' (downstream of treatment), 'contested' (causal role disputed), 'unknown'.

► **Ground truth at the leaves** 3 the raw values this ultimately depends on

TRUE when AffectsTreatmentAssignment AND AffectsOutcome AND CausalRole = 'confounder'. A confounder is the classic driver of Simpson's Paradox.



Mechanism Note

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Plain-language description of how this variable creates the confounding: why it affects both assignment and outcome.



Conditioning Risk

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

The causal hazard posed by conditioning on this variable. Values: confounder (safe to condition — removes bias), mediator (conditioning blocks the causal path of interest — harmful), collider (conditioning opens a spurious backdoor — harmful), proxy (conditioning on a noisy proxy of the true confounder — risky), none (no conditioning risk identified). NULL means unclassified.



Confounder Identity

Lookup

1 hop from ground truth

string

DR-175

IN RULESPEAK®

A stratum variable's [confounder identity](#) is the value of `LOOKUP(StratumVariableId, StratumVariableIdentityMaps[StratumVariable], StratumVariableIdentityMaps[ConfounderIdentity])`.

▼ **Inputs** 1 fields this rule reads

● Stratum Variable ID

Stratum Variable

Unique slug (e.g. kidney-1986-stone-size).

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Canonical cross-study identity via StratumVariableIdentityMaps (loop-80).



Annotation Commit Hash

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Full git SHA of the commit that first introduced this row's CausalRole/MechanismNote — derived via ``git log -S`` pickaxe search on this row's StratumVariableId (loop-98 audit-trail backfill). Real repo history, not a hand-entered label.



Annotation Commit Date

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

ISO date (YYYY-MM-DD) of AnnotationCommitHash.

Annotation Commit Subject Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Git commit subject line of AnnotationCommitHash — the loop/change that landed this row's causal-role judgment call.

Causal Role Ever Revised Ground truth boolean

Ground truth. Written directly to the database (datatype `boolean`); not computed from anything else. Everything derived ultimately traces back to values like this one.

TRUE if this row's CausalRole value differs across any two points in its git history (a real reclassification); FALSE if the value has been identical since AnnotationCommitHash. Computed once at loop-97 backfill via full-history diff, not re-derived live — re-run the backfill script after any future CausalRole edit to keep this accurate.

Treatment Ranking

A treatment ranking is identified by its [name](#) and is related to a [study](#).

f Name Calculated 1 hop from ground truth string DR-176

IN RULESPEAK®

A treatment ranking's [name](#) is the same as its treatment ranking ID.

▼ **Inputs** 1 fields this rule reads

● **Treatment Ranking ID** Treatment Ranking
Unique slug (e.g. kidney-1986-A-vs-B).

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Display label.

... Study Relationship string

A pointer to [Study](#). It is not derived from anything else — it is the link the rest of the DAG hangs off.

FK to the parent Study.

● Treatment a Ground truth string

Ground truth. Written directly to the database (datatype string); not computed from anything else. Everything derived ultimately traces back to values like this one.

Label for the first treatment in the comparison (matches `Treatments.TreatmentLabel`).

● Treatment B Ground truth string

Ground truth. Written directly to the database (datatype string); not computed from anything else. Everything derived ultimately traces back to values like this one.

Label for the second treatment in the comparison.

Σ Total Cases a Aggregation 1 hop from ground truth integer DR-177

IN RULESPEAK®

A treatment ranking's total cases a is the total cases across the case cells related to the treatment ranking.

▼ Inputs 5 fields this rule reads

● **Cases** Case Cell
Total number of patients in this cell. RAW INPUT
— never derived.

↗ **Study** Case Cell
FK to the parent Study.

↗ **Study** Treatment Ranking
FK to the parent Study.

● **Treatment Label** Case Cell
The treatment applied in this cell (matches `Treatments.TreatmentLabel`).

● **Treatment a** Treatment Ranking
Label for the first treatment in the comparison
(matches `Treatments.TreatmentLabel`).

► Ground truth at the leaves 5 the raw values this ultimately depends on

Total cases for TreatmentA across all strata in this study.

Σ Total Successes a Aggregation 1 hop from ground truth integer DR-178

IN RULESPEAK®

A treatment ranking's total successes a is the total successes across the case cells related to the treatment ranking.

▼ **Inputs** 5 fields this rule reads

● **Successes**

Case Cell

Number of patients in this cell who had a successful outcome. RAW INPUT — never derived.

→ **Study**

Case Cell

FK to the parent Study.

→ **Study**

Treatment Ranking

FK to the parent Study.

● **Treatment Label**

Case Cell

The treatment applied in this cell (matches Treatments.TreatmentLabel).

● **Treatment a**

Treatment Ranking

Label for the first treatment in the comparison (matches Treatments.TreatmentLabel).

► **Ground truth at the leaves** 5 the raw values this ultimately depends on

Total successes for TreatmentA across all strata.

f **Pooled Rate a** Calculated 2 hops from ground truth number DR-179

IN RULESPEAK®

The treatment ranking's pooled rate a is determined by the following priority:

1. **an empty string** if the total cases a is 0
2. **the total successes a divided by the total cases a** in all other cases

▼ **Inputs** 2 fields this rule reads

Σ **Total Cases a**

Treatment Ranking

Sum of cases across all case cells whose study is study and treatment label is treatment a.

Σ **Total Successes a**

Treatment Ranking

Sum of successes across all case cells whose study is study and treatment label is treatment a.

► **Ground truth at the leaves** 6 the raw values this ultimately depends on

Pooled success rate for TreatmentA: TotalSuccessesA / TotalCasesA.

Σ **Total Cases B** Aggregation 1 hop from ground truth integer DR-180

IN RULESPEAK®

A treatment ranking's total cases b is the total cases across the case cells related to the treatment ranking.

▼ **Inputs** 5 fields this rule reads

● **Cases**

Case Cell

Total number of patients in this cell. RAW INPUT
— never derived.

→ **Study**

Case Cell

FK to the parent Study.

→ **Study**

Treatment Ranking

FK to the parent Study.

● **Treatment Label**

Case Cell

The treatment applied in this cell (matches
Treatments.TreatmentLabel).

● **Treatment B**

Treatment Ranking

Label for the second treatment in the comparison.

► **Ground truth at the leaves** 5 the raw values this ultimately depends on

Total cases for TreatmentB across all strata.

Σ Total Successes B Aggregation 1 hop from ground truth integer DR-181

IN RULESPEAK®

A treatment ranking's total successes b is the total successes across the case cells related to the treatment ranking.

▼ **Inputs** 5 fields this rule reads

● **Successes**

Case Cell

Number of patients in this cell who had a
successful outcome. RAW INPUT — never
derived.

→ **Study**

Case Cell

FK to the parent Study.

→ **Study**

Treatment Ranking

FK to the parent Study.

● **Treatment Label**

Case Cell

The treatment applied in this cell (matches
Treatments.TreatmentLabel).

● **Treatment B**

Treatment Ranking

Label for the second treatment in the comparison.

► **Ground truth at the leaves** 5 the raw values this ultimately depends on

Total successes for TreatmentB across all strata.

f Pooled Rate B Calculated 2 hops from ground truth number DR-182

IN RULESPEAK®

The treatment ranking's pooled rate b is determined by the following priority:

1. **an empty string** if the total cases b is 0
2. **the total successes b divided by the total cases b** in all other cases

▼ **Inputs** 2 fields this rule reads

Σ **Total Cases B**

Treatment Ranking

Sum of cases across all case cells whose study is study and treatment label is treatment b.

Σ **Total Successes B**

Treatment Ranking

Sum of successes across all case cells whose study is study and treatment label is treatment b.

► **Ground truth at the leaves** 6 the raw values this ultimately depends on

Pooled success rate for TreatmentB: $TotalSuccessesB / TotalCasesB$.

f

Pooled Winner

Calculated

3 hops from ground truth

string

DR-183

IN RULESPEAK®

The treatment ranking's pooled winner is determined by the following priority:

1. **an empty string** if **any of these hold**:
 - the pooled rate a is blank
 - the pooled rate b is blank
2. **"tie"** if the absolute value of the pooled rate a minus the pooled rate b is less than 0.0000001
3. **the treatment a** if the pooled rate a is greater than the pooled rate b
4. **the treatment b** in all other cases

▼ **Inputs** 4 fields this rule reads

f **Pooled Rate a**

Treatment Ranking

If total cases a equals 0 then "", otherwise (total successes a divided by total cases a).

f **Pooled Rate B**

Treatment Ranking

If total cases b equals 0 then "", otherwise (total successes b divided by total cases b).

● **Treatment a**

Treatment Ranking

Label for the first treatment in the comparison (matches Treatments.TreatmentLabel).

● **Treatment B**

Treatment Ranking

Label for the second treatment in the comparison.

► **Ground truth at the leaves** 7 the raw values this ultimately depends on

Which treatment wins when strata are ignored. Returns tie when pooled rates are equal within floating-point tolerance.

Σ

Stratum Count

Aggregation

1 hop from ground truth

integer

DR-184

IN RULESPEAK®

A treatment ranking's [stratum count](#) is the number of [stratum summaries](#) related to the treatment ranking.

▼ **Inputs** 4 fields this rule reads

Study Stratum Summary FK to the parent Study.	Study Treatment Ranking FK to the parent Study.
Treatment Label Stratum Summary Which treatment (matches Treatments.TreatmentLabel).	Treatment a Treatment Ranking Label for the first treatment in the comparison (matches Treatments.TreatmentLabel).

► **Ground truth at the leaves** 4 the raw values this ultimately depends on

Number of strata in this study. Counted from StratumSummaries filtered to TreatmentA rows (one row per stratum).

Σ **Strata Won by a** Aggregation 4 hops from ground truth integer DR-185

IN RULESPEAK®

A treatment ranking's [strata won by a](#) is the number of [stratum summaries](#) related to the treatment ranking.

▼ **Inputs** 5 fields this rule reads

Study Stratum Summary FK to the parent Study.	Study Treatment Ranking FK to the parent Study.
Treatment Label Stratum Summary Which treatment (matches Treatments.TreatmentLabel).	Treatment a Treatment Ranking Label for the first treatment in the comparison (matches Treatments.TreatmentLabel).
Stratum Winner Stratum Summary If stratum rate a is greater than stratum rate b then "A", otherwise "B".	

► **Ground truth at the leaves** 10 the raw values this ultimately depends on

Number of strata where A wins: count StratumSummaries rows whose TreatmentLabel=TreatmentA AND StratumWinner=TreatmentA. These are the one-per-stratum sentinel rows.

Σ **Strata Won by B** Aggregation 4 hops from ground truth integer DR-186

IN RULESPEAK®

A treatment ranking's strata won by b is the number of stratum summaries related to the treatment ranking.

▼ **Inputs** 6 fields this rule reads

Study Stratum Summary FK to the parent Study.	Study Treatment Ranking FK to the parent Study.
Treatment Label Stratum Summary Which treatment (matches Treatments.TreatmentLabel).	Treatment a Treatment Ranking Label for the first treatment in the comparison (matches Treatments.TreatmentLabel).
Stratum Winner Stratum Summary If stratum rate a is greater than stratum rate b then "A", otherwise "B".	Treatment B Treatment Ranking Label for the second treatment in the comparison.

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

Number of strata where B wins: count StratumSummaries rows whose TreatmentLabel=TreatmentA AND StratumWinner=TreatmentB.

f **Per Stratum Winner** Calculated 5 hops from ground truth string DR-187

IN RULESPEAK®

The treatment ranking's per stratum winner is determined by the following priority:

1. the treatment a if the strata won by a is the stratum count
2. the treatment b if the strata won by b is the stratum count
3. **"none"** in all other cases

▼ **Inputs** 5 fields this rule reads

Σ Strata Won by a Treatment Ranking Count of stratum summaries whose study is study, treatment label is treatment a, and stratum winner is treatment a.	Σ Stratum Count Treatment Ranking Count of stratum summaries whose study is study and treatment label is treatment a.
Treatment a Treatment Ranking Label for the first treatment in the comparison (matches Treatments.TreatmentLabel).	Σ Strata Won by B Treatment Ranking Count of stratum summaries whose study is study, treatment label is treatment a, and stratum winner is treatment b.
Treatment B Treatment Ranking	

Label for the second treatment in the comparison.

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

The treatment that wins in every stratum — 'A' if $\text{StrataWonByA} = \text{StratumCount}$, 'B' if $\text{StrataWonByB} = \text{StratumCount}$, else 'none' (no unanimous per-stratum winner).



Is Stratum Unanimous

Calculated

5 hops from ground truth

boolean

DR-188

IN RULESPEAK®

A treatment ranking is considered stratum-unanimous if all of the following:

- the strata won by a is the stratum count
- the strata won by b is the stratum count

▼ **Inputs** 3 fields this rule reads

Σ Strata Won by a

Treatment Ranking

Count of stratum summaries whose study is study, treatment label is treatment a, and stratum winner is treatment a.

Σ Stratum Count

Treatment Ranking

Count of stratum summaries whose study is study and treatment label is treatment a.

Σ Strata Won by B

Treatment Ranking

Count of stratum summaries whose study is study, treatment label is treatment a, and stratum winner is treatment b.

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

TRUE when one treatment wins every stratum ($\text{StrataWonByA} = \text{StratumCount}$ OR $\text{StrataWonByB} = \text{StratumCount}$). Maps A/B policy split: unanimous → stratify-immediately geometry; heterogeneous → investigate-confounder.



Is Reversal

Calculated

7 hops from ground truth

boolean

DR-189

IN RULESPEAK®

The treatment ranking's is reversal is determined by the following priority:

1. an empty string if the is sign flip is blank
2. the is sign flip in all other cases

▼ **Inputs** 1 fields this rule reads

f Is Sign Flip

Treatment Ranking

If weighted stratum gap sum is "" then "",
otherwise if weighted stratum gap sum is greater

than 0 then signed pooled gap is less than 0,
otherwise signed pooled gap is greater than 0.

- **Ground truth at the leaves** 11 the raw values this ultimately depends on

TRUE when the equal-weight pooled signal (WeightedStratumGapSum) and the actual pooled signal (SignedPooledGap) point in opposite directions — allocation has flipped the direction of the aggregate conclusion. This is Simpson's Paradox as a derived fact.

Σ **Confounders in Study** Aggregation 2 hops from ground truth integer DR-190

IN RULESPEAK®

A treatment ranking's confounders in study is the number of the treatment ranking's stratum variables that are confounders.

- ▼ **Inputs** 3 fields this rule reads

... **Study** Stratum Variable
FK to the parent Study.

... **Study** Treatment Ranking
FK to the parent Study.

f **Is Confounder** Stratum Variable
Affects treatment assignment, affects outcome,
and causal role is "confounder".

- **Ground truth at the leaves** 5 the raw values this ultimately depends on

Count of StratumVariables in this study whose IsConfounder = TRUE. When > 0, the paradox has a causal explanation.

f **Is Paradox Explained** Calculated 8 hops from ground truth boolean DR-191

IN RULESPEAK®

A treatment ranking is considered paradox-explained if all of the following:

- the reversal flag is set
- the confounders in study is greater than 0

- ▼ **Inputs** 2 fields this rule reads

f **Is Reversal** Treatment Ranking
If is sign flip is "" then "", otherwise is sign flip.

Σ **Confounders in Study** Treatment Ranking
Count of stratum variables whose study is study
and is confounder is true().

- **Ground truth at the leaves** 15 the raw values this ultimately depends on

TRUE when *IsReversal* is present AND at least one confirmed confounder exists in the study.



Pooled Gap

Calculated

3 hops from ground truth

number

DR-192

IN RULESPEAK®

The treatment ranking's pooled gap is determined by the following priority:

1. an empty string if the pooled rate a is blank
2. the absolute value of the pooled rate a minus the pooled rate b in all other cases

▼ Inputs 2 fields this rule reads

f Pooled Rate a

Treatment Ranking

If total cases a equals 0 then "", otherwise (total successes a divided by total cases a).

f Pooled Rate B

Treatment Ranking

If total cases b equals 0 then "", otherwise (total successes b divided by total cases b).

► Ground truth at the leaves 7 the raw values this ultimately depends on

Absolute difference between the two pooled rates: $|PooledRateA - PooledRateB|$. The size of the aggregate misleading signal.



Strata Won by Loser

Calculated

5 hops from ground truth

integer

DR-193

IN RULESPEAK®

The treatment ranking's strata won by loser is determined by the following priority:

1. the strata won by b if the pooled winner is the treatment a
2. the strata won by a in all other cases

▼ Inputs 4 fields this rule reads

f Pooled Winner

Treatment Ranking

If pooled rate a is "" or pooled rate b is "" then "", otherwise if the absolute value of (pooled rate a minus pooled rate b) is less than 0.0000001 then "tie", otherwise if pooled rate a is greater than pooled rate b then treatment a, otherwise treatment b.

● Treatment a

Treatment Ranking

Label for the first treatment in the comparison (matches `Treatments.TreatmentLabel`).

Σ Strata Won by B

Treatment Ranking

Count of stratum summaries whose study is study, treatment label is treatment a, and stratum winner is treatment b.

Σ Strata Won by a

Treatment Ranking

Count of stratum summaries whose study is study, treatment label is treatment a, and stratum winner is treatment a.

► Ground truth at the leaves 11 the raw values this ultimately depends on

Number of strata won by the pooled loser — the counter-signal. For full reversals this equals StratumCount; for partial paradoxes it is between 0 and StratumCount.

f Paradox Strength Calculated 6 hops from ground truth number DR-194

IN RULESPEAK®

The treatment ranking's paradox strength is determined by the following priority:

1. an empty string if the stratum count is 0
2. the pooled gap times the strata won by loser divided by the stratum count in all other cases

▼ Inputs 3 fields this rule reads

Σ **Stratum Count** Treatment Ranking
Count of stratum summaries whose study is study and treatment label is treatment a.

f **Pooled Gap** Treatment Ranking
If pooled rate a is "" then "", otherwise the absolute value of (pooled rate a minus pooled rate b).

f **Strata Won by Loser** Treatment Ranking
If pooled winner equals treatment a then strata won by b, otherwise strata won by a.

► Ground truth at the leaves 11 the raw values this ultimately depends on

Scalar severity of the paradox: $\text{PooledGap} \times (\text{StrataWonByLoser} / \text{StratumCount})$. Zero when no strata go against the pooled winner. Positive for partial paradoxes. Maximum when every stratum contradicts the pooled result.

Σ Pooled Rate From Weights a Aggregation 4 hops from ground truth number DR-195

IN RULESPEAK®

A treatment ranking's pooled rate from weights a is the total weighted stratum rate across the stratum summaries related to the treatment ranking.

▼ Inputs 5 fields this rule reads

f **Weighted Stratum Rate** Stratum Summary
If stratum fraction is "" then "", otherwise (stratum fraction times stratum success rate).

→ **Study** Stratum Summary
FK to the parent Study.

→ **Study** Treatment Ranking
FK to the parent Study.

● **Treatment Label** Stratum Summary
Which treatment (matches Treatments.TreatmentLabel).

● **Treatment a** Treatment Ranking

Label for the first treatment in the comparison
(matches `Treatments.TreatmentLabel`).

- **Ground truth at the leaves** 10 the raw values this ultimately depends on

*Pooled rate for TreatmentA reconstructed as a weighted average of stratum rates:
 $SUM(WeightedStratumRate)$ across all stratum rows for TreatmentA. Must match `PooledRateA` — this witnesses the mechanism equation: the pooled rate IS a weighted average of stratum rates, weighted by `StratumFraction`.*

Σ **Pooled Rate From Weights B** Aggregation 4 hops from ground truth number DR-196

IN RULESPEAK®

A treatment ranking's pooled rate from weights b is the total weighted stratum rate across the stratum summaries related to the treatment ranking.

- ▼ **Inputs** 5 fields this rule reads

f **Weighted Stratum Rate** Stratum Summary

If stratum fraction is "" then "", otherwise
(stratum fraction times stratum success rate).

→ **Study** Stratum Summary

FK to the parent Study.

→ **Study** Treatment Ranking

FK to the parent Study.

● **Treatment Label** Stratum Summary

Which treatment (matches
`Treatments.TreatmentLabel`).

● **Treatment B** Treatment Ranking

Label for the second treatment in the comparison.

- **Ground truth at the leaves** 10 the raw values this ultimately depends on

Pooled rate for TreatmentB reconstructed as a weighted average. Must match `PooledRateB`. If it does, the mechanism equation is verified: reversal is purely a consequence of differential weighting.

f **Reversal Intensity** Calculated 6 hops from ground truth number DR-197

IN RULESPEAK®

The treatment ranking's reversal intensity is determined by the following priority:

1. an empty string if the stratum count is 0
2. the strata won by loser divided by the stratum count in all other cases

- ▼ **Inputs** 2 fields this rule reads

Σ **Stratum Count** Treatment Ranking

Count of stratum summaries whose study is study and treatment label is treatment a.

f **Strata Won by Loser** Treatment Ranking

If pooled winner equals treatment a then strata won by b, otherwise strata won by a.

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

StrataWonByLoser / StratumCount: the fraction of strata that go against the pooled winner. Zero for no paradox, 1.0 for full reversal, between 0 and 1 for partial. This is the allocation-side measure of how deeply the confounding has penetrated the stratified view.

f **Threshold Margin** Calculated 7 hops from ground truth number DR-198

IN RULESPEAK®

The treatment ranking's threshold margin is determined by the following priority:

1. **an empty string** if the stratum count is 0
2. **the reversal intensity minus 0.5** in all other cases

▼ **Inputs** 2 fields this rule reads

Σ **Stratum Count** Treatment Ranking

Count of stratum summaries whose study is study and treatment label is treatment a.

f **Reversal Intensity** Treatment Ranking

If stratum count equals 0 then "", otherwise (strata won by loser divided by stratum count).

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

ReversalIntensity minus 0.5: positive when more than half the strata go against the pooled winner (reversal is robust), zero at the tipping point (exactly half), negative when fewer than half oppose the pooled winner (reversal is fragile or absent). A study with ThresholdMargin > 0 is robustly paradoxical; one with ThresholdMargin < 0 has a weak or absent paradox.

f **Signed Pooled Gap** Calculated 3 hops from ground truth number DR-199

IN RULESPEAK®

The treatment ranking's signed pooled gap is determined by the following priority:

1. **an empty string** if the pooled rate a is blank
2. **the pooled rate a minus the pooled rate b** in all other cases

▼ **Inputs** 2 fields this rule reads

f **Pooled Rate a** Treatment Ranking

If total cases a equals 0 then "", otherwise (total successes a divided by total cases a).

f **Pooled Rate B** Treatment Ranking

If total cases b equals 0 then "", otherwise (total successes b divided by total cases b).

► **Ground truth at the leaves** 7 the raw values this ultimately depends on

Signed difference between pooled rates: $\text{PooledRateA} - \text{PooledRateB}$. Positive = A wins pooled; negative = B wins pooled. Unlike *PooledGap* (which is absolute), this preserves direction — essential for detecting sign flips between the equal-weight and actual pooled signals.

Σ **Weighted Stratum Gap Sum** Aggregation 5 hops from ground truth number DR-200

IN RULESPEAK®

A treatment ranking's weighted stratum gap sum is the total weighted stratum gap across the stratum summaries related to the treatment ranking.

▼ **Inputs** 5 fields this rule reads

f **Weighted Stratum Gap** Stratum Summary
If stratum gap is "" then "", otherwise (stratum gap times stratum fraction).

→ **Study** Stratum Summary
FK to the parent Study.

→ **Study** Treatment Ranking
FK to the parent Study.

● **Treatment Label** Stratum Summary
Which treatment (matches `Treatments.TreatmentLabel`).

● **Treatment a** Treatment Ranking
Label for the first treatment in the comparison (matches `Treatments.TreatmentLabel`).

► **Ground truth at the leaves** 10 the raw values this ultimately depends on

*Sum of $\text{WeightedStratumGap}$ ($= \text{StratumGap} \times \text{StratumFraction}$) across all `StratumSummaries` rows where $\text{TreatmentLabel} = \text{TreatmentA}$ for this study. This is the equal-weight pooled gap — the gap the pooled analysis would produce if allocation were perfectly balanced across strata. When this has the opposite sign to *SignedPooledGap*, the allocation has not merely compressed the signal: it has flipped it entirely.*

f **Is Sign Flip** Calculated 6 hops from ground truth boolean DR-201

IN RULESPEAK®

The treatment ranking's is sign flip is determined by the following priority:

1. **an empty string** if the weighted stratum gap sum is blank
2. **the signed pooled gap is less than 0** if the weighted stratum gap sum is greater than 0
3. **the signed pooled gap is greater than 0** in all other cases

▼ **Inputs** 2 fields this rule reads

Σ **Weighted Stratum Gap Sum** Treatment Ranking

f **Signed Pooled Gap** Treatment Ranking
If pooled rate a is "" then "", otherwise (pooled rate a minus pooled rate b).

Sum of weighted stratum gap across all stratum summaries whose study is study and treatment label is treatment a.

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

TRUE when WeightedStratumGapSum and SignedPooledGap have opposite signs. Same condition as IsReversal; retained as the coordinate-level name in the DAG.



Allocation Distortion

Calculated

6 hops from ground truth

number

DR-202

IN RULESPEAK®

The treatment ranking's allocation distortion is determined by the following priority:

1. **an empty string** if the weighted stratum gap sum is blank
2. **the absolute value of the weighted stratum gap sum minus the signed pooled gap** in all other cases

▼ **Inputs** 2 fields this rule reads

Σ **Weighted Stratum Gap Sum**

Treatment Ranking

Sum of weighted stratum gap across all stratum summaries whose study is study and treatment label is treatment a.

f **Signed Pooled Gap**

Treatment Ranking

If pooled rate a is "" then "", otherwise (pooled rate a minus pooled rate b).

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

The magnitude of how far the allocation has bent the pooled signal: $|WeightedStratumGapSum - SignedPooledGap|$. Zero means the allocation is neutral — the pooled analysis faithfully represents the equal-weight stratum evidence. A large AllocationDistortion means the allocation is doing most of the work: the pooled number is mostly noise from how cases were assigned, not signal about which treatment is better. This measure does NOT require a sign flip to be nonzero — it captures any allocation-induced distortion, not just reversals.



Distortion Type

Calculated

7 hops from ground truth

string

DR-203

IN RULESPEAK®

The treatment ranking's distortion type is determined by the following priority:

1. **an empty string** if the allocation distortion is blank
2. **"A" if the stratum unanimous flag is set, in all other cases "B" if the sign flip flag is set**
3. **"C+" if all of these hold:**
 - the sign flip flag is not set
 - the allocation distortion is greater than 0.01

- the absolute value of the signed pooled gap is greater than the absolute value of the corrected gap plus 0.001

4. **"C-"** if all of these hold:

- the sign flip flag is not set
- the allocation distortion is greater than 0.01
- the absolute value of the signed pooled gap is less than the absolute value of the corrected gap minus 0.001

5. **"D"** in all other cases

▼ Inputs 5 fields this rule reads

f **Allocation Distortion** Treatment Ranking
If weighted stratum gap sum is "" then "", otherwise the absolute value of (weighted stratum gap sum minus signed pooled gap).

f **Is Sign Flip** Treatment Ranking
If weighted stratum gap sum is "" then "", otherwise if weighted stratum gap sum is greater than 0 then signed pooled gap is less than 0, otherwise signed pooled gap is greater than 0.

f **Is Stratum Unanimous** Treatment Ranking
Strata won by a equals stratum count or strata won by b equals stratum count.

f **Signed Pooled Gap** Treatment Ranking
If pooled rate a is "" then "", otherwise (pooled rate a minus pooled rate b).

f **Corrected Gap** Treatment Ranking
Weighted stratum gap sum.

► Ground truth at the leaves 11 the raw values this ultimately depends on

Geometric classification (five types A/B/C+/C-/D). Sign-flip types: A = unanimous stratum direction (IsStratumUnanimous); B = heterogeneous stratum directions. Non-flip: C+/C-/D by distortion ratio.

f **Policy Implication** Calculated 8 hops from ground truth string DR-204

IN RULESPEAK®

The treatment ranking's policy implication is determined by the following priority:

1. **an empty string** if the distortion type is blank
2. **"stratify-immediately"** if the distortion type is "A"
3. **"investigate-confounder"** if the distortion type is "B"
4. **"check-allocation-bias"** if any of these hold:
 - the distortion type is "C+"
 - the distortion type is "C-"
5. **"pooled-analysis-trustworthy"** in all other cases

▼ Inputs 1 fields this rule reads

f Distortion Type

Treatment Ranking

If allocation distortion is "" then "", otherwise if is sign flip then if is stratum unanimous then "A", otherwise "B", otherwise if not is sign flip, allocation distortion is greater than 0.01, and the absolute value of signed pooled gap is greater than (the absolute value of corrected gap plus 0.001) then "C+", otherwise if not is sign flip, allocation distortion is greater than 0.01, and the absolute value of signed pooled gap is less than (the absolute value of corrected gap minus 0.001) then "C-", otherwise "D".

- **Ground truth at the leaves** 11 the raw values this ultimately depends on

The researcher action implied by the geometric classification. Derived from DistortionType: type A (full reversal) → 'stratify-immediately' — the pooled conclusion is directionally wrong; do not act on it without stratifying. Type B (partial sign-flip) → 'investigate-confounder' — the equal-weight signal opposes the pooled signal, but the per-stratum picture is mixed; the causal mechanism needs examination before trusting either estimate. Type C (compression without flip) → 'check-allocation-bias' — the pooled winner is probably correct but the effect size is distorted; report adjusted or standardized margins rather than the raw pooled gap. Type D (neutral) → 'pooled-analysis-trustworthy' — allocation is not materially distorting the signal; the pooled conclusion is safe with respect to this stratification.

f Corrected Gap

Calculated

6 hops from ground truth

number

DR-205

IN RULESPEAK®

A treatment ranking's corrected gap is the same as its weighted stratum gap sum.

- ▼ **Inputs** 1 fields this rule reads

Σ Weighted Stratum Gap Sum

Treatment
Ranking

Sum of weighted stratum gap across all stratum summaries whose study is study and treatment label is treatment a.

- **Ground truth at the leaves** 10 the raw values this ultimately depends on

The allocation-corrected treatment gap: WeightedStratumGapSum, i.e. what SignedPooledGap would be if allocation were equal across strata. Positive means A is favoured in the corrected world; negative means B is favoured.

f Corrected Winner

Calculated

7 hops from ground truth

string

DR-206

IN RULESPEAK®

The treatment ranking's corrected winner is determined by the following priority:

1. **an empty string** if the corrected gap is blank
2. **"tie"** if the absolute value of the corrected gap is less than 0.0001
3. **the treatment a** if the corrected gap is greater than 0
4. **the treatment b** if the corrected gap is less than 0
5. **"tie"** in all other cases

▼ **Inputs** 3 fields this rule reads

f **Corrected Gap** Treatment Ranking
Weighted stratum gap sum.

● **Treatment a** Treatment Ranking
Label for the first treatment in the comparison
(matches `Treatments.TreatmentLabel`).

● **Treatment B** Treatment Ranking
Label for the second treatment in the comparison.

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

Which treatment wins once allocation bias is removed. Derived from `CorrectedGap`: positive → `TreatmentA`; negative → `TreatmentB`; near-zero → `tie`.

f **Corrected Vs Pooled Agreement** Calculated 8 hops from ground truth boolean DR-207

IN RULESPEAK®

The treatment ranking's corrected vs pooled agreement is determined by the following priority:

1. **an empty string** if the corrected winner is blank
2. **the corrected winner is the pooled winner** in all other cases

▼ **Inputs** 2 fields this rule reads

f **Corrected Winner** Treatment Ranking
If corrected gap is "" then "", otherwise if the absolute value of corrected gap is less than 0.0001 then "tie", otherwise if corrected gap is greater than 0 then treatment a, otherwise if corrected gap is less than 0 then treatment b, otherwise "tie".

f **Pooled Winner** Treatment Ranking
If pooled rate a is "" or pooled rate b is "" then "", otherwise if the absolute value of (pooled rate a minus pooled rate b) is less than 0.0000001 then "tie", otherwise if pooled rate a is greater than pooled rate b then treatment a, otherwise treatment b.

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

TRUE when the allocation-corrected winner matches the pooled winner. FALSE when removing allocation bias would change which treatment appears to win — the machine-readable definition of a reversal recovery.



Corrected Policy Implication

Calculated

8 hops from ground truth

string

DR-208

IN RULESPEAK®

The treatment ranking's corrected policy implication is determined by the following priority:

1. **an empty string** if the distortion type is blank
2. **"use-corrected-winner"** if the distortion type is "A"
3. **"use-corrected-winner-with-caution"** if the distortion type is "B"
4. **"check-allocation-bias"** if any of these hold:
 - the distortion type is "C+"
 - the distortion type is "C-"
5. **"pooled-analysis-trustworthy"** in all other cases

▼ Inputs 1 fields this rule reads

f Distortion Type

Treatment Ranking

If allocation distortion is "" then "", otherwise if is sign flip then if is stratum unanimous then "A", otherwise "B", otherwise if not is sign flip, allocation distortion is greater than 0.01, and the absolute value of signed pooled gap is greater than (the absolute value of corrected gap plus 0.001) then "C+", otherwise if not is sign flip, allocation distortion is greater than 0.01, and the absolute value of signed pooled gap is less than (the absolute value of corrected gap minus 0.001) then "C-", otherwise "D".

► Ground truth at the leaves 11 the raw values this ultimately depends on

The allocation-aware researcher action: what to do when the corrected verdict is available. Derived from DistortionType and CorrectedWinner together, making the instrument self-consistent with Reversal Recovery (loop-27). Type A (full sign-flip, unanimous per-stratum reversal): CorrectedWinner is the true signal — 'use-corrected-winner'. The pooled analysis was directionally wrong; act on CorrectedWinner, not PooledWinner. Type B (partial sign-flip, non-unanimous): CorrectedWinner points against the pooled signal but strata disagree among themselves — 'use-corrected-winner-with-caution'. Trust the corrected direction but acknowledge residual uncertainty. Type C (compression, no sign flip): PooledWinner is directionally correct; the distortion is in the magnitude, not the direction — 'check-allocation-bias'. Same as PolicyImplication. Type D (neutral): allocation is not materially distorting the signal — 'pooled-analysis-trustworthy'. Same as PolicyImplication. For Types C and D, CorrectedPolicyImplication and PolicyImplication always agree; for Types A and B, CorrectedPolicyImplication supersedes PolicyImplication with the allocation-corrected verdict.



Allocation Direction

Calculated

7 hops from ground truth

string

DR-209

IN RULESPEAK®

The treatment ranking's allocation direction is determined by the following priority:

1. **an empty string** if the corrected gap is blank
2. **"reversal"** if the sign flip flag is set
3. **"amplification"** if the absolute value of the signed pooled gap is greater than the absolute value of the corrected gap plus 0.001
4. **"compression"** if the absolute value of the signed pooled gap is less than the absolute value of the corrected gap minus 0.001
5. **"neutral"** in all other cases

▼ Inputs 3 fields this rule reads

f **Corrected Gap** Treatment Ranking
Weighted stratum gap sum.

f **Is Sign Flip** Treatment Ranking
If weighted stratum gap sum is "" then "",
otherwise if weighted stratum gap sum is greater
than 0 then signed pooled gap is less than 0,
otherwise signed pooled gap is greater than 0.

f **Signed Pooled Gap** Treatment Ranking
If pooled rate a is "" then "", otherwise (pooled
rate a minus pooled rate b).

► Ground truth at the leaves 11 the raw values this ultimately depends on

Whether the allocation is reversing, amplifying, or compressing the true treatment signal (CorrectedGap). 'reversal': allocation changed the sign of the pooled gap relative to CorrectedGap — worst case; pooled winner is wrong (types A and B). 'amplification': allocation inflated the pooled gap beyond CorrectedGap in the same direction — pooled winner is correct but effect size is overstated (type C+). 'compression': allocation shrank the pooled gap below CorrectedGap in the same direction — pooled winner is correct but effect size is understated (type C-). 'neutral': allocation had negligible effect on the pooled gap (type D). Policy consequences: reversal → wrong decision; amplification → overconfidence; compression → underconfidence / premature abandonment; neutral → trustworthy.

f **Signal Purity** Calculated 7 hops from ground truth number DR-210

IN RULESPEAK®

The treatment ranking's signal purity is determined by the following priority:

1. **an empty string** if the corrected gap is blank
2. **1** if the absolute value of the corrected gap plus the allocation distortion is 0
3. **the absolute value of the corrected gap divided by the absolute value of the corrected gap plus the allocation distortion** in all other cases

▼ Inputs 2 fields this rule reads

f **Corrected Gap** Treatment Ranking

f **Allocation Distortion** Treatment Ranking

Weighted stratum gap sum.

If weighted stratum gap sum is "" then "", otherwise the absolute value of (weighted stratum gap sum minus signed pooled gap).

- **Ground truth at the leaves** 11 the raw values this ultimately depends on

The fraction of total observed variation ($|CorrectedGap| + AllocationDistortion$) that is genuine treatment signal rather than allocation noise. $SignalPurity = |CorrectedGap| / (|CorrectedGap| + AllocationDistortion)$. Range $[0,1]$: 1.0 = allocation contributed nothing (type D); 0 = corrected gap is zero (pure noise). Critical threshold is 0.5: $SignalPurity < 0.5$ means allocation noise exceeds true signal. The signal-purity theorem (loop-44): $AllocationDirection='reversal' \rightarrow SignalPurity < 0.5$. Equivalently: a sign flip requires the confound to produce more distortion than the treatment produced signal. This is the mechanistic account of why reversals happen: not study-design accidents but algebraic necessities.

f **Pooled Gap Crosses Zero** Calculated 1 hop from ground truth boolean DR-211

IN RULESPEAK®

A treatment ranking is flagged pooled gap crosses zero if the value of `LOOKUP(Study, SweepStudySummary[SweepStudyId], SweepStudySummary[PooledGapCrossesZero])`.

- ▼ **Inputs** 1 fields this rule reads

→ **Study** Treatment Ranking
FK to the parent Study.

- **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: SweepStudySummary.PooledGapCrossesZero via Study. TRUE when reweighting allocation would change the sign of the pooled gap.

f **Sweep Pooled Gap Range** Calculated 1 hop from ground truth number DR-212

IN RULESPEAK®

A treatment ranking's sweep pooled gap range is computed as the value of `LOOKUP(Study, SweepStudySummary[SweepStudyId], SweepStudySummary[SweepPooledGapRange])`.

- ▼ **Inputs** 1 fields this rule reads

→ **Study** Treatment Ranking
FK to the parent Study.

- **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: SweepStudySummary.SweepPooledGapRange via Study. Width of pooled-gap wander under allocation sweep.



Latent Flip Potential

Calculated

8 hops from ground truth

boolean

DR-213

IN RULESPEAK®

The treatment ranking's latent flip potential is determined by the following priority:

1. **an empty string** if the distortion type is blank
2. **the AND of the distortion type is "D" and the pooled gap crosses zero is true** in all other cases

▼ Inputs 2 fields this rule reads

f Distortion Type

Treatment Ranking

If allocation distortion is "" then "", otherwise if is sign flip then if is stratum unanimous then "A", otherwise "B", otherwise if not is sign flip, allocation distortion is greater than 0.01, and the absolute value of signed pooled gap is greater than (the absolute value of corrected gap plus 0.001) then "C+", otherwise if not is sign flip, allocation distortion is greater than 0.01, and the absolute value of signed pooled gap is less than (the absolute value of corrected gap minus 0.001) then "C-", otherwise "D".

f Pooled Gap Crosses Zero

Treatment Ranking

LOOKUP({{Study}},
SweepStudySummary[SweepStudyId],
SweepStudySummary[PooledGapCrossesZero]).

► Ground truth at the leaves 11 the raw values this ultimately depends on

TRUE when DistortionType=D but PooledGapCrossesZero=TRUE — SAFE at observed allocation but flip-capable under counterfactual reweighting.



Allocation Fragility

Calculated

4 hops from ground truth

number

DR-214

IN RULESPEAK®

The treatment ranking's allocation fragility is determined by the following priority:

1. **0** if **any of these hold**:
 - the sweep pooled gap range is blank
 - the pooled gap is blank
 - the pooled gap is 0
2. **the sweep pooled gap range divided by the absolute value of the pooled gap** in all other cases

▼ Inputs 2 fields this rule reads

f Sweep Pooled Gap Range

Treatment Ranking

LOOKUP({{Study}},
SweepStudySummary[SweepStudyId],

f Pooled Gap

Treatment Ranking

If pooled rate a is "" then "", otherwise the absolute value of (pooled rate a minus pooled rate

SweepStudySummary[SweepPooledGapRange]).

b).

- **Ground truth at the leaves** 7 the raw values this ultimately depends on

SweepPooledGapRange divided by |PooledGap|. High values mean small observed effect but wide counterfactual wander. Fixed loop-90: this rulebook's IF(x="", "", ...) blank-on-degenerate idiom transpiles to a bare "":numeric cast that errors at runtime the moment a live row genuinely has PooledGap=0 (surfaced only once _erb_tr_metrics, which masked this row, was eliminated). Returns 0 for the degenerate case instead — see TreatmentRankings.DistortionRatio for the same fix pattern.

Σ **Max Stratum Imbalance** Aggregation 5 hops from ground truth number DR-215

IN RULESPEAK®

A treatment ranking's max stratum imbalance is the largest abs allocation bias across the treatment ranking's stratum summaries that have a treatment label of "A".

- ▼ **Inputs** 4 fields this rule reads

f **Abs Allocation Bias** Stratum Summary
If allocation bias is "" then "", otherwise the absolute value of allocation bias.

→ **Study** Stratum Summary
FK to the parent Study.

→ **Study** Treatment Ranking
FK to the parent Study.

● **Treatment Label** Stratum Summary
Which treatment (matches Treatments.TreatmentLabel).

- **Ground truth at the leaves** 8 the raw values this ultimately depends on

MAXIFS(StratumSummaries.AbsAllocationBias) over rows where Study matches and TreatmentLabel='A' — the largest allocation-share skew any single stratum shows for this study. Live aggregation; replaces the former hand-written _erb_tr_metrics cache column of the same name.

Σ **Max Stratum Gap** Aggregation 5 hops from ground truth number DR-216

IN RULESPEAK®

A treatment ranking's max stratum gap is the largest abs stratum gap across the treatment ranking's stratum summaries that have a treatment label of "A".

- ▼ **Inputs** 4 fields this rule reads

f **Abs Stratum Gap** Stratum Summary
If stratum gap is "" then "", otherwise the absolute value of stratum gap.

→ **Study** Stratum Summary
FK to the parent Study.

→ **Study** Treatment Ranking

● **Treatment Label** Stratum Summary

FK to the parent Study.

Which treatment (matches
Treatments.TreatmentLabel).

- **Ground truth at the leaves** 9 the raw values this ultimately depends on

MAXIFS(StratumSummaries.AbsStratumGap) over rows where Study matches and TreatmentLabel='A' — the largest within-stratum rate gap for this study. Live aggregation; replaces the former hand-written _erb_tr_metrics cache column of the same name.



Is Sweep Fragile

Calculated

8 hops from ground truth

boolean

DR-217

IN RULESPEAK®

A treatment ranking is considered a sweep fragile if all of the following:

- the distortion type is "D"
- the signal purity is 1
- the allocation distortion is 0
- the sweep pooled gap range is greater than 0.3
- the allocation fragility is at least 10

- ▼ **Inputs** 5 fields this rule reads

f Distortion Type

Treatment Ranking

If allocation distortion is "" then "", otherwise if is sign flip then if is stratum unanimous then "A", otherwise "B", otherwise if not is sign flip, allocation distortion is greater than 0.01, and the absolute value of signed pooled gap is greater than (the absolute value of corrected gap plus 0.001) then "C+", otherwise if not is sign flip, allocation distortion is greater than 0.01, and the absolute value of signed pooled gap is less than (the absolute value of corrected gap minus 0.001) then "C-", otherwise "D".

f Signal Purity

Treatment Ranking

If corrected gap is "" then "", otherwise if (the absolute value of corrected gap plus allocation distortion) equals 0 then 1, otherwise (the absolute value of corrected gap divided by (the absolute value of corrected gap plus allocation distortion)).

f Allocation Distortion

Treatment Ranking

If weighted stratum gap sum is "" then "", otherwise the absolute value of (weighted stratum gap sum minus signed pooled gap).

f Sweep Pooled Gap Range

Treatment Ranking

LOOKUP({{Study}},
SweepStudySummary[SweepStudyId],
SweepStudySummary[SweepPooledGapRange]).

f Allocation Fragility

Treatment Ranking

If sweep pooled gap range is "", pooled gap is "", or pooled gap equals 0 then 0, otherwise (sweep pooled gap range divided by the absolute value of pooled gap).

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

Type-D study that is pooled-safe but ultra-fragile under allocation reweighting: signal_purity=1, allocation_distortion=0, sweep_pooled_gap_range>0.3, allocation_fragility>=10.

f Study Domain Calculated 1 hop from ground truth string DR-218

IN RULESPEAK®

A treatment ranking's study domain is computed as the value of LOOKUP(Study, Studies[StudyId], Studies[Domain]).

▼ **Inputs** 1 fields this rule reads

...→ **Study** Treatment Ranking
FK to the parent Study.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: Studies.Domain via Study.

↗ Stratum Causal Role Lookup 1 hop from ground truth string DR-219

IN RULESPEAK®

A treatment ranking's stratum causal role is the value of LOOKUP(Study, StratumVariables[Study], StratumVariables[CausalRole]).

▼ **Inputs** 1 fields this rule reads

...→ **Study** Treatment Ranking
FK to the parent Study.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: StratumVariables.CausalRole for this study — confounder, mediator, collider, selection, contested, etc.

f Is Latent Only Flip Calculated 7 hops from ground truth boolean DR-220

IN RULESPEAK®

The treatment ranking's is latent only flip is determined by the following priority:

1. **an empty string** if the pooled gap crosses zero is blank
2. **the AND of the pooled gap crosses zero is true and the is sign flip is false** in all other cases

▼ **Inputs** 2 fields this rule reads

f **Pooled Gap Crosses Zero** Treatment Ranking
LOOKUP({{Study}},
SweepStudySummary[SweepStudyId],
SweepStudySummary[PooledGapCrossesZero]).

f **Is Sign Flip** Treatment Ranking
If weighted stratum gap sum is "" then "",
otherwise if weighted stratum gap sum is greater
than 0 then signed pooled gap is less than 0,
otherwise signed pooled gap is greater than 0.

► **Ground truth at the leaves** 11 the raw values this ultimately depends on

TRUE when allocation sweep crosses zero but observed allocation shows no sign flip — latent-only reversal.

Σ **Confirmed Causal Role Count** Aggregation 1 hop from ground truth integer DR-221

IN RULESPEAK®

A treatment ranking's confirmed causal role count is the number of the treatment ranking's stratum variables that have a causal role of "confounder".

▼ **Inputs** 3 fields this rule reads

→ **Study** Stratum Variable
FK to the parent Study.

→ **Study** Treatment Ranking
FK to the parent Study.

● **Causal Role** Stratum Variable
The variable's causal role: 'confounder' (affects
both assignment and outcome), 'mediator'
(downstream of treatment), 'contested' (causal
role disputed), 'unknown'.

► **Ground truth at the leaves** 3 the raw values this ultimately depends on

Count of StratumVariables in this study whose CausalRole is "confounder" — includes variables that confound the outcome regardless of whether allocation was balanced.

Σ **Mediator Risk Count** Aggregation 1 hop from ground truth integer DR-222

IN RULESPEAK®

A treatment ranking's mediator risk count is the number of the treatment ranking's mediator risks.

▼ **Inputs** 3 fields this rule reads

→ **Study** Stratum Variable
FK to the parent Study.

→ **Study** Treatment Ranking
FK to the parent Study.

● Conditioning Risk

Stratum Variable

The causal hazard posed by conditioning on this variable. Values: confounder (safe to condition — removes bias), mediator (conditioning blocks the causal path of interest — harmful), collider (conditioning opens a spurious backdoor — harmful), proxy (conditioning on a noisy proxy of the true confounder — risky), none (no conditioning risk identified). NULL means unclassified.

- **Ground truth at the leaves** 3 the raw values this ultimately depends on

Count of StratumVariables in this study whose ConditioningRisk is mediator or collider — the hazardous conditioning types.



Contested Stratum Count

Aggregation

1 hop from ground truth

integer

DR-223

IN RULESPEAK®

A treatment ranking's contested stratum count is the number of the treatment ranking's stratum variables that have a causal role of "contested".

- ▼ **Inputs** 3 fields this rule reads

→ **Study**

Stratum Variable

FK to the parent Study.

→ **Study**

Treatment Ranking

FK to the parent Study.

● Causal Role

Stratum Variable

The variable's causal role: 'confounder' (affects both assignment and outcome), 'mediator' (downstream of treatment), 'contested' (causal role disputed), 'unknown'.

- **Ground truth at the leaves** 3 the raw values this ultimately depends on

Count of StratumVariables in this study whose CausalRole is contested — the causal interpretation is disputed.



Unknown Causal Role Count

Aggregation

1 hop from ground truth

integer

DR-224

IN RULESPEAK®

A treatment ranking's unknown causal role count is the number of the treatment ranking's stratum variables that have a causal role of "unknown".

- ▼ **Inputs** 3 fields this rule reads

→ **Study**

Stratum Variable

FK to the parent Study.

→ **Study**

Treatment Ranking

FK to the parent Study.

● **Causal Role**

Stratum Variable

The variable's causal role: 'confounder' (affects both assignment and outcome), 'mediator' (downstream of treatment), 'contested' (causal role disputed), 'unknown'.

- **Ground truth at the leaves** 3 the raw values this ultimately depends on

Count of StratumVariables in this study whose CausalRole is unknown — no causal account has been established.



Causal Claim Status

Calculated

2 hops from ground truth

string

DR-225

IN RULESPEAK®

The treatment ranking's causal claim status is determined by the following priority:

1. **"contested"** if the contested stratum count is greater than 0
2. **"geometric-only"** if the unknown causal role count is greater than 0
3. **"established"** if the confirmed causal role count is greater than 0
4. **"geometric-only"** in all other cases

- ▼ **Inputs** 3 fields this rule reads

Σ **Contested Stratum Count**

Treatment Ranking

Count of stratum variables whose study is study and causal role is "contested".

Σ **Unknown Causal Role Count**

Treatment Ranking

Count of stratum variables whose study is study and causal role is "unknown".

Σ **Confirmed Causal Role Count**

Treatment Ranking

Count of stratum variables whose study is study and causal role is "confounder".

- **Ground truth at the leaves** 3 the raw values this ultimately depends on

The epistemic status of the causal interpretation for this study. Values: established (all stratum variables are confirmed confounders with known mechanisms), contested (at least one stratum variable has a disputed causal role — e.g. Berkeley 1973 where department choice may be endogenous), geometric-only (no stratum variable has an established causal role — the distortion is a geometric fact, not a causal claim). Derived from StratumVariables.CausalRole and ConditioningRisk.

f Adjustment Appropriate Calculated 3 hops from ground truth boolean DR-226

IN RULESPEAK®

A treatment ranking is flagged adjustment appropriate if all of the following:

- the mediator risk count is 0
- the causal claim status is "established"

▼ Inputs 2 fields this rule reads

Σ Mediator Risk Count Treatment Ranking

Count of stratum variables whose study is study and conditioning risk is "mediator" plus count of stratum variables whose study is study and conditioning risk is "collider".

f Causal Claim Status Treatment Ranking

If contested stratum count is greater than 0 then "contested", otherwise if unknown causal role count is greater than 0 then "geometric-only", otherwise if confirmed causal role count is greater than 0 then "established", otherwise "geometric-only".

► Ground truth at the leaves 4 the raw values this ultimately depends on

TRUE only when it is epistemically safe to act on CorrectedGap as the causal treatment effect estimate: no mediator/collider conditioning risk AND causal claim is established (not contested or geometric-only). FALSE for the birth-weight paradox (mediator), berkeley-1973 (contested), and synthetic/degenerate studies (geometric-only).

● Adjustment Rationale Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Human-authored note explaining why AdjustmentAppropriate is TRUE or FALSE for this study. Covers mediator/collider hazards, contested causal roles, and cases where geometric correction is valid but causal interpretation is uncertain.

f Purity Trap Flag Calculated boolean DR-227

IN RULESPEAK®

A treatment ranking's purity trap flag is a defined attribute of the treatment ranking.

TRUE when SignalPurity >= 0.90 AND the study's confounder identity has PolicyDefault IN ('do-not-condition','do-not-naively-adjust'). Marks the purity trap: data appears cleanest where naive stratification is most harmful.

f Distortion Ratio Calculated 6 hops from ground truth number DR-228

IN RULESPEAK®

The treatment ranking's distortion ratio is determined by the following priority:

1. **0** if the weighted stratum gap sum is 0
2. **the signed pooled gap divided by the weighted stratum gap sum** in all other cases

▼ Inputs 2 fields this rule reads

Σ **Weighted Stratum Gap Sum**

Treatment Ranking

Sum of weighted stratum gap across all stratum summaries whose study is study and treatment label is treatment a.

f **Signed Pooled Gap**

Treatment Ranking

If pooled rate a is "" then "", otherwise (pooled rate a minus pooled rate b).

► Ground truth at the leaves 11 the raw values this ultimately depends on

SignedPooledGap / WeightedStratumGapSum. Promoted loop-90 from a 03b-customize-views.sql presentation-layer bolt-on to a first-class rulebook field. The IF(x=0,"",...) blank-on-zero-denominator idiom used elsewhere in this rulebook is unsafe here: NULLIF is not a supported transpiler function, and the transpiler's translation of the "" branch for a numeric-typed field is a bare "":numeric cast that errors at runtime the moment a live row actually hits it (4 of 242 TreatmentRankings rows have WeightedStratumGapSum=0). Returns 0 in that case instead — WeightedStratumGapSum=0 means there is no equal-weight baseline signal to compare against, so a 0 ratio (no distortion multiplier) is the defensible convention, not a guess.



Screening Tier

Calculated

8 hops from ground truth

string

DR-229

IN RULESPEAK®

The treatment ranking's screening tier is determined by the following priority:

1. **"DANGER"** if any of these hold:
 - the distortion type is "A"
 - the distortion type is "B"
2. **"CAUTION"** if any of these hold:
 - the distortion type is "C+"
 - the distortion type is "C-"
3. **"SAFE"** if the distortion type is "D"
4. **an empty string** in all other cases

▼ Inputs 1 fields this rule reads

f **Distortion Type**

Treatment Ranking

If allocation distortion is "" then "", otherwise if is sign flip then if is stratum unanimous then "A", otherwise "B", otherwise if not is sign flip, allocation distortion is greater than 0.01, and the absolute value of signed pooled gap is greater

than (the absolute value of corrected gap plus 0.001) then "C+", otherwise if not is sign flip, allocation distortion is greater than 0.01, and the absolute value of signed pooled gap is less than (the absolute value of corrected gap minus 0.001) then "C-", otherwise "D".

- **Ground truth at the leaves** 11 the raw values this ultimately depends on

DANGER when DistortionType is A or B, CAUTION when C+/C-, SAFE when D. Promoted loop-90 (see DistortionRatio).

f **Arm Size Ratio** Calculated 2 hops from ground truth number DR-230

IN RULESPEAK®

The treatment ranking's arm size ratio is determined by the following priority:

1. 0 if the total cases a plus the total cases b is 0
2. the total cases a divided by the total cases a plus the total cases b in all other cases

- ▼ **Inputs** 2 fields this rule reads

Σ **Total Cases a** Treatment Ranking
Sum of cases across all case cells whose study is study and treatment label is treatment a.

Σ **Total Cases B** Treatment Ranking
Sum of cases across all case cells whose study is study and treatment label is treatment b.

- **Ground truth at the leaves** 6 the raw values this ultimately depends on

TotalCasesA / (TotalCasesA + TotalCasesB) — the share of total cases allocated to TreatmentA. Promoted loop-90 (see DistortionRatio for why the empty-string zero-denominator idiom is unsafe on a numeric field). Returns 0 for the degenerate zero-case-count edge (never occurs in this corpus, kept for defensive symmetry with DistortionRatio).

f **Symmetry Departure** Calculated 3 hops from ground truth number DR-231

IN RULESPEAK®

A treatment ranking's symmetry departure is computed as the absolute value of the arm size ratio minus 0.5.

- ▼ **Inputs** 1 fields this rule reads

f **Arm Size Ratio** Treatment Ranking
If (total cases a plus total cases b) equals 0 then 0, otherwise (total cases a divided by (total cases a plus total cases b)).

- **Ground truth at the leaves** 6 the raw values this ultimately depends on

ABS(ArmSizeRatio - 0.5) — how far the case allocation departs from a balanced 50/50 split. Promoted loop-90 (see DistortionRatio).

Invariant Check

An invariant check is identified by its name.

f **Name** Calculated 1 hop from ground truth string DR-232

IN RULESPEAK®

An invariant check's name is the same as its invariant check ID.

- ▼ **Inputs** 1 fields this rule reads

● **Invariant Check ID** Invariant Check
Unique slug for this invariant (e.g. inv-balanced-alloc).

- **Ground truth at the leaves** 1 the raw values this ultimately depends on

Display label.

Algebraic Statement Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

The mathematical assertion in compact symbolic form.

Natural Language Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Plain-English statement of what property must hold and why it is an invariant of the model.

Source Table Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

The DAG table whose rows are checked (TreatmentRankings or StratumSummaries).



Filter Expression

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Condition selecting the universe of rows this invariant applies to. Empty means all rows.



Assertion Expression

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

The per-row condition that must be TRUE for every row in scope.



SQL Filter

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Optional SQL WHERE clause that selects the universe of rows in scope for this invariant (e.g. `distortion_type = 'D'`). Empty string means all rows.



SQL Assertion

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

The SQL WHERE clause (without WHERE keyword) that counts PASSING rows from `vw_treatment_rankings` for this invariant. Used by `run_invariant_checks()` to evaluate each assertion dynamically from the rulebook table.



Pass Count

Ground truth

integer

Ground truth. Written directly to the database (datatype `integer`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Rows where the invariant holds. Populated by `05b-customize-data.sql` from a live query against `vw_treatment_rankings`.



Fail Count

Ground truth

integer

Ground truth. Written directly to the database (datatype `integer`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Rows where the invariant is violated. Any `FailCount > 0` is a build-breaking bug.

f Universe Count Calculated 1 hop from ground truth integer DR-233

IN RULESPEAK®

An invariant check's universe count is computed as the pass count plus the fail count.

▼ **Inputs** 2 fields this rule reads

● **Pass Count**

Invariant Check

Rows where the invariant holds. Populated by 05b-customize-data.sql from a live query against vw_treatment_rankings.

● **Fail Count**

Invariant Check

Rows where the invariant is violated. Any FailCount > 0 is a build-breaking bug.

► **Ground truth at the leaves** 2 the raw values this ultimately depends on

Total rows in scope (PassCount + FailCount).

f Is Green Calculated 1 hop from ground truth boolean DR-234

IN RULESPEAK®

An invariant check is considered a green if the fail count is 0.

▼ **Inputs** 1 fields this rule reads

● **Fail Count**

Invariant Check

Rows where the invariant is violated. Any FailCount > 0 is a build-breaking bug.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

TRUE when FailCount = 0.

f Status Label Calculated 2 hops from ground truth string DR-235

IN RULESPEAK®

The invariant check's status label is determined by the following priority:

1. **"PASS"** if the green flag is set
2. **"FAIL(", followed by the fail count, followed by ")"** in all other cases

▼ **Inputs** 2 fields this rule reads

f **Is Green**

Invariant Check

Fail count equals 0.

● **Fail Count**

Invariant Check

Rows where the invariant is violated. Any FailCount > 0 is a build-breaking bug.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

PASS or FAIL(n) summary for display.

Severity Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

critical | warning — critical means any FailCount > 0 is a build-breaking bug.

Tradition ID Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

FK → ResearchTraditions.TraditionId. The tradition whose theoretical commitments this algebraic invariant protects — a failed invariant would falsify that tradition's key claim.

Protects Conclusion Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

FK → Conclusions.ConclusionId. The conclusion whose witness depends on this invariant holding. If this invariant fails, that conclusion is undermined.

Tradition Name Lookup 1 hop from ground truth string DR-236

IN RULESPEAK®

An invariant check's tradition name is the value of `LOOKUP(TraditionId, ResearchTraditions[TraditionId], ResearchTraditions[Name])`.

▼ **Inputs** 1 fields this rule reads

Tradition ID Invariant Check

FK → ResearchTraditions.TraditionId. The tradition whose theoretical commitments this algebraic invariant protects — a failed invariant would falsify that tradition's key claim.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: ResearchTraditions.Name via TraditionId.

Tradition Key Claim Summary Lookup 1 hop from ground truth string DR-237

IN RULESPEAK®

An invariant check's [tradition key claim summary](#) is the value of LOOKUP(TraditionId, ResearchTraditions[TraditionId], ResearchTraditions[KeyClaim]).

▼ **Inputs** 1 fields this rule reads

● Tradition ID Invariant Check

FK → ResearchTraditions.TraditionId. The tradition whose theoretical commitments this algebraic invariant protects — a failed invariant would falsify that tradition's key claim.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: ResearchTraditions.KeyClaim via TraditionId — a failed invariant would falsify this tradition's key claim.

Protects Conclusion Title Lookup 1 hop from ground truth string DR-238

IN RULESPEAK®

An invariant check's [protects conclusion title](#) is the value of LOOKUP(ProtectsConclusion, Conclusions[ConclusionId], Conclusions[Title]).

▼ **Inputs** 1 fields this rule reads

● Protects Conclusion Invariant Check

FK → Conclusions.ConclusionId. The conclusion whose witness depends on this invariant holding. If this invariant fails, that conclusion is undermined.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: Conclusions.Title via ProtectsConclusion.

Protects Conclusion Category Lookup 1 hop from ground truth string DR-239

IN RULESPEAK®

An invariant check's [protects conclusion category](#) is the value of LOOKUP(ProtectsConclusion, Conclusions[ConclusionId], Conclusions[Category]).

▼ **Inputs** 1 fields this rule reads

● Protects Conclusion

Invariant Check

FK → Conclusions.ConclusionId. The conclusion whose witness depends on this invariant holding. If this invariant fails, that conclusion is undermined.

- **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: Conclusions.Category via ProtectsConclusion.

Materialized Entity

A materialized entity is identified by its name.



Name

Calculated

1 hop from ground truth

string

DR-240

IN RULESPEAK®

A materialized entity's name is the same as its materialized entity ID.

- ▼ **Inputs** 1 fields this rule reads

● Materialized Entity ID

Materialized Entity

Unique slug (e.g. mat-sweep-study-summary).

- **Ground truth at the leaves** 1 the raw values this ultimately depends on

Display label, mirrors MaterializedEntityId.



Entity Table

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

The rulebook table whose fields are materialized (e.g. SweepStudySummary).



Refresh Contract

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

eager-on-write | on-demand | scheduled — when the refresh function is invoked relative to writes on the upstream data.



Refresh Function

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

The SQL function (in the customize- override files) that recomputes this entity's materialized fields from its upstream SSoT tables.*

Upstream Tables Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Comma-separated list of the tables this materialization is derived from — its sealed/stable upstream.

Justification Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

The measured cost or evidence that justified materializing this entity instead of leaving it as a live calculated/aggregation field.

Methodology

A methodology is identified by its name.

f Name Calculated 1 hop from ground truth string DR-241

IN RULESPEAK®

A methodology's name is the same as its methodology ID.

▼ **Inputs** 1 fields this rule reads

● **Methodology ID** Methodology
Unique slug (e.g. method-01-cmcc, method-02-dag-first).

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Display label, mirrors MethodologyId.

Phase Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.



Title

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Short title of the methodological principle or step.



Statement

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

One-to-three sentence statement of the principle, precise enough to be applied or falsified.



Why It Matters

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Why this step or principle is load-bearing for this model — what breaks if you skip it.



Illustrated by Loop

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

LoopId(s) where this principle was concretely demonstrated (comma-separated).



Tradition ID

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

FK → ResearchTraditions.TraditionId. The scholarly tradition that introduced or most relies on this methodological approach.



Pioneering Researcher

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

FK → Researchers.ResearcherId. The researcher most associated with developing or popularising this methodological principle in the Simpson's Paradox context.

Tradition Name Lookup 1 hop from ground truth string DR-242

IN RULESPEAK®

A methodology's tradition name is the value of LOOKUP(TraditionId, ResearchTraditions[TraditionId], ResearchTraditions[Name]).

▼ **Inputs** 1 fields this rule reads

● Tradition ID Methodology

FK → ResearchTraditions.TraditionId. The scholarly tradition that introduced or most relies on this methodological approach.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: ResearchTraditions.Name via TraditionId.

Tradition Primary Venue Lookup 1 hop from ground truth string DR-243

IN RULESPEAK®

A methodology's tradition primary venue is the value of LOOKUP(TraditionId, ResearchTraditions[TraditionId], ResearchTraditions[PrimaryVenue]).

▼ **Inputs** 1 fields this rule reads

● Tradition ID Methodology

FK → ResearchTraditions.TraditionId. The scholarly tradition that introduced or most relies on this methodological approach.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: ResearchTraditions.PrimaryVenue via TraditionId.

Pioneering Researcher Name Lookup 1 hop from ground truth string DR-244

IN RULESPEAK®

A methodology's pioneering researcher name is the value of LOOKUP(PioneeringResearcher, Researchers[ResearcherId], Researchers[Name]).

▼ **Inputs** 1 fields this rule reads

● Pioneering Researcher Methodology

FK → Researchers.ResearcherId. The researcher most associated with developing or popularising

this methodological principle in the Simpson's Paradox context.

- **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: Researchers.Name via PioneeringResearcher.

Pioneering Researcher Affiliation Lookup 1 hop from ground truth string DR-245

IN RULESPEAK®

A methodology's pioneering researcher affiliation is the value of `LOOKUP(PioneeringResearcher, Researchers[ResearcherId], Researchers[Affiliation])`.

- ▼ **Inputs** 1 fields this rule reads

● **Pioneering Researcher** Methodology

FK → Researchers.ResearcherId. The researcher most associated with developing or popularising this methodological principle in the Simpson's Paradox context.

- **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: Researchers.Affiliation via PioneeringResearcher.

Loop Count Aggregation 1 hop from ground truth integer DR-246

IN RULESPEAK®

A methodology's loop count is the number of loops related to the methodology.

- ▼ **Inputs** 2 fields this rule reads

● **Tradition ID** Loop

FK → ResearchTraditions.TraditionId. The scholarly tradition whose conceptual vocabulary this loop primarily draws on — tracks which tradition has driven each stage of the instrument's development.

● **Tradition ID** Methodology

FK → ResearchTraditions.TraditionId. The scholarly tradition that introduced or most relies on this methodological approach.

- **Ground truth at the leaves** 2 the raw values this ultimately depends on

Count of Loops whose TraditionId matches this methodology entry's TraditionId — how many build iterations this methodological tradition has driven.

Conclusion

A conclusion is identified by its name.

f **Name** Calculated 1 hop from ground truth string DR-247

IN RULESPEAK®

A conclusion's name is the same as its conclusion ID.

▾ **Inputs** 1 fields this rule reads

● **Conclusion ID** Conclusion

Unique slug (e.g. conc-01-paradox-is-derived).

▸ **Ground truth at the leaves** 1 the raw values this ultimately depends on

Display label, mirrors ConclusionId.

Category Ground truth string

Ground truth. Written directly to the database (datatype string); not computed from anything else. Everything derived ultimately traces back to values like this one.

domain | instrument | methodology | open-question | corpus-theorem

Status Ground truth string

Ground truth. Written directly to the database (datatype string); not computed from anything else. Everything derived ultimately traces back to values like this one.

witnessed | planned | open

Report Tier Ground truth string

Ground truth. Written directly to the database (datatype string); not computed from anything else. Everything derived ultimately traces back to values like this one.

critical | supporting | transient — controls how much space this conclusion gets in the generated PDF/summary report. critical = full section with evidence + chart; supporting = 1-2 line bullet; transient = folded into a single compact list.

Validating Hypothesis Ground truth string

Ground truth. Written directly to the database (datatype string); not computed from anything else. Everything derived ultimately traces back to values like this one.

FK → DiscoveryHypotheses.HypothesisId. The pre-registered hypothesis whose live DiscoveryFindings.IsConfirmed value is the current-truth check for this conclusion — set on theorem/corpus-theorem conclusions so a report can distinguish a conclusion's archived prose (may say 'Proved') from its live status (may since be contradicted by corpus growth). Null when no single hypothesis tests this exact claim.



Live Status Confirmed

Lookup

1 hop from ground truth

boolean

DR-248

IN RULESPEAK®

A conclusion's live status confirmed is the value of LOOKUP(ValidatingHypothesis, DiscoveryFindings[HypothesisId], DiscoveryFindings[IsConfirmed]).

▼ Inputs 1 fields this rule reads

● Validating Hypothesis

Conclusion

FK → DiscoveryHypotheses.HypothesisId. The pre-registered hypothesis whose live DiscoveryFindings.IsConfirmed value is the current-truth check for this conclusion — set on theorem/corpus-theorem conclusions so a report can distinguish a conclusion's archived prose (may say 'Proved') from its live status (may since be contradicted by corpus growth). Null when no single hypothesis tests this exact claim.

► Ground truth at the leaves 1 the raw values this ultimately depends on

Lookup: DiscoveryFindings.IsConfirmed via ValidatingHypothesis — the current pass/fail state of the hypothesis backing this conclusion, recomputed on every build from live corpus data. NULL when ValidatingHypothesis is not set (no live check exists for this conclusion).



Live Status Observed Metric

Lookup

1 hop from ground truth

string

DR-249

IN RULESPEAK®

A conclusion's live status observed metric is the value of LOOKUP(ValidatingHypothesis, DiscoveryFindings[HypothesisId], DiscoveryFindings[ObservedMetric]).

▼ Inputs 1 fields this rule reads

● Validating Hypothesis

Conclusion

FK → DiscoveryHypotheses.HypothesisId. The pre-registered hypothesis whose live DiscoveryFindings.IsConfirmed value is the current-truth check for this conclusion — set on theorem/corpus-theorem conclusions so a report can distinguish a conclusion's archived prose

(may say 'Proved') from its live status (may since be contradicted by corpus growth). Null when no single hypothesis tests this exact claim.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: `DiscoveryFindings.ObservedMetric` via `ValidatingHypothesis` — the current observed values for the hypothesis backing this conclusion, so a reader can see exactly why it currently passes or fails.

● **Title** Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

One-line statement of the conclusion.

● **Evidence** Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

The specific DB query result, loop outcome, or argument that establishes or will establish this conclusion.

● **Witnessed in Loop** Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

LoopId where this conclusion was first witnessed (if status=witnessed).

● **Target Loop** Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

LoopId that is designed to witness this conclusion (if status=planned).

● **Tradition ID** Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

FK → `ResearchTraditions.TraditionId`. The tradition whose theoretical framework this conclusion primarily vindicates, refines, or challenges.

● **Researcher ID** Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

FK → Researchers.ResearcherId. The researcher whose specific claim, definition, or framework this conclusion directly affirms.

● Challenges Researcher Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

FK → Researchers.ResearcherId. A researcher whose position this conclusion refutes or qualifies (e.g. conc-03 refutes Blyth's unanimity criterion as the primary definition).

↗ Tradition Name Lookup 1 hop from ground truth string DR-250

IN RULESPEAK®

A conclusion's tradition name is the value of LOOKUP(TraditionId, ResearchTraditions[TraditionId], ResearchTraditions[Name]).

▼ **Inputs** 1 fields this rule reads

● **Tradition ID** Conclusion
FK → ResearchTraditions.TraditionId. The tradition whose theoretical framework this conclusion primarily vindicates, refines, or challenges.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: ResearchTraditions.Name via TraditionId.

↗ Tradition Key Claim Summary Lookup 1 hop from ground truth string DR-251

IN RULESPEAK®

A conclusion's tradition key claim summary is the value of LOOKUP(TraditionId, ResearchTraditions[TraditionId], ResearchTraditions[KeyClaim]).

▼ **Inputs** 1 fields this rule reads

● **Tradition ID** Conclusion
FK → ResearchTraditions.TraditionId. The tradition whose theoretical framework this conclusion primarily vindicates, refines, or challenges.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: ResearchTraditions.KeyClaim via TraditionId — shows which tradition's central claim this conclusion validates or refines.

➤ **Researcher Name** Lookup 1 hop from ground truth string DR-252

IN RULESPEAK®

A conclusion's researcher name is the value of LOOKUP(ResearcherId, Researchers[ResearcherId], Researchers[Name]).

▼ **Inputs** 1 fields this rule reads

- **Researcher ID** Conclusion
FK → Researchers.ResearcherId. The researcher whose specific claim, definition, or framework this conclusion directly affirms.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: Researchers.Name via ResearcherId.

➤ **Researcher Affiliation** Lookup 1 hop from ground truth string DR-253

IN RULESPEAK®

A conclusion's researcher affiliation is the value of LOOKUP(ResearcherId, Researchers[ResearcherId], Researchers[Affiliation]).

▼ **Inputs** 1 fields this rule reads

- **Researcher ID** Conclusion
FK → Researchers.ResearcherId. The researcher whose specific claim, definition, or framework this conclusion directly affirms.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: Researchers.Affiliation via ResearcherId.

➤ **Challenges Researcher Name** Lookup 1 hop from ground truth string DR-254

IN RULESPEAK®

A conclusion's challenges researcher name is the value of LOOKUP(ChallengesResearcher, Researchers[ResearcherId], Researchers[Name]).

▼ **Inputs** 1 fields this rule reads

● Challenges Researcher Conclusion

FK → Researchers.ResearcherId. A researcher whose position this conclusion refutes or qualifies (e.g. conc-03 refutes Blyth's unanimity criterion as the primary definition).

- **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: Researchers.Name via ChallengesResearcher.

↗ **Witnessed in Loop Title** Lookup 1 hop from ground truth string DR-255

IN RULESPEAK®

A conclusion's witnessed in loop title is the value of LOOKUP(WitnessedInLoop, Loops[LoopId], Loops[Title]).

- ▼ **Inputs** 1 fields this rule reads

● Witnessed in Loop Conclusion

LoopId where this conclusion was first witnessed (if status=witnessed).

- **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: Loops.Title via WitnessedInLoop.

↗ **Witnessed in Loop Commit Hash** Lookup 1 hop from ground truth string DR-256

IN RULESPEAK®

A conclusion's witnessed in loop commit hash is the value of LOOKUP(WitnessedInLoop, Loops[LoopId], Loops[CommitHash]).

- ▼ **Inputs** 1 fields this rule reads

● Witnessed in Loop Conclusion

LoopId where this conclusion was first witnessed (if status=witnessed).

- **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: Loops.CommitHash via WitnessedInLoop — git checkout this SHA to replay the instrument state when this conclusion was first witnessed.

↗ **Witnessed in Loop Commit Short** Lookup 1 hop from ground truth string DR-257

IN RULESPEAK®

A conclusion's [witnessed in loop commit short](#) is the value of LOOKUP(WitnessedInLoop, Loops[LoopId], Loops[CommitShort]).

▼ **Inputs** 1 fields this rule reads

- **Witnessed in Loop** Conclusion
LoopId where this conclusion was first witnessed
(if status=witnessed).

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: Loops.CommitShort via WitnessedInLoop.

↗ **Witnessed in Loop Commit Date** Lookup 1 hop from ground truth string DR-258

IN RULESPEAK®

A conclusion's [witnessed in loop commit date](#) is the value of LOOKUP(WitnessedInLoop, Loops[LoopId], Loops[CommitDate]).

▼ **Inputs** 1 fields this rule reads

- **Witnessed in Loop** Conclusion
LoopId where this conclusion was first witnessed
(if status=witnessed).

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: Loops.CommitDate via WitnessedInLoop.

↗ **Witnessed in Loop Git Tag** Lookup 1 hop from ground truth string DR-259

IN RULESPEAK®

A conclusion's [witnessed in loop git tag](#) is the value of LOOKUP(WitnessedInLoop, Loops[LoopId], Loops[GitTag]).

▼ **Inputs** 1 fields this rule reads

- **Witnessed in Loop** Conclusion
LoopId where this conclusion was first witnessed
(if status=witnessed).

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: Loops.GitTag via WitnessedInLoop — preferred replay entry point when a milestone tag exists.



Invariant Protecting Count

Aggregation

1 hop from ground truth

integer

DR-260

IN RULESPEAK®

A conclusion's invariant protecting count is the number of invariant checks related to the conclusion.

▼ **Inputs** 2 fields this rule reads

● Protects Conclusion

Invariant Check

FK → Conclusions.ConclusionId. The conclusion whose witness depends on this invariant holding. If this invariant fails, that conclusion is undermined.

● Conclusion ID

Conclusion

Unique slug (e.g. conc-01-paradox-is-derived).

► **Ground truth at the leaves** 2 the raw values this ultimately depends on

Count of InvariantChecks whose ProtectsConclusion FK points to this conclusion — how many algebraic invariants stand behind this finding.

UI Screen

A UI screen is identified by its name.



Name

Calculated

1 hop from ground truth

string

DR-261

IN RULESPEAK®

A UI screen's name is the same as its screen ID.

▼ **Inputs** 1 fields this rule reads

● Screen ID

UI Screen

Unique slug (e.g. screen-01-study-list).

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Display label, mirrors ScreenId.



Title

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Human-readable screen title shown in the UI.



Route

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

URL path pattern (e.g. `/`, `/study/:studyId`, `/compare`).



Primary Entity

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

The main rulebook entity this screen reads from (the `vw_*` view name without prefix).



Purpose

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

One sentence: what a user can learn or do on this screen.



Key Fields

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Comma-separated list of the most important fields displayed on this screen.



User Action

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Primary user action available on this screen (e.g. `select a study`, `drill into a stratum`).



Tradition ID

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

`FK` → `ResearchTraditions.TraditionId`. The scholarly tradition whose lens this screen primarily serves — determines which framing language and which fields are most prominent.



Primary Conclusion

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

FK → Conclusions.ConclusionId. The conclusion this screen most directly visualises or communicates to the user.

➤ Tradition Name Lookup 1 hop from ground truth string DR-262

IN RULESPEAK®

A UI screen's tradition name is the value of LOOKUP(TraditionId, ResearchTraditions[TraditionId], ResearchTraditions[Name]).

▼ **Inputs** 1 fields this rule reads

- **Tradition ID** UI Screen
FK → ResearchTraditions.TraditionId. The scholarly tradition whose lens this screen primarily serves — determines which framing language and which fields are most prominent.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: ResearchTraditions.Name via TraditionId.

➤ Tradition Core Concern Lookup 1 hop from ground truth string DR-263

IN RULESPEAK®

A UI screen's tradition core concern is the value of LOOKUP(TraditionId, ResearchTraditions[TraditionId], ResearchTraditions[CoreConcern]).

▼ **Inputs** 1 fields this rule reads

- **Tradition ID** UI Screen
FK → ResearchTraditions.TraditionId. The scholarly tradition whose lens this screen primarily serves — determines which framing language and which fields are most prominent.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: ResearchTraditions.CoreConcern via TraditionId — the lens this screen applies.

➤ Primary Conclusion Title Lookup 1 hop from ground truth string DR-264

IN RULESPEAK®

A UI screen's [primary conclusion title](#) is the value of LOOKUP(PrimaryConclusion, Conclusions[ConclusionId], Conclusions[Title]).

▼ **Inputs** 1 fields this rule reads

● **Primary Conclusion** UI Screen

FK → Conclusions.ConclusionId. The conclusion this screen most directly visualises or communicates to the user.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: Conclusions.Title via PrimaryConclusion.

↗ **Primary Conclusion Evidence** Lookup 1 hop from ground truth string DR-265

IN RULESPEAK®

A UI screen's [primary conclusion evidence](#) is the value of LOOKUP(PrimaryConclusion, Conclusions[ConclusionId], Conclusions[Evidence]).

▼ **Inputs** 1 fields this rule reads

● **Primary Conclusion** UI Screen

FK → Conclusions.ConclusionId. The conclusion this screen most directly visualises or communicates to the user.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: Conclusions.Evidence via PrimaryConclusion.

UI Component

A UI component is identified by its [name](#).

f **Name** Calculated 1 hop from ground truth string DR-266

IN RULESPEAK®

A UI component's [name](#) is the same as its component ID.

▼ **Inputs** 1 fields this rule reads

● **Component ID** UI Component

Unique slug (e.g. comp-01-distortion-badge).

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Display label, mirrors ComponentId.

● **Component Name** Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Code-level component name (PascalCase, e.g. DistortionBadge).

● **Used on Screens** Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Comma-separated ScreenIds where this component appears.

● **Primary Entity** Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

The rulebook entity this component reads (the vw_ view).*

● **Description** Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

What this component renders and what decisions a developer must make to implement it.

● **Required Fields** Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Comma-separated rulebook fields this component must receive as props.

● **Visual Spec** Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Plain-English description of the visual output: shape, color coding, interactivity.

Instrument Spec

An instrument spec is identified by its name.

f **Name** Calculated 1 hop from ground truth string DR-267

IN RULESPEAK®

An instrument spec's name is the same as its spec ID.

▾ **Inputs** 1 fields this rule reads

● **Spec ID** Instrument Spec
Unique identifier for this instrument component.

▸ **Ground truth at the leaves** 1 the raw values this ultimately depends on

Display label.

Component Type Ground truth string

Ground truth. Written directly to the database (datatype string); not computed from anything else. Everything derived ultimately traces back to values like this one.

input | derived | classification | adapter-requirement

Field Name Ground truth string

Ground truth. Written directly to the database (datatype string); not computed from anything else. Everything derived ultimately traces back to values like this one.

The field or variable name in the DAG.

Data Type Ground truth string

Ground truth. Written directly to the database (datatype string); not computed from anything else. Everything derived ultimately traces back to values like this one.

integer | decimal | boolean | string | categorical

Source Table Ground truth string

Ground truth. Written directly to the database (datatype string); not computed from anything else. Everything derived ultimately traces back to values like this one.

The DAG table where this field is computed or stored.



Formula

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

The ERB formula or derivation rule for this component.



Natural Language

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Plain-English statement of what this component represents or requires.



Adapter Note

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

What a researcher must supply or verify to satisfy this component when hydrating a new study.

Allocation Sweep

An allocation sweep is identified by its [name](#).



Name

Calculated

1 hop from ground truth

string

DR-268

IN RULESPEAK®

An allocation sweep's [name](#) is the same as its sweep ID.

▼ **Inputs** 1 fields this rule reads

● Sweep ID

Allocation Sweep

Unique row identifier: <study>-f<pct> where pct is AllocFractionA*100.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Display label.



Study ID

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

The study being swept — all 238 studies in the corpus.

Alloc Fraction a Ground truth number

Ground truth. Written directly to the database (datatype number); not computed from anything else. Everything derived ultimately traces back to values like this one.

Fraction of the sweep-stratum cases assigned to treatment A. Ranges from 0.05 to 0.95. The sweep stratum is the one whose allocation was most confounded in the original study.

f Is Original Calculated 1 hop from ground truth boolean DR-269

IN RULESPEAK®

An allocation sweep is considered an original if the value of $\text{ABS}(\text{AllocFractionA} - \text{LOOKUP}(\text{StudyId}, \text{SweepStudyConfig}[\text{StudyId}], \text{SweepStudyConfig}[\text{OriginalAllocFractionA}])) < 0.03$.

▼ **Inputs** 2 fields this rule reads

● **Alloc Fraction a** Allocation Sweep

Fraction of the sweep-stratum cases assigned to treatment A. Ranges from 0.05 to 0.95. The sweep stratum is the one whose allocation was most confounded in the original study.

● **Study ID** Allocation Sweep

The study being swept — all 238 studies in the corpus.

► **Ground truth at the leaves** 2 the raw values this ultimately depends on

TRUE for the row closest to the actual original study allocation.

f N Sweep Stratum Total Calculated 1 hop from ground truth number DR-270

IN RULESPEAK®

An allocation sweep's [n sweep stratum total](#) is computed as the value of $\text{LOOKUP}(\text{StudyId}, \text{SweepStudyConfig}[\text{StudyId}], \text{SweepStudyConfig}[\text{NSweepStratumTotal}])$.

▼ **Inputs** 1 fields this rule reads

● **Study ID** Allocation Sweep

The study being swept — all 238 studies in the corpus.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Total cases in the sweep stratum (the stratum whose allocation is being varied). Fixed per study.

f **N Sweep a** Calculated 2 hops from ground truth number DR-271

IN RULESPEAK®

An allocation sweep's n sweep a is computed as the n sweep stratum total times the alloc fraction a rounded to the nearest whole number.

▼ **Inputs** 2 fields this rule reads

f **N Sweep Stratum Total** Allocation Sweep

LOOKUP({{StudyId}},
SweepStudyConfig[StudyId],
SweepStudyConfig[NSweepStratumTotal]).

● **Alloc Fraction a** Allocation Sweep

Fraction of the sweep-stratum cases assigned to treatment A. Ranges from 0.05 to 0.95. The sweep stratum is the one whose allocation was most confounded in the original study.

► **Ground truth at the leaves** 2 the raw values this ultimately depends on

*Cases in the sweep stratum assigned to treatment A = ROUND(NSweepStratumTotal * AllocFractionA).*

f **N Sweep B** Calculated 3 hops from ground truth number DR-272

IN RULESPEAK®

An allocation sweep's n sweep b is computed as the n sweep stratum total minus the n sweep a.

▼ **Inputs** 2 fields this rule reads

f **N Sweep Stratum Total** Allocation Sweep

LOOKUP({{StudyId}},
SweepStudyConfig[StudyId],
SweepStudyConfig[NSweepStratumTotal]).

f **N Sweep a** Allocation Sweep

(n sweep stratum total times alloc fraction a)
rounded to the nearest whole number.

► **Ground truth at the leaves** 2 the raw values this ultimately depends on

Cases in the sweep stratum assigned to treatment B = NSweepStratumTotal - NSweepA.

f **N Fixed a** Calculated 1 hop from ground truth number DR-273

IN RULESPEAK®

An allocation sweep's n fixed a is computed as the value of LOOKUP(StudyId, SweepStudyConfig[StudyId], SweepStudyConfig[NFixedA]).

▼ **Inputs** 1 fields this rule reads

● **Study ID** Allocation Sweep

The study being swept — all 238 studies in the corpus.

- **Ground truth at the leaves** 1 the raw values this ultimately depends on

Total A-treatment cases in all NON-swept strata (held fixed across sweep rows). For kidney-1986 = 87 (small-stratum A). For berkeley-1973 = 1302 (depts B+C+D male). For compressed-synthetic = 80 (easy-stratum A). For balanced-synthetic = 100 (hard-stratum A).



N Fixed B

Calculated

1 hop from ground truth

number

DR-274

IN RULESPEAK®

An allocation sweep's n fixed b is computed as the value of LOOKUP(StudyId, SweepStudyConfig[StudyId], SweepStudyConfig[NFixedB]).

- ▼ **Inputs** 1 fields this rule reads

● Study ID

Allocation Sweep

The study being swept — all 238 studies in the corpus.

- **Ground truth at the leaves** 1 the raw values this ultimately depends on

Total B-treatment cases in all NON-swept strata (held fixed across sweep rows). For kidney-1986 = 270 (small-stratum B). For berkeley-1973 = 993 (depts B+C+D female). For compressed-synthetic = 120 (easy-stratum B). For balanced-synthetic = 100 (hard-stratum B).



Sweep Rate a

Calculated

1 hop from ground truth

number

DR-275

IN RULESPEAK®

An allocation sweep's sweep rate a is computed as the value of LOOKUP(StudyId, SweepStudyConfig[StudyId], SweepStudyConfig[SweepRateA]).

- ▼ **Inputs** 1 fields this rule reads

● Study ID

Allocation Sweep

The study being swept — all 238 studies in the corpus.

- **Ground truth at the leaves** 1 the raw values this ultimately depends on

Per-cell success rate for treatment A in the sweep stratum. Fixed per study.



Sweep Rate B

Calculated

1 hop from ground truth

number

DR-276

IN RULESPEAK®

An allocation sweep's [sweep rate b](#) is computed as the value of LOOKUP(StudyId, SweepStudyConfig[StudyId], SweepStudyConfig[SweepRateB]).

▼ **Inputs** 1 fields this rule reads

● **Study ID** Allocation Sweep

The study being swept — all 238 studies in the corpus.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Per-cell success rate for treatment B in the sweep stratum. Fixed per study.

f **Fixed Rate a** Calculated 1 hop from ground truth number DR-277

IN RULESPEAK®

An allocation sweep's [fixed rate a](#) is computed as the value of LOOKUP(StudyId, SweepStudyConfig[StudyId], SweepStudyConfig[FixedRateA]).

▼ **Inputs** 1 fields this rule reads

● **Study ID** Allocation Sweep

The study being swept — all 238 studies in the corpus.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Weighted success rate for treatment A across all fixed (non-swept) strata. Constant per study.

f **Fixed Rate B** Calculated 1 hop from ground truth number DR-278

IN RULESPEAK®

An allocation sweep's [fixed rate b](#) is computed as the value of LOOKUP(StudyId, SweepStudyConfig[StudyId], SweepStudyConfig[FixedRateB]).

▼ **Inputs** 1 fields this rule reads

● **Study ID** Allocation Sweep

The study being swept — all 238 studies in the corpus.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Weighted success rate for treatment B across all fixed (non-swept) strata. Constant per study.



Sweep Pooled Rate a

Calculated

3 hops from ground truth

number

DR-279

IN RULESPEAK®

The allocation sweep's sweep pooled rate a is determined by the following priority:

1. **an empty string** if the n sweep a plus the n fixed a is 0
2. **the n sweep a times the sweep rate a plus the n fixed a times the fixed rate a divided by the n sweep a plus the n fixed a** in all other cases

▼ Inputs 4 fields this rule reads

f N Sweep a

Allocation Sweep

(n sweep stratum total times alloc fraction a)
rounded to the nearest whole number.

f N Fixed a

Allocation Sweep

LOOKUP({{StudyId}},
SweepStudyConfig[StudyId],
SweepStudyConfig[NFixedA]).

f Sweep Rate a

Allocation Sweep

LOOKUP({{StudyId}},
SweepStudyConfig[StudyId],
SweepStudyConfig[SweepRateA]).

f Fixed Rate a

Allocation Sweep

LOOKUP({{StudyId}},
SweepStudyConfig[StudyId],
SweepStudyConfig[FixedRateA]).

► Ground truth at the leaves 2 the raw values this ultimately depends on

*Pooled success rate for treatment A = (NSweepA * SweepRateA + NFixedA * FixedRateA) / (NSweepA + NFixedA). Changes as AllocFractionA varies.*



Sweep Pooled Rate B

Calculated

4 hops from ground truth

number

DR-280

IN RULESPEAK®

The allocation sweep's sweep pooled rate b is determined by the following priority:

1. **an empty string** if the n sweep b plus the n fixed b is 0
2. **the n sweep b times the sweep rate b plus the n fixed b times the fixed rate b divided by the n sweep b plus the n fixed b** in all other cases

▼ Inputs 4 fields this rule reads

f N Sweep B

Allocation Sweep

N sweep stratum total minus n sweep a.

f N Fixed B

Allocation Sweep

LOOKUP({{StudyId}},
SweepStudyConfig[StudyId],
SweepStudyConfig[NFixedB]).

f Sweep Rate B

Allocation Sweep

f Fixed Rate B

Allocation Sweep

```
LOOKUP({{StudyId}},
SweepStudyConfig[StudyId],
SweepStudyConfig[SweepRateB]).
```

```
LOOKUP({{StudyId}},
SweepStudyConfig[StudyId],
SweepStudyConfig[FixedRateB]).
```

- **Ground truth at the leaves** 2 the raw values this ultimately depends on

*Pooled success rate for treatment B = (NSweepB * SweepRateB + NFixedB * FixedRateB) / (NSweepB + NFixedB). Changes as AllocFractionA varies.*



Sweep Pooled Gap

Calculated

5 hops from ground truth

number

DR-281

IN RULESPEAK®

The allocation sweep's sweep pooled gap is determined by the following priority:

1. **an empty string** if the sweep pooled rate a is blank
2. **the sweep pooled rate a minus the sweep pooled rate b** in all other cases

- ▼ **Inputs** 2 fields this rule reads

f Sweep Pooled Rate a Allocation Sweep

If (n sweep a plus n fixed a) equals 0 then "",
otherwise (((n sweep a times sweep rate a) plus
(n fixed a times fixed rate a)) divided by (n sweep
a plus n fixed a)).

f Sweep Pooled Rate B Allocation Sweep

If (n sweep b plus n fixed b) equals 0 then "",
otherwise (((n sweep b times sweep rate b) plus
(n fixed b times fixed rate b)) divided by (n sweep
b plus n fixed b)).

- **Ground truth at the leaves** 2 the raw values this ultimately depends on

SweepPooledRateA minus SweepPooledRateB. Wanders as AllocFractionA varies because pooled rates mix stratum sizes unequally. Crosses zero at some allocation (the paradox reversal point).



Sweep Corrected Gap

Calculated

1 hop from ground truth

number

DR-282

IN RULESPEAK®

An allocation sweep's sweep corrected gap is computed as the value of LOOKUP(StudyId, SweepStudyConfig[StudyId], SweepStudyConfig[SweepCorrectedGap]).

- ▼ **Inputs** 1 fields this rule reads

● Study ID Allocation Sweep

The study being swept — all 238 studies in the corpus.

- **Ground truth at the leaves** 1 the raw values this ultimately depends on

Allocation-free signal: sum over strata of $(n_s/N) * (rate_{As} - rate_{Bs})$. Constant across the sweep by construction — the allocation variable does not appear. This is the conserved quantity under re-stratification.

f Allocation Distortion Witness

Calculated

6 hops from ground truth

number

DR-283

IN RULESPEAK®

The allocation sweep's allocation distortion witness is determined by the following priority:

1. **an empty string** if the sweep pooled gap is blank
2. **the sweep pooled gap minus the sweep corrected gap** in all other cases

▼ Inputs 2 fields this rule reads

f Sweep Pooled Gap

Allocation Sweep

If sweep pooled rate a is "" then "", otherwise
(sweep pooled rate a minus sweep pooled rate b).

f Sweep Corrected Gap

Allocation Sweep

LOOKUP({{StudyId}},
SweepStudyConfig[StudyId],
SweepStudyConfig[SweepCorrectedGap]).

► Ground truth at the leaves 2 the raw values this ultimately depends on

SweepPooledGap minus SweepCorrectedGap. The allocation distortion: how much the pooled signal has been inflated or deflated by differential stratum allocation. Zero only when allocation is balanced.

Sweep Study Summary

A sweep study summary is identified by its name.

f Name

Calculated

1 hop from ground truth

string

DR-284

IN RULESPEAK®

A sweep study summary's name is the same as its sweep study ID.

▼ Inputs 1 fields this rule reads

● Sweep Study ID

Sweep Study Summary

FK to Studies.StudyId. One row per study in the AllocationSweep.

► Ground truth at the leaves 1 the raw values this ultimately depends on

Display label.

f Distortion Type Label

Calculated

1 hop from ground truth

string

DR-285

A sweep study summary's [distortion type label](#) is computed as the value of `LOOKUP(SweepStudyId, TreatmentRankings[Study], TreatmentRankings[DistortionType])`.

▼ **Inputs** 1 fields this rule reads

● **Sweep Study ID** Sweep Study Summary

FK to Studies.StudyId. One row per study in the AllocationSweep.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

DistortionType for this study (A, B, C, or D) — provides context for what the sweep should show.

● **Corrected Gap Constant** Ground truth number

Ground truth. Written directly to the database (datatype `number`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Principled materialization (refresh_sweep_study_summary(), 03b/05b customize SQL). The allocation-free corrected gap for this study = MINIFS(AllocationSweep[SweepCorrectedGap], AllocationSweep[StudyId], SweepStudyId). Upstream (AllocationSweep, SweepStudyConfig) is sealed/stable per build; recomputing this per-row via nested calc_ calls was measured to fan out to 22.5s for a 242-study corpus scan with zero memoization — materialized here instead of left as a live calculated field. See MaterializedEntities.*

● **Sweep Corrected Gap Max** Ground truth number

Ground truth. Written directly to the database (datatype `number`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Principled materialization (refresh_sweep_study_summary()). Maximum SweepCorrectedGap across all allocation fractions for this study = MAXIFS(AllocationSweep[SweepCorrectedGap], AllocationSweep[StudyId], SweepStudyId). See CorrectedGapConstant for why this is materialized rather than calculated.

● **Sweep Corrected Gap Min** Ground truth number

Ground truth. Written directly to the database (datatype `number`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Principled materialization (refresh_sweep_study_summary()). Minimum SweepCorrectedGap across all allocation fractions for this study = MINIFS(AllocationSweep[SweepCorrectedGap], AllocationSweep[StudyId], SweepStudyId). See CorrectedGapConstant for why this is materialized rather than calculated.

● **Sweep Corrected Gap Range** Ground truth number

Ground truth. Written directly to the database (datatype `number`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Principled materialization (refresh_sweep_study_summary()). Range of SweepCorrectedGap across the sweep = SweepCorrectedGapMax - SweepCorrectedGapMin. Should be effectively zero (< 0.0001) for all studies if CorrectedGap is truly allocation-free. See CorrectedGapConstant for why this is materialized rather than calculated.

Sweep Pooled Gap Max Ground truth number

Ground truth. Written directly to the database (datatype `number`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Principled materialization (refresh_sweep_study_summary()). Maximum SweepPooledGap across all allocation fractions for this study = MAXIFS(AllocationSweep[SweepPooledGap], AllocationSweep[StudyId], SweepStudyId). See CorrectedGapConstant for why this is materialized rather than calculated.

Sweep Pooled Gap Min Ground truth number

Ground truth. Written directly to the database (datatype `number`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Principled materialization (refresh_sweep_study_summary()). Minimum SweepPooledGap across all allocation fractions for this study = MINIFS(AllocationSweep[SweepPooledGap], AllocationSweep[StudyId], SweepStudyId). See CorrectedGapConstant for why this is materialized rather than calculated.

Sweep Pooled Gap Range Ground truth number

Ground truth. Written directly to the database (datatype `number`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Principled materialization (refresh_sweep_study_summary()). Range of SweepPooledGap across the sweep = SweepPooledGapMax - SweepPooledGapMin. Should be large for Type A/B studies (pooled gap wanders) and smaller for Type D (balanced allocation = less wandering). See CorrectedGapConstant for why this is materialized rather than calculated.

Pooled Gap Crosses Zero Ground truth boolean

Ground truth. Written directly to the database (datatype `boolean`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Principled materialization (refresh_sweep_study_summary()). TRUE when SweepPooledGapMin < 0 and SweepPooledGapMax > 0 — the pooled gap changes sign somewhere in the sweep, meaning the paradox can be created or destroyed by pure allocation. See CorrectedGapConstant for why this is materialized rather than calculated.



Invariant Witness

Calculated

1 hop from ground truth

string

DR-286

IN RULESPEAK®

The sweep study summary's [invariant witness](#) is determined by the following priority:

1. **"PASS: CorrectedGap invariant across allocation sweep"** if the [sweep corrected gap range](#) is less than 0.0001
2. **"FAIL: CorrectedGap varies — formula error"** in all other cases

▼ Inputs 1 fields this rule reads



Sweep Corrected Gap Range

Sweep Study Summary

Principled materialization (refresh_sweep_study_summary()). Range of SweepCorrectedGap across the sweep = SweepCorrectedGapMax - SweepCorrectedGapMin. Should be effectively zero (< 0.0001) for all studies if CorrectedGap is truly allocation-free. See CorrectedGapConstant for why this is materialized rather than calculated.

► Ground truth at the leaves 1 the raw values this ultimately depends on

Summary witness statement: PASS if CorrectedGap is invariant (range < 0.0001), FAIL otherwise.



Sweep Stratum Label

Calculated

1 hop from ground truth

string

DR-287

IN RULESPEAK®

A sweep study summary's [sweep stratum label](#) is computed as the value of LOOKUP(SweepStudyId, SweepStudyConfig[StudyId], SweepStudyConfig[SweepStratumLabel]).

▼ Inputs 1 fields this rule reads



Sweep Study ID

Sweep Study Summary

FK to Studies.StudyId. One row per study in the AllocationSweep.

► Ground truth at the leaves 1 the raw values this ultimately depends on

Which stratum is being swept for this study.

A research tradition is identified by its name.

Name Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Short display name for the tradition.

Era Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Approximate active period, e.g. "1899–1972".

Core Concern Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

The central question this tradition asks about the paradox.

Primary Venue Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Where this tradition publishes (journal, conference, book series).

Key Claim Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

The tradition's signature thesis or resolution strategy.

Open Question Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

What this tradition leaves unresolved or contested.

Verified by Deep Research Ground truth boolean

Ground truth. Written directly to the database (datatype `boolean`); not computed from anything else. Everything derived ultimately traces back to values like this one.

True if the 2026-06-28 adversarial deep-research pass confirmed the tradition at high confidence.

Illustrated by Study Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

FK → Studies.StudyId. The study in this model that most vividly illustrates this tradition's characteristic concern or resolution strategy.

Supporting Conclusion Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

FK → Conclusions.ConclusionId. The conclusion in this model that most directly validates or advances this tradition's central claim about Simpson's Paradox.

Illustrated by Study Title Lookup 1 hop from ground truth string DR-288

IN RULESPEAK®

A research tradition's illustrated by study title is the value of LOOKUP(IllustratedByStudy, Studies[StudyId], Studies[Title]).

▼ **Inputs** 1 fields this rule reads

● **Illustrated by Study** Research Tradition
FK → Studies.StudyId. The study in this model that most vividly illustrates this tradition's characteristic concern or resolution strategy.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: Studies.Title via IllustratedByStudy.

Illustrated by Study Source Lookup 1 hop from ground truth string DR-289

IN RULESPEAK®

A research tradition's illustrated by study source is the value of LOOKUP(IllustratedByStudy, Studies[StudyId], Studies[Source]).

▼ **Inputs** 1 fields this rule reads

● Illustrated by Study

Research Tradition

FK → Studies.StudyId. The study in this model that most vividly illustrates this tradition's characteristic concern or resolution strategy.

- **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: Studies.Source via IllustratedByStudy — the citation for the study that best illustrates this tradition.



Illustrated by Study Distortion Type

Lookup

1 hop from ground truth

string

DR-290

IN RULESPEAK®

A research tradition's illustrated by study distortion type is the value of LOOKUP(IllustratedByStudy, TreatmentRankings[Study], TreatmentRankings[DistortionType]).

- ▼ **Inputs** 1 fields this rule reads

● Illustrated by Study

Research Tradition

FK → Studies.StudyId. The study in this model that most vividly illustrates this tradition's characteristic concern or resolution strategy.

- **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: TreatmentRankings.DistortionType for the illustrated study.



Supporting Conclusion Title

Lookup

1 hop from ground truth

string

DR-291

IN RULESPEAK®

A research tradition's supporting conclusion title is the value of LOOKUP(SupportingConclusion, Conclusions[ConclusionId], Conclusions[Title]).

- ▼ **Inputs** 1 fields this rule reads

● Supporting Conclusion

Research Tradition

FK → Conclusions.ConclusionId. The conclusion in this model that most directly validates or advances this tradition's central claim about Simpson's Paradox.

- **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: Conclusions.Title via SupportingConclusion.



Supporting Conclusion Category

Lookup

1 hop from ground truth

string

DR-292

IN RULESPEAK®

A research tradition's supporting conclusion category is the value of `LOOKUP(SupportingConclusion, Conclusions[ConclusionId], Conclusions[Category])`.

▼ Inputs 1 fields this rule reads

● Supporting Conclusion Research Tradition

FK → Conclusions.ConclusionId. The conclusion in this model that most directly validates or advances this tradition's central claim about Simpson's Paradox.

► Ground truth at the leaves 1 the raw values this ultimately depends on

Lookup: Conclusions.Category via SupportingConclusion.



Researcher Count

Aggregation

1 hop from ground truth

integer

DR-293

IN RULESPEAK®

A research tradition's researcher count is the number of researchers related to the research tradition.

▼ Inputs 2 fields this rule reads

● Tradition ID Researcher

FK → ResearchTraditions.TraditionId

● Tradition ID Research Tradition

Slug identifier, e.g. tradition-historical.

► Ground truth at the leaves 2 the raw values this ultimately depends on

Count of researchers in this tradition.



Study Count

Aggregation

1 hop from ground truth

integer

DR-294

IN RULESPEAK®

A research tradition's study count is the number of studies related to the research tradition.

▼ Inputs 2 fields this rule reads

● Tradition ID Study

FK → ResearchTraditions.TraditionId. The scholarly tradition whose framing best characterizes this study's paradox type. Synthetic

● Tradition ID Research Tradition

Slug identifier, e.g. tradition-historical.

studies inherit the tradition of their conceptual origin.

- **Ground truth at the leaves** 2 the raw values this ultimately depends on

Count of studies in this model whose TraditionId matches this tradition.

Σ Loop Count Aggregation 1 hop from ground truth integer DR-295

IN RULESPEAK®

A research tradition's loop count is the number of loops related to the research tradition.

- ▼ **Inputs** 2 fields this rule reads

● **Tradition ID** Loop
FK → ResearchTraditions.TraditionId. The scholarly tradition whose conceptual vocabulary this loop primarily draws on — tracks which tradition has driven each stage of the instrument's development.

● **Tradition ID** Research Tradition
Slug identifier, e.g. tradition-historical.

- **Ground truth at the leaves** 2 the raw values this ultimately depends on

Count of Effortless Loops that drew primarily on this tradition's conceptual vocabulary.

Researcher

A researcher is identified by its name.

● **Name** Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Full name as commonly cited.

● **Tradition ID** Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

FK → ResearchTraditions.TraditionId

● **Active Period** Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Approximate years of relevant work, e.g. "1899".

Affiliation Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Primary institutional affiliation for the relevant work.

Key Contribution Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

One-sentence description of their specific contribution to the paradox literature.

Signature Work Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

The canonical paper or book most associated with their contribution.

Relation to Hannan Pearl Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

How this researcher's line of work relates to or differs from Hannan (aggregation bias) and Pearl (DAG/do-calculus).

Verified by Deep Research Ground truth boolean

Ground truth. Written directly to the database (datatype `boolean`); not computed from anything else. Everything derived ultimately traces back to values like this one.

True if the 2026-06-28 adversarial deep-research pass confirmed this person at high confidence.

Confidence Level Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

high | medium | unverified — matches deep-research vote outcome.



Canonical Study ID

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

FK → Studies.StudyId. The study in this model that best illustrates this researcher's specific contribution to the Simpson's Paradox literature.



Illustrates Conclusion

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

FK → Conclusions.ConclusionId. The conclusion in this model that this researcher's work directly supports, challenges, or frames.



Tradition Name

Lookup

1 hop from ground truth

string

DR-296

IN RULESPEAK®

A researcher's tradition name is the value of LOOKUP(TraditionId, ResearchTraditions[TraditionId], ResearchTraditions[Name]).

▼ **Inputs** 1 fields this rule reads



Tradition ID

Researcher

FK → ResearchTraditions.TraditionId

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: ResearchTraditions.Name via TraditionId.



Tradition Era

Lookup

1 hop from ground truth

string

DR-297

IN RULESPEAK®

A researcher's tradition era is the value of LOOKUP(TraditionId, ResearchTraditions[TraditionId], ResearchTraditions[Era]).

▼ **Inputs** 1 fields this rule reads



Tradition ID

Researcher

FK → ResearchTraditions.TraditionId

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: ResearchTraditions.Era via TraditionId.



Canonical Study Title

Lookup

1 hop from ground truth

string

DR-298

IN RULESPEAK®

A researcher's canonical study title is the value of LOOKUP(CanonicalStudyId, Studies[StudyId], Studies[Title]).

▼ Inputs 1 fields this rule reads

● Canonical Study ID

Researcher

FK → Studies.StudyId. The study in this model that best illustrates this researcher's specific contribution to the Simpson's Paradox literature.

► Ground truth at the leaves 1 the raw values this ultimately depends on

Lookup: Studies.Title via CanonicalStudyId.



Canonical Study Distortion Type

Lookup

1 hop from ground truth

string

DR-299

IN RULESPEAK®

A researcher's canonical study distortion type is the value of LOOKUP(CanonicalStudyId, TreatmentRankings[Study], TreatmentRankings[DistortionType]).

▼ Inputs 1 fields this rule reads

● Canonical Study ID

Researcher

FK → Studies.StudyId. The study in this model that best illustrates this researcher's specific contribution to the Simpson's Paradox literature.

► Ground truth at the leaves 1 the raw values this ultimately depends on

Lookup: TreatmentRankings.DistortionType for the canonical study — the geometric type of paradox this researcher's work exemplifies.



Canonical Study Policy Implication

Lookup

1 hop from ground truth

string

DR-300

IN RULESPEAK®

A researcher's canonical study policy implication is the value of LOOKUP(CanonicalStudyId, TreatmentRankings[Study], TreatmentRankings[CorrectedPolicyImplication]).

▼ Inputs 1 fields this rule reads

● Canonical Study ID

Researcher

FK → Studies.StudyId. The study in this model that best illustrates this researcher's specific contribution to the Simpson's Paradox literature.

- **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: *TreatmentRankings.CorrectedPolicyImplication for the canonical study.*



Illustrates Conclusion Title

Lookup

1 hop from ground truth

string

DR-301

IN RULESPEAK®

A researcher's illustrates conclusion title is the value of LOOKUP(IllustratesConclusion, Conclusions[ConclusionId], Conclusions[Title]).

- ▼ **Inputs** 1 fields this rule reads

● Illustrates Conclusion

Researcher

FK → Conclusions.ConclusionId. The conclusion in this model that this researcher's work directly supports, challenges, or frames.

- **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: *Conclusions.Title via IllustratesConclusion.*



Tradition Key Claim Summary

Lookup

1 hop from ground truth

string

DR-302

IN RULESPEAK®

A researcher's tradition key claim summary is the value of LOOKUP(TraditionId, ResearchTraditions[TraditionId], ResearchTraditions[KeyClaim]).

- ▼ **Inputs** 1 fields this rule reads

● Tradition ID

Researcher

FK → ResearchTraditions.TraditionId

- **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: *ResearchTraditions.KeyClaim via TraditionId — the tradition's signature thesis this researcher contributed to.*

Synthetic Phase

A synthetic phase is identified by its name.

f **Name** Calculated 1 hop from ground truth string DR-303

IN RULESPEAK®

A synthetic phase's name is the same as its phase ID.

▼ **Inputs** 1 fields this rule reads

● **Phase ID** Synthetic Phase
Unique grid-point identifier encoding (w, g1, g2, b) coordinates.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

| *Display label.*

Param Stratum Fraction Ground truth number

Ground truth. Written directly to the database (datatype number); not computed from anything else. Everything derived ultimately traces back to values like this one.

| *w — fraction of total cases in the hard (stratum-1) subgroup. Range 0.30–0.75 in this grid.*

Param Stratum Gap1 Ground truth number

Ground truth. Written directly to the database (datatype number); not computed from anything else. Everything derived ultimately traces back to values like this one.

| *g1 — signed treatment advantage (rate_A – rate_B) in the hard stratum. Positive means A wins stratum 1.*

Param Stratum Gap2 Ground truth number

Ground truth. Written directly to the database (datatype number); not computed from anything else. Everything derived ultimately traces back to values like this one.

| *g2 — signed treatment advantage in the easy stratum. Negative g2 enables type-B (partial reversal) cells.*

Param Allocation Bias Ground truth number

Ground truth. Written directly to the database (datatype number); not computed from anything else. Everything derived ultimately traces back to values like this one.

| *b — allocation bias in stratum 1: extra fraction of stratum-1 cases assigned to treatment A above 50/50. Range 0.00–0.95.*

f **Phase S1 Total** **Calculated** 1 hop from ground truth number DR-304

IN RULESPEAK®

A synthetic phase's phase s1 total is computed as 200 times the param stratum fraction rounded to the nearest whole number.

▼ **Inputs** 1 fields this rule reads

- **Param Stratum Fraction** Synthetic Phase
w — fraction of total cases in the hard (stratum-1) subgroup. Range 0.30–0.75 in this grid.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Total cases in stratum 1 = ROUND(200 × ParamStratumFraction).

f **Phase NA1** **Calculated** 2 hops from ground truth number DR-305

IN RULESPEAK®

A synthetic phase's phase NA1 is computed as the phase s1 total times 0.5 plus the param allocation bias divided by 2 rounded to the nearest whole number.

▼ **Inputs** 2 fields this rule reads

- f Phase S1 Total** Synthetic Phase
(200 times param stratum fraction) rounded to the nearest whole number.

- **Param Allocation Bias** Synthetic Phase
b — allocation bias in stratum 1: extra fraction of stratum-1 cases assigned to treatment A above 50/50. Range 0.00–0.95.

► **Ground truth at the leaves** 2 the raw values this ultimately depends on

Treatment-A cases in stratum 1.

f **Phase NB1** **Calculated** 3 hops from ground truth number DR-306

IN RULESPEAK®

A synthetic phase's phase NB1 is computed as the phase s1 total minus the phase NA1.

▼ **Inputs** 2 fields this rule reads

- f Phase S1 Total** Synthetic Phase
(200 times param stratum fraction) rounded to the nearest whole number.

- f Phase NA1** Synthetic Phase
(phase s1 total times (0.5 plus (param allocation bias divided by 2))) rounded to the nearest whole number.

► **Ground truth at the leaves** 2 the raw values this ultimately depends on

| *Treatment-B cases in stratum 1.*

f **Phase NA2** **Calculated** 3 hops from ground truth number DR-307

IN RULESPEAK®

A synthetic phase's phase NA2 is computed as 100 minus the phase NA1.

▼ **Inputs** 1 fields this rule reads

f **Phase NA1** Synthetic Phase
(phase s1 total times (0.5 plus (param allocation bias divided by 2))) rounded to the nearest whole number.

► **Ground truth at the leaves** 2 the raw values this ultimately depends on

| *Treatment-A cases in stratum 2.*

f **Phase NB2** **Calculated** 4 hops from ground truth number DR-308

IN RULESPEAK®

A synthetic phase's phase NB2 is computed as 100 minus the phase NB1.

▼ **Inputs** 1 fields this rule reads

f **Phase NB1** Synthetic Phase
Phase s1 total minus phase NA1.

► **Ground truth at the leaves** 2 the raw values this ultimately depends on

| *Treatment-B cases in stratum 2.*

f **Phase Rate A1** **Calculated** 1 hop from ground truth number DR-309

IN RULESPEAK®

A synthetic phase's phase rate a1 is computed as 0.45 plus the param stratum gap1 divided by 2.

▼ **Inputs** 1 fields this rule reads

● **Param Stratum Gap1** Synthetic Phase
g1 — signed treatment advantage (rate_A – rate_B) in the hard stratum. Positive means A wins

stratum 1.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

| *Success rate for A in hard stratum.*

f **Phase Rate B1** Calculated 1 hop from ground truth number DR-310

IN RULESPEAK®

A synthetic phase's phase rate b1 is computed as 0.45 minus the param stratum gap1 divided by 2.

▼ **Inputs** 1 fields this rule reads

● **Param Stratum Gap1** Synthetic Phase
g1 — signed treatment advantage (rate_A – rate_B) in the hard stratum. Positive means A wins stratum 1.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

| *Success rate for B in hard stratum.*

f **Phase Rate A2** Calculated 1 hop from ground truth number DR-311

IN RULESPEAK®

A synthetic phase's phase rate a2 is computed as 0.75 plus the param stratum gap2 divided by 2.

▼ **Inputs** 1 fields this rule reads

● **Param Stratum Gap2** Synthetic Phase
g2 — signed treatment advantage in the easy stratum. Negative g2 enables type-B (partial reversal) cells.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

| *Success rate for A in easy stratum.*

f **Phase Rate B2** Calculated 1 hop from ground truth number DR-312

IN RULESPEAK®

A synthetic phase's phase rate b2 is computed as 0.75 minus the param stratum gap2 divided by 2.

▼ **Inputs** 1 fields this rule reads

● **Param Stratum Gap2** Synthetic Phase
g2 — signed treatment advantage in the easy stratum. Negative g2 enables type-B (partial reversal) cells.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

| Success rate for B in easy stratum.

f **Phase Pooled Rate a** Calculated 4 hops from ground truth number DR-313

IN RULESPEAK®

A synthetic phase's phase pooled rate a is computed as the phase NA1 times the phase rate a1 plus the phase NA2 times the phase rate a2 divided by 100.

▼ **Inputs** 4 fields this rule reads

f **Phase NA1** Synthetic Phase
(phase s1 total times (0.5 plus (param allocation bias divided by 2))) rounded to the nearest whole number.

f **Phase Rate A1** Synthetic Phase
0.45 plus (param stratum gap1 divided by 2).

f **Phase NA2** Synthetic Phase
100 minus phase NA1.

f **Phase Rate A2** Synthetic Phase
0.75 plus (param stratum gap2 divided by 2).

► **Ground truth at the leaves** 4 the raw values this ultimately depends on

| Pooled success rate for treatment A.

f **Phase Pooled Rate B** Calculated 5 hops from ground truth number DR-314

IN RULESPEAK®

A synthetic phase's phase pooled rate b is computed as the phase NB1 times the phase rate b1 plus the phase NB2 times the phase rate b2 divided by 100.

▼ **Inputs** 4 fields this rule reads

f **Phase NB1** Synthetic Phase
Phase s1 total minus phase NA1.

f **Phase Rate B1** Synthetic Phase
0.45 minus (param stratum gap1 divided by 2).

f **Phase NB2** Synthetic Phase
100 minus phase NB1.

f **Phase Rate B2** Synthetic Phase
0.75 minus (param stratum gap2 divided by 2).

► **Ground truth at the leaves** 4 the raw values this ultimately depends on

Pooled success rate for treatment B.

f Phase Signed Pooled Gap Calculated 6 hops from ground truth number DR-315

IN RULESPEAK®

A synthetic phase's phase signed pooled gap is computed as the phase pooled rate a minus the phase pooled rate b.

▼ **Inputs** 2 fields this rule reads

f **Phase Pooled Rate a** Synthetic Phase
((phase NA1 times phase rate a1) plus (phase NA2 times phase rate a2)) divided by 100.

f **Phase Pooled Rate B** Synthetic Phase
((phase NB1 times phase rate b1) plus (phase NB2 times phase rate b2)) divided by 100.

► **Ground truth at the leaves** 4 the raw values this ultimately depends on

Pooled rate gap (A minus B).

f Phase Corrected Gap Calculated 1 hop from ground truth number DR-316

IN RULESPEAK®

A synthetic phase's phase corrected gap is computed as the param stratum fraction times the param stratum gap1 plus 1 minus the param stratum fraction times the param stratum gap2.

▼ **Inputs** 3 fields this rule reads

● **Param Stratum Fraction** Synthetic Phase
w — fraction of total cases in the hard (stratum-1) subgroup. Range 0.30–0.75 in this grid.

● **Param Stratum Gap1** Synthetic Phase
g1 — signed treatment advantage (rate_A – rate_B) in the hard stratum. Positive means A wins stratum 1.

● **Param Stratum Gap2** Synthetic Phase
g2 — signed treatment advantage in the easy stratum. Negative g2 enables type-B (partial reversal) cells.

► **Ground truth at the leaves** 3 the raw values this ultimately depends on

Allocation-free equal-weight gap = $w \times g1 + (1-w) \times g2$.

f Phase Strata Won by Loser Calculated 7 hops from ground truth number DR-317

IN RULESPEAK®

The synthetic phase's phase strata won by loser is determined by the following priority:

1. the count of the following that hold: the param stratum gap1 is less than 0 and the param stratum gap2 is less than 0 if the phase signed pooled gap is greater than 0
2. the count of the following that hold: the param stratum gap1 is greater than 0 and the param stratum gap2 is greater than 0 in all other cases

▼ **Inputs** 3 fields this rule reads

f **Phase Signed Pooled Gap** Synthetic Phase
Phase pooled rate a minus phase pooled rate b.

● **Param Stratum Gap1** Synthetic Phase
g1 — signed treatment advantage (rate_A – rate_B) in the hard stratum. Positive means A wins stratum 1.

● **Param Stratum Gap2** Synthetic Phase
g2 — signed treatment advantage in the easy stratum. Negative g2 enables type-B (partial reversal) cells.

► **Ground truth at the leaves** 4 the raw values this ultimately depends on

| *Count of strata whose per-stratum winner disagrees with the pooled winner.*

f **Phase Reversal Intensity** Calculated 8 hops from ground truth number DR-318

IN RULESPEAK®

A synthetic phase's phase reversal intensity is computed as the phase strata won by loser divided by 2.

▼ **Inputs** 1 fields this rule reads

f **Phase Strata Won by Loser** Synthetic Phase
If phase signed pooled gap is greater than 0 then
(if param stratum gap1 is less than 0 then 1,
otherwise 0 plus if param stratum gap2 is less
than 0 then 1, otherwise 0), otherwise (if param
stratum gap1 is greater than 0 then 1, otherwise 0
plus if param stratum gap2 is greater than 0 then
1, otherwise 0).

► **Ground truth at the leaves** 4 the raw values this ultimately depends on

| *PhaseStrataWonByLoser / 2 for this 2-stratum template.*

f **Phase is Sign Flip** Calculated 7 hops from ground truth boolean DR-319

The synthetic phase's phase is sign flip is determined by the following priority:

1. the phase signed pooled gap is less than 0 if the phase corrected gap is greater than 0
2. the phase signed pooled gap is greater than 0 in all other cases

▼ **Inputs** 2 fields this rule reads

f Phase Corrected Gap Synthetic Phase
(param stratum fraction times param stratum gap1) plus ((1 minus param stratum fraction) times param stratum gap2).

f Phase Signed Pooled Gap Synthetic Phase
Phase pooled rate a minus phase pooled rate b.

► **Ground truth at the leaves** 4 the raw values this ultimately depends on

| *TRUE when equal-weight and pooled gaps have opposite signs.*



Phase Allocation Distortion

Calculated

7 hops from ground truth

number

DR-320

A synthetic phase's phase allocation distortion is computed as the absolute value of the phase corrected gap minus the phase signed pooled gap.

▼ **Inputs** 2 fields this rule reads

f Phase Corrected Gap Synthetic Phase
(param stratum fraction times param stratum gap1) plus ((1 minus param stratum fraction) times param stratum gap2).

f Phase Signed Pooled Gap Synthetic Phase
Phase pooled rate a minus phase pooled rate b.

► **Ground truth at the leaves** 4 the raw values this ultimately depends on

| *|PhaseCorrectedGap - PhaseSignedPooledGap|.*



Phase Distortion Type

Calculated

9 hops from ground truth

string

DR-321

The synthetic phase's phase distortion type is determined by the following priority:

1. **"A"** if all of these hold:
 - the phase is sign flip flag is set
 - the phase reversal intensity is 1
2. **"B"** if all of these hold:
 - the phase is sign flip flag is set
 - the phase reversal intensity is less than 1

3. **"C+"** if all of these hold:

- the [phase is sign flip](#) flag is not set
- the [phase allocation distortion](#) is greater than 0.01
- the absolute value of the [phase signed pooled gap](#) is greater than the absolute value of the [phase corrected gap](#) plus 0.001

4. **"C-"** if all of these hold:

- the [phase is sign flip](#) flag is not set
- the [phase allocation distortion](#) is greater than 0.01
- the absolute value of the [phase signed pooled gap](#) is less than the absolute value of the [phase corrected gap](#) minus 0.001

5. **"D"** in all other cases

▼ Inputs 5 fields this rule reads

f **Phase is Sign Flip** Synthetic Phase

If phase corrected gap is greater than 0 then
phase signed pooled gap is less than 0, otherwise
phase signed pooled gap is greater than 0.

f **Phase Reversal Intensity** Synthetic Phase

Phase strata won by loser divided by 2.

f **Phase Allocation Distortion** Synthetic Phase

The absolute value of (phase corrected gap minus
phase signed pooled gap).

f **Phase Signed Pooled Gap** Synthetic Phase

Phase pooled rate a minus phase pooled rate b.

f **Phase Corrected Gap** Synthetic Phase

(param stratum fraction times param stratum
gap1) plus ((1 minus param stratum fraction)
times param stratum gap2).

► Ground truth at the leaves 4 the raw values this ultimately depends on

Five-type geometric classification at this parameter point.

Phase Diagram Summary

A phase diagram summary is identified by its [name](#).

f **Name** Calculated 1 hop from ground truth string DR-322

IN RULESPEAK®

A phase diagram summary's [name](#) is the same as its phase diagram ID.

▼ Inputs 1 fields this rule reads

● Phase Diagram ID

Phase Diagram Summary

Unique identifier for this phase diagram instance.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Display label.



Phase Point Count

Calculated

1 hop from ground truth

integer

DR-323

IN RULESPEAK®

A phase diagram summary's phase point count is computed as the phase type a count plus the phase type b count plus the phase type c plus count plus the phase type c minus count plus the phase type d count.

▼ **Inputs** 5 fields this rule reads

● Phase Type a Count

Phase Diagram Summary

Grid points classified as type A. Populated from vw_synthetic_phase by 05b-customize-data.sql.

● Phase Type B Count

Phase Diagram Summary

Grid points classified as type B. Populated from vw_synthetic_phase by 05b-customize-data.sql.

● Phase Type C Plus Count

Phase Diagram
Summary

Grid points classified as type C+. Populated from vw_synthetic_phase by 05b-customize-data.sql.

● Phase Type C Minus Count

Phase Diagram
Summary

Grid points classified as type C-. Populated from vw_synthetic_phase by 05b-customize-data.sql.

● Phase Type D Count

Phase Diagram Summary

Grid points classified as type D. Populated from vw_synthetic_phase by 05b-customize-data.sql.

► **Ground truth at the leaves** 5 the raw values this ultimately depends on

Total grid points in SyntheticPhase (sum of type counts). Refreshed from vw_synthetic_phase by 05b-customize-data.sql.



Phase Type a Count

Ground truth

integer

Ground truth. Written directly to the database (datatype integer); not computed from anything else. Everything derived ultimately traces back to values like this one.

Grid points classified as type A. Populated from vw_synthetic_phase by 05b-customize-data.sql.



Phase Type B Count

Ground truth

integer

Ground truth. Written directly to the database (datatype `integer`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Grid points classified as type B. Populated from vw_synthetic_phase by 05b-customize-data.sql.

Phase Type C Plus Count Ground truth integer

Ground truth. Written directly to the database (datatype `integer`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Grid points classified as type C+. Populated from vw_synthetic_phase by 05b-customize-data.sql.

Phase Type C Minus Count Ground truth integer

Ground truth. Written directly to the database (datatype `integer`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Grid points classified as type C-. Populated from vw_synthetic_phase by 05b-customize-data.sql.

Phase Type D Count Ground truth integer

Ground truth. Written directly to the database (datatype `integer`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Grid points classified as type D. Populated from vw_synthetic_phase by 05b-customize-data.sql.

All Five Types Present Calculated 1 hop from ground truth boolean DR-324

IN RULESPEAK®

A phase diagram summary is flagged all five types present if all of the following:

- the [phase type a count](#) is greater than 0
- the [phase type b count](#) is greater than 0
- the [phase type c plus count](#) is greater than 0
- the [phase type c minus count](#) is greater than 0
- the [phase type d count](#) is greater than 0

▼ **Inputs** 5 fields this rule reads

● **Phase Type a Count** Phase Diagram Summary
Grid points classified as type A. Populated from vw_synthetic_phase by 05b-customize-data.sql.

● **Phase Type B Count** Phase Diagram Summary
Grid points classified as type B. Populated from vw_synthetic_phase by 05b-customize-data.sql.

● **Phase Type C Plus Count** Phase Diagram Summary

● **Phase Type C Minus Count** Phase Diagram Summary

Grid points classified as type C+. Populated from vw_synthetic_phase by 05b-customize-data.sql.

Grid points classified as type C-. Populated from vw_synthetic_phase by 05b-customize-data.sql.

● **Phase Type D Count** Phase Diagram Summary

Grid points classified as type D. Populated from vw_synthetic_phase by 05b-customize-data.sql.

► **Ground truth at the leaves** 5 the raw values this ultimately depends on

TRUE when every DistortionType cell has at least one grid point.

f **Phase Taxonomy Witness** Calculated 1 hop from ground truth string DR-325

IN RULESPEAK®

A phase diagram summary's [phase taxonomy witness](#) is computed as "A:", followed by the [phase type a count](#), followed by " B:", followed by the [phase type b count](#), followed by " C+:", followed by the [phase type c plus count](#), followed by " C-:", followed by the [phase type c minus count](#), followed by " D:", followed by the [phase type d count](#).

▼ **Inputs** 5 fields this rule reads

● **Phase Type a Count** Phase Diagram Summary

Grid points classified as type A. Populated from vw_synthetic_phase by 05b-customize-data.sql.

● **Phase Type B Count** Phase Diagram Summary

Grid points classified as type B. Populated from vw_synthetic_phase by 05b-customize-data.sql.

● **Phase Type C Plus Count** Phase Diagram Summary

Grid points classified as type C+. Populated from vw_synthetic_phase by 05b-customize-data.sql.

● **Phase Type C Minus Count** Phase Diagram Summary

Grid points classified as type C-. Populated from vw_synthetic_phase by 05b-customize-data.sql.

● **Phase Type D Count** Phase Diagram Summary

Grid points classified as type D. Populated from vw_synthetic_phase by 05b-customize-data.sql.

► **Ground truth at the leaves** 5 the raw values this ultimately depends on

Human-readable witness string of type counts across the grid.

f **Phase Witness Note** Calculated 2 hops from ground truth string DR-326

IN RULESPEAK®

The phase diagram summary's [phase witness note](#) is determined by the following priority:

1. **"PASS: all five distortion types populated in parameter space"** if the [all five types present](#) flag is set

2. "FAIL: missing distortion type in grid" in all other cases

▼ **Inputs** 1 fields this rule reads

f All Five Types Present

Phase Diagram
Summary

Phase type a count is greater than 0, phase type b count is greater than 0, phase type c plus count is greater than 0, phase type c minus count is greater than 0, and phase type d count is greater than 0.

► **Ground truth at the leaves** 5 the raw values this ultimately depends on

PASS/FAIL summary for taxonomy completeness.

Ingestion Protocol

An ingestion protocol is identified by its name.

f **Name** Calculated 1 hop from ground truth string DR-327

IN RULESPEAK®

An ingestion protocol's name is the same as its protocol ID.

▼ **Inputs** 1 fields this rule reads

● **Protocol ID**

Ingestion Protocol

Unique slug for this contract item.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Display label.

○ **Rule Type** Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

required-input / structural / inclusion / exclusion / adapter-metadata

○ **Target Table** Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Rulebook table this item applies to.

Field Name Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Column name when RuleType=required-input.

Requirement Statement Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Plain-English statement of the requirement.

Mechanical Check Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

How to verify compliance before classification.

Sort Order Ground truth integer

Ground truth. Written directly to the database (datatype `integer`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Display order in the ingestion checklist.

Ingestion Summary

An ingestion summary is identified by its [name](#).

Name Calculated 1 hop from ground truth string DR-328

IN RULESPEAK®

An ingestion summary's [name](#) is the same as its ingestion summary ID.

▼ **Inputs** 1 fields this rule reads

Ingestion Summary ID Ingestion Summary

Unique identifier.

- **Ground truth at the leaves** 1 the raw values this ultimately depends on

Display label.

Σ **Protocol Item Count** Aggregation 1 hop from ground truth integer DR-329

IN RULESPEAK®

An ingestion summary's protocol item count is the number of ingestion protocol related to the ingestion summary.

- ▼ **Inputs** 1 fields this rule reads

● **Protocol ID** Ingestion Protocol
Unique slug for this contract item.

- **Ground truth at the leaves** 1 the raw values this ultimately depends on

Total contract items. Fixed loop-90: COUNTIFS(...,"<>") mistranslates to a literal '<>' string-equality filter (a known transpiler bug) instead of IS NOT NULL, always returning 0; replaced with bare COUNT(), which transpiles correctly.

Σ **Corpus Cell Count** Aggregation 1 hop from ground truth integer DR-330

IN RULESPEAK®

An ingestion summary's corpus cell count is the number of case cells related to the ingestion summary.

- ▼ **Inputs** 1 fields this rule reads

● **Case Cell ID** Case Cell
Unique slug (e.g. kidney-1986-small-A).

- **Ground truth at the leaves** 1 the raw values this ultimately depends on

Total CaseCell rows. Fixed loop-90 (see ProtocolItemCount).

Σ **Valid Cell Count** Aggregation 2 hops from ground truth integer DR-331

IN RULESPEAK®

An ingestion summary's valid cell count is the number of case cells related to the ingestion summary.

- ▼ **Inputs** 1 fields this rule reads

f Is Valid Ingestion Cell

Case Cell

Cases is at least 0, successes is at least 0, and successes is at most cases.

- **Ground truth at the leaves** 2 the raw values this ultimately depends on

Valid CaseCells.



All Cells Valid

Calculated

3 hops from ground truth

boolean

DR-332

IN RULESPEAK®

An ingestion summary is flagged all cells valid if the valid cell count is the corpus cell count.

- ▼ **Inputs** 2 fields this rule reads

Σ Valid Cell Count

Ingestion Summary

Count of case cells whose is valid ingestion cell is true().

Σ Corpus Cell Count

Ingestion Summary

Count(case cell ID).

- **Ground truth at the leaves** 3 the raw values this ultimately depends on

Every cell passes validity predicate.



Study Count

Aggregation

1 hop from ground truth

integer

DR-333

IN RULESPEAK®

An ingestion summary's study count is the number of studies related to the ingestion summary.

- ▼ **Inputs** 1 fields this rule reads

● Is Control Study

Study

TRUE if this study is a control (no Simpson's paradox expected/found); FALSE for the main Simpson's Paradox corpus.

- **Ground truth at the leaves** 1 the raw values this ultimately depends on

Count of studies subject to the ingestion contract (excludes control studies, loop-92 -- control studies were never encoded with paired treatment arms, so IngestionCellParity is structurally undefined for them, not a real ingestion defect).



Structural Compliant Count

Aggregation

3 hops from ground truth

integer

DR-334

IN RULESPEAK®

An ingestion summary's structural compliant count is the number of studies related to the ingestion summary.

▼ **Inputs** 1 fields this rule reads

f Ingestion Cell Parity Study

```
AND(LOOKUP({{StudyId}},
TreatmentRankings[Study],
TreatmentRankings[StratumCount]) >= 2,
{{CellCount}} = LOOKUP({{StudyId}},
TreatmentRankings[Study],
TreatmentRankings[StratumCount]) * 2).
```

► **Ground truth at the leaves** 2 the raw values this ultimately depends on

Studies with IngestionCellParity.

f All Studies Structural Calculated 4 hops from ground truth boolean DR-335

IN RULESPEAK®

An ingestion summary is flagged all studies structural if the structural compliant count is the study count.

▼ **Inputs** 2 fields this rule reads

Σ Structural Compliant Count Ingestion Summary

Count of studies whose ingestion cell parity is true().

Σ Study Count Ingestion Summary

Count of studies whose is control study is false().

► **Ground truth at the leaves** 3 the raw values this ultimately depends on

All studies structurally valid.

Σ Real Study Count Aggregation 1 hop from ground truth integer DR-336

IN RULESPEAK®

An ingestion summary's real study count is the number of the ingestion summary's studies that are not control studies.

▼ **Inputs** 2 fields this rule reads

● **Is Synthetic** Study

● **Is Control Study** Study

TRUE for constructed/counterfactual studies;
FALSE for real published studies. Filters the
corpus for empirical claims.

TRUE if this study is a control (no Simpson's
paradox expected/found); FALSE for the main
Simpson's Paradox corpus.

- **Ground truth at the leaves** 2 the raw values this ultimately depends on

Non-synthetic, non-control studies (loop-92 -- control studies were never encoded with paired treatment arms, so they are outside the ingestion contract's scope, same as StructuralCompliantCount's StudyCount).

Σ Real Fully Compliant Count

Aggregation

4 hops from ground truth

integer

DR-337

IN RULESPEAK®

An ingestion summary's real fully compliant count is the number of studies related to the ingestion summary.

- ▼ **Inputs** 1 fields this rule reads

f **Ingestion Compliance** Study

```
IF({{IsSynthetic}}, "exempt-synthetic",  
IF(AND({{IngestionCellParity}},  
LOOKUP({{StudyId}}, StratumVariables[Study],  
StratumVariables[VariableName]) <> ""), "all",  
"partial")).
```

- **Ground truth at the leaves** 3 the raw values this ultimately depends on

Real studies with IngestionCompliance=all.

f All Real Studies Compliant

Calculated

5 hops from ground truth

boolean

DR-338

IN RULESPEAK®

An ingestion summary is flagged all real studies compliant if the real fully compliant count is the real study count.

- ▼ **Inputs** 2 fields this rule reads

Σ **Real Fully Compliant Count** Ingestion Summary

Count of studies whose ingestion compliance is "all".

Σ **Real Study Count** Ingestion Summary

Count of studies whose is synthetic is false() and is control study is false().

- **Ground truth at the leaves** 4 the raw values this ultimately depends on

All real studies fully compliant.



Ingestion Contract Passes

Calculated

6 hops from ground truth

boolean

DR-339

IN RULESPEAK®

An ingestion summary is flagged ingestion contract passes if all of the following:

- the [all cells valid](#) flag is set
- the [all studies structural](#) flag is set
- the [all real studies compliant](#) flag is set

▼ Inputs 3 fields this rule reads



All Cells Valid

Ingestion Summary

Valid cell count equals corpus cell count.



All Studies Structural

Ingestion Summary

Structural compliant count equals study count.



All Real Studies Compliant

Ingestion Summary

Real fully compliant count equals real study count.

► Ground truth at the leaves 7 the raw values this ultimately depends on

Full contract passes.



Ingestion Witness Note

Calculated

7 hops from ground truth

string

DR-340

IN RULESPEAK®

The ingestion summary's [ingestion witness note](#) is determined by the following priority:

1. **"PASS: ", followed by the [protocol item count](#), followed by " contract items; ", followed by the [corpus cell count](#), followed by " cells valid; ", followed by the [real fully compliant count](#), followed by a slash, followed by the [real study count](#), followed by " real studies fully compliant"** if the [ingestion contract passes](#) flag is set
2. **"FAIL: ingestion contract violated"** in all other cases

▼ Inputs 5 fields this rule reads



Ingestion Contract Passes

Ingestion Summary

All cells valid, all studies structural, and all real studies compliant.



Protocol Item Count

Ingestion Summary

Count(protocol ID).



Corpus Cell Count

Ingestion Summary

Count(case cell ID).



Real Fully Compliant Count

Ingestion Summary

Count of studies whose ingestion compliance is "all".

Σ **Real Study Count** Ingestion Summary

Count of studies whose is synthetic is false() and is control study is false().

► **Ground truth at the leaves** **8** the raw values this ultimately depends on

PASS/FAIL summary.

Candidate Study Catalog

A candidate study catalog is identified by its name.

f **Name** Calculated 1 hop from ground truth string DR-341

IN RULESPEAK®

A candidate study catalog's name is the same as its candidate ID.

▼ **Inputs** **1** fields this rule reads

● **Candidate ID** Candidate Study Catalog

Unique slug for this catalog entry.

► **Ground truth at the leaves** **1** the raw values this ultimately depends on

Display label.

○ **Proposed Study ID** Ground truth string

Ground truth. Written directly to the database (datatype string); not computed from anything else. Everything derived ultimately traces back to values like this one.

Stable slug to assign in Studies when imported.

○ **Title** Ground truth string

Ground truth. Written directly to the database (datatype string); not computed from anything else. Everything derived ultimately traces back to values like this one.

Human-readable study title.

○ **Citation** Ground truth string

Ground truth. Written directly to the database (datatype string); not computed from anything else. Everything derived ultimately traces back to values like this one.

Primary publication citation.

Source URL Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

URL to source paper or data.

Domain Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Domain tag: medicine, epidemiology, legal, sports, education, economics, social-science.

Stratum Variable Name Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Pre-registered stratifying variable name for encoding.

Expected Distortion Type Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Hypothesized `DistortionType` after classification: A, B, C+, C-, D, or unknown.

Ingestion Status Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

imported | candidate | blocked | rejected

Priority Ground truth integer

Ground truth. Written directly to the database (datatype `integer`); not computed from anything else. Everything derived ultimately traces back to values like this one.

1=high priority encode next; 5=blocked/duplicate.

Stratum Count Estimate Ground truth integer

Ground truth. Written directly to the database (datatype `integer`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Expected number of strata (minimum 2).

Data Source Note Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Where to find cell counts and encoding notes.

Linked Study ID Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Studies.StudyId when IngestionStatus=imported.

Publication Year Ground truth integer

Ground truth. Written directly to the database (datatype `integer`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Publication year when known.

Is Ready to Encode Calculated 1 hop from ground truth boolean DR-342

IN RULESPEAK®

A candidate study catalog is considered a ready to encode if all of the following:

- the [ingestion status](#) is "candidate"
- the [proposed study ID](#) has a value
- the [citation](#) has a value
- the [stratum variable name](#) has a value

▼ **Inputs** 4 fields this rule reads

● **Ingestion Status** Candidate Study Catalog
imported | candidate | blocked | rejected

● **Proposed Study ID** Candidate Study Catalog
Stable slug to assign in Studies when imported.

● **Citation** Candidate Study Catalog
Primary publication citation.

● **Stratum Variable Name** Candidate Study Catalog
Pre-registered stratifying variable name for encoding.

► **Ground truth at the leaves** 4 the raw values this ultimately depends on

TRUE when candidate has citation, proposed id, and stratum variable — ready for an import session.

f Is Imported Calculated 1 hop from ground truth boolean DR-343

IN RULESPEAK®

A candidate study catalog is considered imported if the ingestion status is "imported".

▼ **Inputs** 1 fields this rule reads

● **Ingestion Status** Candidate Study Catalog

imported | candidate | blocked | rejected

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

TRUE when already in corpus.

↗ Observed Distortion Type Lookup 1 hop from ground truth string DR-344

IN RULESPEAK®

A candidate study catalog's observed distortion type is the value of LOOKUP(LinkedStudyId, TreatmentRankings[Study], TreatmentRankings[DistortionType]).

▼ **Inputs** 1 fields this rule reads

● **Linked Study ID** Candidate Study Catalog

Studies.StudyId when IngestionStatus=imported.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Actual DistortionType from TreatmentRankings when imported.

f Type Prediction Match Calculated 2 hops from ground truth boolean DR-345

IN RULESPEAK®

The candidate study catalog's type prediction match is determined by the following priority:

1. the observed distortion type is the expected distortion type if the imported flag is set
2. an empty string in all other cases

▼ **Inputs** 3 fields this rule reads

f Is Imported Candidate Study Catalog

Ingestion status is "imported".

↗ Observed Distortion Type Candidate Study Catalog

```
LOOKUP({{LinkedStudyId}},
TreatmentRankings[Study],
TreatmentRankings[DistortionType]).
```

- **Expected Distortion Type** Candidate Study Catalog
Hypothesized DistortionType after classification:
A, B, C+, C-, D, or unknown.

- **Ground truth at the leaves** **3** the raw values this ultimately depends on
- TRUE when observed type matches expected (imported rows only).*

f **Expected Sign Flip** **Calculated** 1 hop from ground truth boolean DR-346

IN RULESPEAK®

A candidate study catalog is flagged expected sign flip if all of the following:

- the expected distortion type is "A"
- the expected distortion type is "B"

- ▾ **Inputs** **1** fields this rule reads

- **Expected Distortion Type** Candidate Study Catalog
Hypothesized DistortionType after classification:
A, B, C+, C-, D, or unknown.

- **Ground truth at the leaves** **1** the raw values this ultimately depends on
- TRUE when curator tagged ExpectedDistortionType as A or B (sign-flip expectation).*

f **Observed Sign Flip Type** **Calculated** 2 hops from ground truth boolean DR-347

IN RULESPEAK®

The candidate study catalog's observed sign flip type is determined by the following priority:

1. **the OR of the observed distortion type is "A" and the observed distortion type is "B"** if the imported flag is set
2. **an empty string** in all other cases

- ▾ **Inputs** **2** fields this rule reads

- f **Is Imported** Candidate Study Catalog
Ingestion status is "imported".

- ↗ **Observed Distortion Type** Candidate Study Catalog

```
LOOKUP({{LinkedStudyId}},
TreatmentRankings[Study],
TreatmentRankings[DistortionType]).
```

► **Ground truth at the leaves** 2 the raw values this ultimately depends on

TRUE when observed DistortionType is A or B (imported rows only).

f Sign Flip Prediction Match Calculated 3 hops from ground truth boolean DR-348

IN RULESPEAK®

The candidate study catalog's sign flip prediction match is determined by the following priority:

1. the observed sign flip type if all of these hold:

- the imported flag is set
- the expected sign flip flag is set

2. **an empty string** in all other cases

▼ **Inputs** 3 fields this rule reads

f Is Imported Candidate Study Catalog
Ingestion status is "imported".

f Expected Sign Flip Candidate Study Catalog
Expected distortion type is "A" or expected distortion type is "B".

f Observed Sign Flip Type Candidate Study Catalog
If is imported then observed distortion type is "A" or observed distortion type is "B", otherwise "".

► **Ground truth at the leaves** 3 the raw values this ultimately depends on

TRUE when Expected A∨B and observed is A∨B — weaker flip-type prediction test.

○ Data Acquisition Status Ground truth string

Ground truth. Written directly to the database (datatype string); not computed from anything else. Everything derived ultimately traces back to values like this one.

not_started | downloaded | manual_only | blocked

○ Reversal Mechanism Ground truth string

Ground truth. Written directly to the database (datatype string); not computed from anything else. Everything derived ultimately traces back to values like this one.

Plain-language reversal mechanism from expansion plan.

Paradox Confirmation Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

confirmed | plausible | pending | not_applicable

Expansion Wave Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Catalog wave tag; loop-67 expansion rows = expansion-wave-1.

Is Data Ready Calculated 1 hop from ground truth boolean DR-349

IN RULESPEAK®

A candidate study catalog is considered a data ready if all of the following:

- the data acquisition status is "downloaded"
- the data acquisition status is "manual_only"

▼ **Inputs** 1 fields this rule reads

Data Acquisition Status Candidate Study Catalog
not_started | downloaded | manual_only | blocked

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

TRUE when stratified data is downloaded or available from published tables.

Corpus Catalog Summary

A corpus catalog summary is identified by its name.

Name Calculated 1 hop from ground truth string DR-350

IN RULESPEAK®

A corpus catalog summary's name is the same as its catalog summary ID.

▼ **Inputs** 1 fields this rule reads

● **Catalog Summary ID**

Corpus Catalog
Summary

Unique identifier.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Display label.



Total Catalog Entries

Aggregation

1 hop from ground truth

integer

DR-351

IN RULESPEAK®

A corpus catalog summary's total catalog entries is the number of candidate study catalog related to the corpus catalog summary.

▼ **Inputs** 1 fields this rule reads

● **Candidate ID**

Candidate Study Catalog

Unique slug for this catalog entry.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

All catalog rows. Fixed loop-90 (see IngestionSummary.ProtocolItemCount for why COUNTIFS(...,"<>") is unsafe).



Imported Count

Aggregation

1 hop from ground truth

integer

DR-352

IN RULESPEAK®

A corpus catalog summary's imported count is the number of candidate study catalog related to the corpus catalog summary.

▼ **Inputs** 1 fields this rule reads

● **Ingestion Status**

Candidate Study Catalog

imported | candidate | blocked | rejected

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Rows already imported.



Candidate Count

Aggregation

1 hop from ground truth

integer

DR-353

IN RULESPEAK®

A corpus catalog summary's candidate count is the number of candidate study catalog related to the corpus catalog summary.

▼ **Inputs** 1 fields this rule reads

● **Ingestion Status** Candidate Study Catalog
imported | candidate | blocked | rejected

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Rows queued for import.

Σ Blocked Count Aggregation 1 hop from ground truth integer DR-354

IN RULESPEAK®

A corpus catalog summary's blocked count is the number of candidate study catalog related to the corpus catalog summary.

▼ **Inputs** 1 fields this rule reads

● **Ingestion Status** Candidate Study Catalog
imported | candidate | blocked | rejected

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Duplicate or blocked rows.

Σ Ready to Encode Count Aggregation 2 hops from ground truth integer DR-355

IN RULESPEAK®

A corpus catalog summary's ready to encode count is the number of candidate study catalog related to the corpus catalog summary.

▼ **Inputs** 1 fields this rule reads

f **Is Ready to Encode** Candidate Study Catalog
Ingestion status is "candidate", proposed study ID is not "", citation is not "", and stratum variable name is not "".

► **Ground truth at the leaves** 4 the raw values this ultimately depends on

Candidates passing IsReadyToEncode.

Σ High Priority Count Aggregation 1 hop from ground truth integer DR-356

IN RULESPEAK®

A corpus catalog summary's [high priority count](#) is the number of the corpus catalog summary's high priorities.

▼ **Inputs** 2 fields this rule reads

● **Ingestion Status** Candidate Study Catalog
imported | candidate | blocked | rejected

● **Priority** Candidate Study Catalog
1=high priority encode next; 5=blocked/duplicate.

► **Ground truth at the leaves** 2 the raw values this ultimately depends on

Candidates with Priority <= 2.

f Import Session Ready Calculated 3 hops from ground truth boolean DR-357

IN RULESPEAK®

A corpus catalog summary is flagged import session ready if all of the following:

- the [candidate count](#) is at least 20
- the [ready to encode count](#) is at least 15

▼ **Inputs** 2 fields this rule reads

Σ **Candidate Count** Corpus Catalog Summary
Count of candidate study catalog whose ingestion status is "candidate".

Σ **Ready to Encode Count** Corpus Catalog Summary
Count of candidate study catalog whose is ready to encode is true().

► **Ground truth at the leaves** 4 the raw values this ultimately depends on

TRUE when backlog is large enough for a bulk import session.

Σ Type Prediction Match Count Aggregation 3 hops from ground truth integer DR-358

IN RULESPEAK®

A corpus catalog summary's [type prediction match count](#) is the number of [candidate study catalog](#) related to the corpus catalog summary.

▼ **Inputs** 1 fields this rule reads

f Type Prediction Match Candidate Study Catalog
If is imported then observed distortion type equals expected distortion type, otherwise "".

► **Ground truth at the leaves** 3 the raw values this ultimately depends on

Imported catalog rows where ObservedDistortionType exactly matches ExpectedDistortionType.

Σ Type Prediction Mismatch Count

Aggregation

3 hops from ground truth

integer

DR-359

IN RULESPEAK®

A corpus catalog summary's [type prediction mismatch count](#) is the number of the corpus catalog summary's [candidate study catalog](#) that [are not type prediction match](#).

▼ Inputs 2 fields this rule reads

f **Is Imported** Candidate Study Catalog
Ingestion status is "imported".

f **Type Prediction Match** Candidate Study Catalog
If is imported then observed distortion type equals expected distortion type, otherwise "".

► Ground truth at the leaves 3 the raw values this ultimately depends on

Imported catalog rows where exact type prediction failed.

Σ Sign Flip Prediction Eligible Count

Aggregation

2 hops from ground truth

integer

DR-360

IN RULESPEAK®

A corpus catalog summary's [sign flip prediction eligible count](#) is the number of the corpus catalog summary's [candidate study catalog](#) that [are expected sign flip](#).

▼ Inputs 2 fields this rule reads

f **Is Imported** Candidate Study Catalog
Ingestion status is "imported".

f **Expected Sign Flip** Candidate Study Catalog
Expected distortion type is "A" or expected distortion type is "B".

► Ground truth at the leaves 2 the raw values this ultimately depends on

Imported rows where ExpectedDistortionType was A or B.

Σ Sign Flip Prediction Match Count

Aggregation

4 hops from ground truth

integer

DR-361

IN RULESPEAK®

A corpus catalog summary's [sign flip prediction match count](#) is the number of [candidate study catalog](#) related to the corpus catalog summary.

▼ **Inputs** 1 fields this rule reads

f **Sign Flip Prediction Match** Candidate Study Catalog

If is imported and expected sign flip then observed sign flip type, otherwise "".

► **Ground truth at the leaves** 3 the raw values this ultimately depends on

| Expected AuB rows where observed type is also AuB.

f **Type Prediction Match Rate** Calculated 4 hops from ground truth number DR-362

IN RULESPEAK®

The corpus catalog summary's type prediction match rate is determined by the following priority:

1. **an empty string** if the imported count is 0
2. **the type prediction match count divided by the imported count** in all other cases

▼ **Inputs** 2 fields this rule reads

Σ **Imported Count** Corpus Catalog Summary
Count of candidate study catalog whose ingestion status is "imported".

Σ **Type Prediction Match Count** Corpus Catalog Summary
Count of candidate study catalog whose type prediction match is true().

► **Ground truth at the leaves** 3 the raw values this ultimately depends on

| Exact-match rate across imported catalog rows.

f **Sign Flip Prediction Match Rate** Calculated 5 hops from ground truth number DR-363

IN RULESPEAK®

The corpus catalog summary's sign flip prediction match rate is determined by the following priority:

1. **an empty string** if the sign flip prediction eligible count is 0
2. **the sign flip prediction match count divided by the sign flip prediction eligible count** in all other cases

▼ **Inputs** 2 fields this rule reads

Σ **Sign Flip Prediction Eligible Count** Corpus Catalog Summary

Σ **Sign Flip Prediction Match Count** Corpus Catalog Summary

Count of candidate study catalog whose is imported is true() and expected sign flip is true().

Count of candidate study catalog whose sign flip prediction match is true().

► **Ground truth at the leaves** 3 the raw values this ultimately depends on

Sign-flip-type match rate among Expected AuB rows.



Catalog Witness Note

Calculated

4 hops from ground truth

string

DR-364

IN RULESPEAK®

The corpus catalog summary's catalog witness note is determined by the following priority:

1. **"READY: ", followed by the ready to encode count, followed by " encode-ready (", followed by the data ready count, followed by " data-ready) of ", followed by the candidate count, followed by " candidates"** if the import session ready flag is set
2. **"BUILDING: ", followed by the candidate count, followed by " candidates — ", followed by the data ready count, followed by " data-ready, ", followed by the ready to encode count, followed by " encode-ready"** in all other cases

▼ **Inputs** 4 fields this rule reads

f Import Session Ready

Corpus Catalog Summary

Candidate count is at least 20 and ready to encode count is at least 15.

Σ Ready to Encode Count

Corpus Catalog Summary

Count of candidate study catalog whose is ready to encode is true().

Σ Data Ready Count

Corpus Catalog Summary

Count of candidate study catalog whose is data ready is true() and ingestion status is "candidate".

Σ Candidate Count

Corpus Catalog Summary

Count of candidate study catalog whose ingestion status is "candidate".

► **Ground truth at the leaves** 5 the raw values this ultimately depends on

Human-readable readiness summary.



Catalog Prediction Witness Note

Calculated

6 hops from ground truth

string

DR-365

IN RULESPEAK®  mechanical

A corpus catalog summary's catalog prediction witness note is computed as "exact=", followed by the type prediction match count, followed by a slash, followed by the imported count, followed by " (", followed by the type prediction match rate times 100 rounded to 1 decimal place(s), followed by "%); flipPred=", followed by the sign flip prediction match count, followed by a slash, followed by the sign flip prediction eligible count, followed by " (", followed by the sign flip prediction match rate times 100 rounded to 1 decimal place(s), followed by "%".

▼ **Inputs** 6 fields this rule reads

Σ Type Prediction Match Count Corpus Catalog Summary

Count of candidate study catalog whose type prediction match is true().

f Type Prediction Match Rate Corpus Catalog Summary

If imported count equals 0 then "", otherwise (type prediction match count divided by imported count).

Σ Sign Flip Prediction Eligible Count Corpus Catalog Summary

Count of candidate study catalog whose is imported is true() and expected sign flip is true().

Σ Imported Count Corpus Catalog Summary

Count of candidate study catalog whose ingestion status is "imported".

Σ Sign Flip Prediction Match Count Corpus Catalog Summary

Count of candidate study catalog whose sign flip prediction match is true().

f Sign Flip Prediction Match Rate Corpus Catalog Summary

If sign flip prediction eligible count equals 0 then "", otherwise (sign flip prediction match count divided by sign flip prediction eligible count).

► **Ground truth at the leaves** 3 the raw values this ultimately depends on

| *Loop-65 type-prediction audit summary.*

Σ Data Ready Count Aggregation 2 hops from ground truth integer DR-366

IN RULESPEAK®

A corpus catalog summary's [data ready count](#) is the number of the corpus catalog summary's [candidate study catalog](#) that have a [ingestion status](#) of "candidate".

▼ **Inputs** 2 fields this rule reads

f Is Data Ready Candidate Study Catalog

Data acquisition status is "downloaded" or data acquisition status is "manual_only".

● Ingestion Status Candidate Study Catalog

imported | candidate | blocked | rejected

► **Ground truth at the leaves** 2 the raw values this ultimately depends on

| *Candidates with IsDataReady=TRUE.*

Σ Encode Pipeline Ready Count Aggregation 2 hops from ground truth integer DR-367

IN RULESPEAK®

A corpus catalog summary's [encode pipeline ready count](#) is the number of the corpus catalog summary's [candidate study catalog](#) that [are data readies](#).

▼ **Inputs** 2 fields this rule reads

f Is Ready to Encode Candidate Study Catalog

Ingestion status is "candidate", proposed study ID is not "", citation is not "", and stratum variable name is not "".

f Is Data Ready Candidate Study Catalog

Data acquisition status is "downloaded" or data acquisition status is "manual_only".

- **Ground truth at the leaves** 5 the raw values this ultimately depends on

Candidates both data-ready and encode-ready.

Σ Expansion Candidate Count Aggregation 1 hop from ground truth integer DR-368

IN RULESPEAK®

A corpus catalog summary's expansion candidate count is the number of candidate study catalog related to the corpus catalog summary.

- ▼ **Inputs** 1 fields this rule reads

● Expansion Wave Candidate Study Catalog

Catalog wave tag; loop-67 expansion rows = expansion-wave-1.

- **Ground truth at the leaves** 1 the raw values this ultimately depends on

Loop-67 expansion-wave-1 candidates.

Domain Expansion Target

A domain expansion target is identified by its name.

f Name Calculated 1 hop from ground truth string DR-369

IN RULESPEAK®

A domain expansion target's name is the same as its domain target ID.

- ▼ **Inputs** 1 fields this rule reads

● Domain Target ID Domain Expansion Target

Unique slug.

- **Ground truth at the leaves** 1 the raw values this ultimately depends on

Display label.



Domain

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Domain tag matching Studies.Domain.



Legacy Domain Alias

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Studies.Domain slug for legacy imported corpus rows.



Target Min Count

Ground truth

integer

Ground truth. Written directly to the database (datatype `integer`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Minimum real studies desired in this domain.



Current Imported Count

Aggregation

1 hop from ground truth

integer

DR-370

IN RULESPEAK®

A domain expansion target's current imported count is the number of the domain expansion target's studies that are not synthetic.

▼ Inputs 4 fields this rule reads

● Domain

Study

Research domain: medicine | epidemiology | social-science | education | legal | sports | economics | synthetic. Enables cross-domain SignalPurity stratification.

● Legacy Domain Alias

Domain Expansion Target

Studies.Domain slug for legacy imported corpus rows.

● Domain

Domain Expansion Target

Domain tag matching Studies.Domain.

● Is Synthetic

Study

TRUE for constructed/counterfactual studies; FALSE for real published studies. Filters the corpus for empirical claims.

► Ground truth at the leaves 4 the raw values this ultimately depends on

Real studies in corpus matching LegacyDomainAlias (or Domain when no alias).



Candidate Queued Count

Aggregation

1 hop from ground truth

integer

DR-371

IN RULESPEAK®

A domain expansion target's candidate queued count is the number of the domain expansion target's candidate study catalog that have a ingestion status of "candidate".

▼ **Inputs** 3 fields this rule reads

● **Domain** Candidate Study Catalog
Domain tag: medicine, epidemiology, legal, sports, education, economics, social-science.

● **Ingestion Status** Candidate Study Catalog
imported | candidate | blocked | rejected

● **Domain** Domain Expansion Target
Domain tag matching Studies.Domain.

► **Ground truth at the leaves** 3 the raw values this ultimately depends on

Catalog candidates in this domain.

f **Projected Count** Calculated 2 hops from ground truth integer DR-372

IN RULESPEAK®

A domain expansion target's projected count is computed as the current imported count plus the candidate queued count.

▼ **Inputs** 2 fields this rule reads

Σ **Current Imported Count** Domain Expansion Target
Count of studies whose domain is if legacy domain alias is not "" then legacy domain alias, otherwise domain and is synthetic is false().

Σ **Candidate Queued Count** Domain Expansion Target
Count of candidate study catalog whose domain is domain and ingestion status is "candidate".

► **Ground truth at the leaves** 6 the raw values this ultimately depends on

Current + queued candidates.

f **Gap Count** Calculated 3 hops from ground truth integer DR-373

IN RULESPEAK®

A domain expansion target's gap count is computed as the largest of 0 and the target min count minus the projected count.

▼ **Inputs** 2 fields this rule reads

● **Target Min Count** Domain Expansion Target
Minimum real studies desired in this domain.

f **Projected Count** Domain Expansion Target

Current imported count plus candidate queued count.

- **Ground truth at the leaves** 7 the raw values this ultimately depends on

Studies still needed to hit target after queued imports.



Is Under Represented

Calculated

3 hops from ground truth

boolean

DR-374

IN RULESPEAK®

A domain expansion target is considered under-represented if the projected count is less than the target min count.

- ▼ **Inputs** 2 fields this rule reads

f Projected Count

Domain Expansion Target

Current imported count plus candidate queued count.

● Target Min Count

Domain Expansion Target

Minimum real studies desired in this domain.

- **Ground truth at the leaves** 7 the raw values this ultimately depends on

TRUE when projected count still below target.

Study Import Template

A study import template is identified by its name.



Name

Calculated

1 hop from ground truth

string

DR-375

IN RULESPEAK®

A study import template's name is the same as its template step ID.

- ▼ **Inputs** 1 fields this rule reads

● Template Step ID

Study Import Template

Unique slug.

- **Ground truth at the leaves** 1 the raw values this ultimately depends on

Display label.



Sort Order

Ground truth

integer

Ground truth. Written directly to the database (datatype `integer`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Order to execute during import.

Target Table Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Rulebook table to write.

Row Description Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

What rows to create.

Required Fields Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Key columns that must be populated.

Mechanical Check Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

How to verify step complete.

Sweep Study Config

A sweep study config is identified by its [name](#).

Name Calculated 1 hop from ground truth string DR-376

IN RULESPEAK®

A sweep study config's [name](#) is the same as its config ID.

▼ **Inputs** 1 fields this rule reads

● **Config ID** Sweep Study Config

Unique slug; mirrors StudyId.

- **Ground truth at the leaves** 1 the raw values this ultimately depends on

Display label.

Study ID Ground truth string

Ground truth. Written directly to the database (datatype string); not computed from anything else. Everything derived ultimately traces back to values like this one.

FK to Studies.StudyId.

Sweep Stratum Label Ground truth string

Ground truth. Written directly to the database (datatype string); not computed from anything else. Everything derived ultimately traces back to values like this one.

Stratum whose A/B allocation is swept (highest |AllocationBias| in original data).

N Sweep Stratum Total Ground truth number

Ground truth. Written directly to the database (datatype number); not computed from anything else. Everything derived ultimately traces back to values like this one.

Total cases in the sweep stratum (fixed across sweep rows).

Sweep Rate a Ground truth number

Ground truth. Written directly to the database (datatype number); not computed from anything else. Everything derived ultimately traces back to values like this one.

Success rate for treatment A in the sweep stratum.

Sweep Rate B Ground truth number

Ground truth. Written directly to the database (datatype number); not computed from anything else. Everything derived ultimately traces back to values like this one.

Success rate for treatment B in the sweep stratum.

N Fixed a Ground truth number

Ground truth. Written directly to the database (datatype number); not computed from anything else. Everything derived ultimately traces back to values like this one.

A cases in all non-swept strata (fixed).

N Fixed B Ground truth number

Ground truth. Written directly to the database (datatype number); not computed from anything else. Everything derived ultimately traces back to values like this one.

B cases in all non-swept strata (fixed).

Fixed Rate a Ground truth number

Ground truth. Written directly to the database (datatype number); not computed from anything else. Everything derived ultimately traces back to values like this one.

Weighted success rate for A in fixed strata.

Fixed Rate B Ground truth number

Ground truth. Written directly to the database (datatype number); not computed from anything else. Everything derived ultimately traces back to values like this one.

Weighted success rate for B in fixed strata.

Original Alloc Fraction a Ground truth number

Ground truth. Written directly to the database (datatype number); not computed from anything else. Everything derived ultimately traces back to values like this one.

Observed fraction of sweep-stratum cases assigned to treatment A.

Sweep Corrected Gap Ground truth number

Ground truth. Written directly to the database (datatype number); not computed from anything else. Everything derived ultimately traces back to values like this one.

*Allocation-free gap: $\sum_s (n_s/N) * (rate_As - rate_Bs)$. Invariant under within-stratum reallocation.*

Substrate Conformance Field

A substrate conformance field is identified by its [name](#).

Name Calculated 1 hop from ground truth string DR-377

IN RULESPEAK®

A substrate conformance field's [name](#) is the same as its field ID.

▼ **Inputs** 1 fields this rule reads

● **Field ID** Substrate Conformance Field
Unique slug.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

| *Display label.*

● **Source Table** Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

| *TreatmentRankings / StratumSummaries / StratumVariables*

● **Field Name** Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

| *PascalCase rulebook field name.*

● **Pg Column** Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

| *snake_case vw_* column.*

● **Owl Local Name** Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

| *camelCase OWL property local name.*

● **Data Type** Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

| *float | int | bool | str*



Assert From Postgres

Ground truth

boolean

Ground truth. Written directly to the database (datatype `boolean`); not computed from anything else. Everything derived ultimately traces back to values like this one.

TRUE when OWL cannot derive this field yet and Postgres value is asserted before SHACL fixpoint.



In Compare Set

Ground truth

boolean

Ground truth. Written directly to the database (datatype `boolean`); not computed from anything else. Everything derived ultimately traces back to values like this one.

TRUE when reason.py diffs this field between substrates.

Discovery Hypothes

A discovery hypothes is identified by its name.



Name

Calculated

1 hop from ground truth

string

DR-378

IN RULESPEAK®

A discovery hypothes's name is the same as its hypothesis ID.

▼ **Inputs** 1 fields this rule reads



Hypothesis ID

Discovery Hypothes

Unique slug (e.g. H-latent-d).

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Display label.



Statement

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Natural-language claim under test.



Expected Outcome

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Pass criterion in plain language.



Registered in Loop

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

LoopId where this hypothesis was registered.



Tradition ID

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

FK → ResearchTraditions.TraditionId.



Epistemic Tier

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

consistency-check (definition-linked regression) | corpus-hypothesis (contingent corpus pattern)

Discovery Finding

A discovery finding is identified by its [name](#).



Name

Calculated

1 hop from ground truth

string

DR-379

IN RULESPEAK®

A discovery finding's [name](#) is the same as its finding ID.

▼ **Inputs** 1 fields this rule reads



Finding ID

Discovery Finding

Unique slug; one row per hypothesis.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Display label.



Hypothesis ID

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

`FK → DiscoveryHypotheses.HypothesisId.`

Hypothesis Statement Lookup 1 hop from ground truth string DR-380

IN RULESPEAK®

A discovery finding's [hypothesis statement](#) is the value of `LOOKUP(HypothesisId, DiscoveryHypotheses[HypothesisId], DiscoveryHypotheses[Statement])`.

▼ **Inputs** 1 fields this rule reads

● **Hypothesis ID** Discovery Finding
`FK → DiscoveryHypotheses.HypothesisId.`

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

`Lookup: DiscoveryHypotheses.Statement.`

Observed Metric Calculated 1 hop from ground truth string DR-381

IN RULESPEAK®

A discovery finding's [observed metric](#) is computed as the value of `IF(HypothesisId = "H-latent-d", CONCAT("fraction=", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[LatentTypeDFraction])), IF(HypothesisId = "H-purity", CONCAT("maxPurity=", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[SignFlipSignalPurityMax])), IF(HypothesisId = "H-small-effect", CONCAT("stable=", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[AvgPooledGapStableD]), " latent=", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[AvgPooledGapLatentD])), IF(HypothesisId = "H-econ-zero", CONCAT("flips=", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[EconomicsSignFlipCount])), IF(HypothesisId = "H-domain-dist", CONCAT("epi=", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[EpidemiologyAvgDistortion]), " edu=", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[EducationAvgDistortion])), IF(HypothesisId = "H-causal-manifest", CONCAT("confFlip=", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[ConfounderSignFlipCount]), " collManifest=", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[ColliderSelectionManifestCount])), IF(HypothesisId = "H-causal-latent", CONCAT("collManifest=", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[ColliderSelectionManifestCount]), " collLatent=", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId],`

ModelSummary[ColliderSelectionLatentOnlyCount]), " collN=", LOOKUP("simpsons-paradox-v1",
 ModelSummary[ModelSummaryId], ModelSummary[ColliderSelectionCount])), IF(HypothesisId =
 "H-explained-confounder", CONCAT("ExplainedConfounderCount=", LOOKUP("simpsons-
 paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[ExplainedConfounderCount]),
 ", ConfounderSignFlipCount=", LOOKUP("simpsons-paradox-v1",
 ModelSummary[ModelSummaryId], ModelSummary[ConfounderSignFlipCount])),
 IF(HypothesisId = "H-unexplained-nonconfounder",
 CONCAT("ContestedOrMediatorExplainedCount=", LOOKUP("simpsons-paradox-v1",
 ModelSummary[ModelSummaryId], ModelSummary[ContestedOrMediatorExplainedCount])),
 IF(HypothesisId = "H-catalog-exact-match", CONCAT("exactRate=", LOOKUP("simpsons-
 paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[TypePredictionMatchRate])),
 IF(HypothesisId = "H-catalog-flip-prediction", CONCAT("flipPredRate=", LOOKUP("simpsons-
 paradox-v1", ModelSummary[ModelSummaryId],
 ModelSummary[SignFlipPredictionMatchRate])), IF(HypothesisId = "H-domain-flip-geometry-
 controlled", CONCAT("econHighImbFlips=", LOOKUP("simpsons-paradox-v1",
 ModelSummary[ModelSummaryId], ModelSummary[EconomicsHighImbalanceSignFlipCount]), ";
 epiRate=", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId],
 ModelSummary[EpidemiologyHighImbalanceSignFlipRate]), "; legal=", LOOKUP("simpsons-
 paradox-v1", ModelSummary[ModelSummaryId],
 ModelSummary[LegalHighImbalanceSignFlipRate]), "; sports=", LOOKUP("simpsons-paradox-
 v1", ModelSummary[ModelSummaryId], ModelSummary[SportsHighImbalanceSignFlipRate])),
 IF(HypothesisId = "H-collider-no-manifest-v2", CONCAT("collManifest=", LOOKUP("simpsons-
 paradox-v1", ModelSummary[ModelSummaryId],
 ModelSummary[ColliderSelectionManifestCount])), IF(HypothesisId = "H-cplus-magnitude",
 CONCAT("C+=", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId],
 ModelSummary[CPlusAvgDistortion]), " C-=", LOOKUP("simpsons-paradox-v1",
 ModelSummary[ModelSummaryId], ModelSummary[CMinusAvgDistortion]), " D=",
 LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId],
 ModelSummary[TypeDAvgDistortion])), IF(HypothesisId = "H-econ-encoding-selection",
 CONCAT("EconExpectedAMismatchRate=", LOOKUP("simpsons-paradox-v1",
 ModelSummary[ModelSummaryId], ModelSummary[EconomicsExpectedAMismatchRate])),
 IF(HypothesisId = "H-domain-profiles-stable", CONCAT("eduLatent=", LOOKUP("simpsons-
 paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[EducationLatentFraction]), "
 sportsLatent=", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId],
 ModelSummary[SportsLatentFraction]), " econFlipRate=", LOOKUP("simpsons-paradox-v1",
 ModelSummary[ModelSummaryId], ModelSummary[EconomicsSignFlipRate])), IF(HypothesisId
 = "H-corrected-gap-invariant", CONCAT("maxRange=", LOOKUP("simpsons-paradox-v1",
 ModelSummary[ModelSummaryId], ModelSummary[MaxStudySweepCorrectedGapRange]), "
 fails=", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId],
 ModelSummary[CorrectedGapInvariantFailCount])), IF(HypothesisId = "H-explained-
 bidirectional", CONCAT("explained=", LOOKUP("simpsons-paradox-v1",
 ModelSummary[ModelSummaryId], ModelSummary[ExplainedConfounderCount]), " confFlip=",

LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId],
ModelSummary[ConfounderSignFlipCount]), " falsePos=", LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId], ModelSummary[FalsePositiveExplainedCount]), "
unexplained=", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId],
ModelSummary[UnexplainedConfounderSignFlipCount])), IF(HypothesisId = "H-collider-no-
manifest-theorem", CONCAT("collN=", LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId], ModelSummary[ColliderSelectionCount]), " collManifest=",
LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId],
ModelSummary[ColliderSelectionManifestCount])), IF(HypothesisId = "H-theorem-portfolio",
CONCAT("theoremCount=", LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId], ModelSummary[TheoremCount])), IF(HypothesisId = "H-
age-identity-flip-rate", CONCAT("rate=", LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId], ModelSummary[AgeIdentityManifestFlipRate]), " n=",
LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId],
ModelSummary[AgeIdentityStudyCount])), IF(HypothesisId = "H-identity-map-coverage",
CONCAT("coverage=", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId],
ModelSummary[IdentityMapCoverageRate])), IF(HypothesisId = "H-severity-medicine-manifest",
CONCAT("medicine_flip_rate=", LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId], ModelSummary[SeverityMedicineManifestFlipRate])),
IF(HypothesisId = "H-severity-epi-latent-only", CONCAT("epi_manifest_count=",
LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId],
ModelSummary[SeverityEpiManifestFlipCount])), IF(HypothesisId = "H-mechanism-other-drain",
CONCAT("other_count=", LOOKUP("cluster-id-mechanism-other",
IdentityClusterSummaries[IdentityClusterId], IdentityClusterSummaries[StudyCount])),
IF(HypothesisId = "H-selection-frailty-zero-manifest", CONCAT("manifest=", LOOKUP("cluster-
id-selection-frailty", IdentityClusterSummaries[IdentityClusterId],
IdentityClusterSummaries[ManifestFlipCount]), " n=", LOOKUP("cluster-id-selection-frailty",
IdentityClusterSummaries[IdentityClusterId], IdentityClusterSummaries[StudyCount])),
IF(HypothesisId = "H-collider-identity-low-manifest", CONCAT("rate=", LOOKUP("simpsons-
paradox-v1", ModelSummary[ModelSummaryId],
ModelSummary[ColliderIdentityManifestFlipRate])), IF(HypothesisId = "H-geographic-stable-
type-d", CONCAT("type_d_frac=", LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId], ModelSummary[GeographicTypeDFraction])),
IF(HypothesisId = "H-unexplained-flips-only-nonconfounders",
CONCAT("unexplained_confounder_flips=", LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId], ModelSummary[UnexplainedConfounderSignFlipCount])),
IF(HypothesisId = "H-confounder-drift-type-d", CONCAT("severity=", LOOKUP("simpsons-
paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[SeverityDriftDirection]), "
geography=", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId],
ModelSummary[GeographicDriftDirection]), " situational=", LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId], ModelSummary[SituationalDriftDirection]), " age=",
LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId],

```
ModelSummary[AgeDriftDirection]), " institutional=", LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId], ModelSummary[InstitutionalDriftDirection]), "
socioeconomic=", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId],
ModelSummary[SocioeconomicDriftDirection])), ("")))))))))))))))))))))))))))))).
```

▼ **Inputs** 1 fields this rule reads

- Hypothesis ID

Discovery Finding

FK → DiscoveryHypotheses.HypothesisId.

- **Ground truth at the leaves** 1 the raw values this ultimately depends on

Witnessed metric value from ModelSummary for this hypothesis.

f

Is Confirmed

Calculated

1 hop from ground truth

boolean

DR-382

IN RULESPEAK®

```
A discovery finding is considered confirmed if the value of IF(HypothesisId = "H-latent-d",
LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId],
ModelSummary[LatentTypeDFraction]) > 0.5, IF(HypothesisId = "H-purity", LOOKUP("simpsons-
paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[SignFlipSignalPurityMax]) <
0.5, IF(HypothesisId = "H-small-effect", LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId], ModelSummary[AvgPooledGapStableD]) >
LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId],
ModelSummary[AvgPooledGapLatentD])), IF(HypothesisId = "H-econ-zero", LOOKUP("simpsons-
paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[EconomicsSignFlipCount]) = 0,
IF(HypothesisId = "H-domain-dist", LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId], ModelSummary[EpidemiologyAvgDistortion]) >
LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId],
ModelSummary[EducationAvgDistortion]), IF(HypothesisId = "H-causal-manifest",
AND(LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId],
ModelSummary[ConfounderSignFlipCount]) > LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId], ModelSummary[ColliderSelectionManifestCount])),
LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId],
ModelSummary[ConfounderSignFlipCount]) >= 10), IF(HypothesisId = "H-causal-latent",
AND(LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId],
ModelSummary[ColliderSelectionManifestCount]) = 0, LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId], ModelSummary[ColliderSelectionLatentOnlyCount]) >= 5),
IF(HypothesisId = "H-explained-confounder", AND(LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId], ModelSummary[ExplainedConfounderCount]) =
LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId],
ModelSummary[ConfounderSignFlipCount]), LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId], ModelSummary[ExplainedConfounderCount]) >= 10),
```


IF(HypothesisId = "H-unexplained-nonconfounder", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[ContestedOrMediatorExplainedCount]) = 0, IF(HypothesisId = "H-catalog-exact-match", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[TypePredictionMatchRate]) < 0.5, IF(HypothesisId = "H-catalog-flip-prediction", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[SignFlipPredictionMatchRate]) < 0.5, IF(HypothesisId = "H-domain-flip-geometry-controlled", AND(LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[EconomicsHighImbalanceSignFlipCount]) = 0, LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[EpidemiologyHighImbalanceSignFlipRate]) > 0.15), IF(HypothesisId = "H-collider-no-manifest-v2", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[ColliderSelectionManifestCount]) = 0, IF(HypothesisId = "H-cplus-magnitude", AND(LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[CPlusAvgDistortion]) > LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[CMinusAvgDistortion])), LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[CPlusAvgDistortion]) > LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[TypeDAvgDistortion])), IF(HypothesisId = "H-econ-encoding-selection", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[EconomicsExpectedAMismatchRate]) > 0.5, IF(HypothesisId = "H-domain-profiles-stable", AND(LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[EducationLatentFraction]) > 0.5, LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[SportsLatentFraction]) > 0.5, LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[EconomicsSignFlipRate]) < 0.05), IF(HypothesisId = "H-corrected-gap-invariant", AND(LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[MaxStudySweepCorrectedGapRange]) < 0.0001, LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[CorrectedGapInvariantFailCount]) = 0), IF(HypothesisId = "H-explained-bidirectional", AND(LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[ExplainedConfounderCount]) = LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[ConfounderSignFlipCount])), LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[FalsePositiveExplainedCount]) = 0, LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[UnexplainedConfounderSignFlipCount]) = 0), IF(HypothesisId = "H-collider-no-manifest-theorem", AND(LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[ColliderSelectionManifestCount]) = 0, LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[ColliderSelectionCount]) >= 10), IF(HypothesisId = "H-theorem-portfolio", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[TheoremCount]) >= 4, IF(HypothesisId = "H-age-identity-flip-rate", AND(LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId],

```

ModelSummary[AgeIdentityManifestFlipRate]) >= 0.40, LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId], ModelSummary[AgeIdentityStudyCount]) >= 10),
IF(HypothesisId = "H-identity-map-coverage", LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId], ModelSummary[IdentityMapCoverageRate]) >= 0.95,
IF(HypothesisId = "H-severity-medicine-manifest", LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId], ModelSummary[SeverityMedicineManifestFlipRate]) >= 0.45,
IF(HypothesisId = "H-severity-epi-latent-only", LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId], ModelSummary[SeverityEpiManifestFlipCount]) = 0,
IF(HypothesisId = "H-mechanism-other-drain", LOOKUP("cluster-id-mechanism-other",
IdentityClusterSummaries[IdentityClusterId], IdentityClusterSummaries[StudyCount]) <= 5,
IF(HypothesisId = "H-selection-frailty-zero-manifest", AND(LOOKUP("cluster-id-selection-
frailty", IdentityClusterSummaries[IdentityClusterId],
IdentityClusterSummaries[ManifestFlipCount]) = 0, LOOKUP("cluster-id-selection-frailty",
IdentityClusterSummaries[IdentityClusterId], IdentityClusterSummaries[StudyCount]) >= 3),
IF(HypothesisId = "H-collider-identity-low-manifest", AND(LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId], ModelSummary[ColliderIdentityManifestFlipRate]) <= 0.15,
LOOKUP("cluster-id-collider-proxy", IdentityClusterSummaries[IdentityClusterId],
IdentityClusterSummaries[StudyCount]) >= 5), IF(HypothesisId = "H-geographic-stable-type-d",
AND(LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId],
ModelSummary[GeographicTypeDFraction]) >= 0.25, LOOKUP("cluster-id-geographic-
composition", IdentityClusterSummaries[IdentityClusterId],
IdentityClusterSummaries[StudyCount]) >= 3), IF(HypothesisId = "H-unexplained-flips-only-
nonconfounders", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId],
ModelSummary[UnexplainedConfounderSignFlipCount]) = 0, IF(HypothesisId = "H-confounder-
drift-type-d", AND(OR(LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId],
ModelSummary[SeverityDriftDirection]) = "rising", LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId], ModelSummary[SeverityDriftDirection]) = "single-decade"),
OR(LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId],
ModelSummary[GeographicDriftDirection]) = "rising", LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId], ModelSummary[GeographicDriftDirection]) = "single-
decade"), OR(LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId],
ModelSummary[SituationalDriftDirection]) = "rising", LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId], ModelSummary[SituationalDriftDirection]) = "single-
decade"), LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId],
ModelSummary[AgeDriftDirection]) <> "rising", LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId], ModelSummary[InstitutionalDriftDirection]) <> "rising",
LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId],
ModelSummary[SocioeconomicDriftDirection]) <> "rising"), FALSE())))))))))))))))))))))))).

```

▼ Inputs 1 fields this rule reads

● Hypothesis ID

Discovery Finding

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

TRUE when the witnessed metric satisfies the pre-registered ExpectedOutcome.

f Evidence Calculated 2 hops from ground truth string DR-383

IN RULESPEAK®

The discovery finding's evidence is determined by the following priority:

1. **"PASS: ", followed by the observed metric** if the is confirmed is true
2. **"FAIL: ", followed by the observed metric** if the is confirmed is false
3. **an empty string** in all other cases

► **Inputs** 2 fields this rule reads

f Is Confirmed

Discovery Finding

```
IF({{HypothesisId}} = "H-latent-d",
  LOOKUP("simpsons-paradox-v1",
    ModelSummary[ModelSummaryId],
    ModelSummary[LatentTypeDFraction]) > 0.5,
  IF({{HypothesisId}} = "H-purity",
    LOOKUP("simpsons-paradox-v1",
      ModelSummary[ModelSummaryId],
      ModelSummary[SignFlipSignalPurityMax]) < 0.5,
    IF({{HypothesisId}} = "H-small-effect",
      LOOKUP("simpsons-paradox-v1",
        ModelSummary[ModelSummaryId],
        ModelSummary[AvgPooledGapStableD]) >
      LOOKUP("simpsons-paradox-v1",
        ModelSummary[ModelSummaryId],
        ModelSummary[AvgPooledGapLatentD]),
      IF({{HypothesisId}} = "H-econ-zero",
        LOOKUP("simpsons-paradox-v1",
          ModelSummary[ModelSummaryId],
          ModelSummary[EconomicsSignFlipCount]) = 0,
        IF({{HypothesisId}} = "H-domain-dist",
          LOOKUP("simpsons-paradox-v1",
            ModelSummary[ModelSummaryId],
            ModelSummary[EpidemiologyAvgDistortion]) >
          LOOKUP("simpsons-paradox-v1",
            ModelSummary[ModelSummaryId],
            ModelSummary[EducationAvgDistortion]),
          IF({{HypothesisId}} = "H-causal-manifest",
            AND(LOOKUP("simpsons-paradox-v1",
              ModelSummary[ModelSummaryId],
              ModelSummary[ConfounderSignFlipCount]) >
```

f Observed Metric

Discovery Finding

```
IF({{HypothesisId}} = "H-latent-d",
  CONCAT("fraction=", LOOKUP("simpsons-
    paradox-v1", ModelSummary[ModelSummaryId],
    ModelSummary[LatentTypeDFraction])),
  IF({{HypothesisId}} = "H-purity",
    CONCAT("maxPurity=", LOOKUP("simpsons-
    paradox-v1", ModelSummary[ModelSummaryId],
    ModelSummary[SignFlipSignalPurityMax])),
    IF({{HypothesisId}} = "H-small-effect",
      CONCAT("stable=", LOOKUP("simpsons-paradox-
    v1", ModelSummary[ModelSummaryId],
    ModelSummary[AvgPooledGapStableD]), "
    latent=", LOOKUP("simpsons-paradox-v1",
    ModelSummary[ModelSummaryId],
    ModelSummary[AvgPooledGapLatentD])),
      IF({{HypothesisId}} = "H-econ-zero",
        CONCAT("flips=", LOOKUP("simpsons-paradox-
    v1", ModelSummary[ModelSummaryId],
    ModelSummary[EconomicsSignFlipCount])),
        IF({{HypothesisId}} = "H-domain-dist",
          CONCAT("epi=", LOOKUP("simpsons-paradox-
    v1", ModelSummary[ModelSummaryId],
    ModelSummary[EpidemiologyAvgDistortion]), "
    edu=", LOOKUP("simpsons-paradox-v1",
    ModelSummary[ModelSummaryId],
    ModelSummary[EducationAvgDistortion])),
          IF({{HypothesisId}} = "H-causal-manifest",
            CONCAT("confFlip=", LOOKUP("simpsons-
    paradox-v1", ModelSummary[ModelSummaryId],
    ModelSummary[ConfounderSignFlipCount]), "
```

```

LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[ColliderSelectionManifestCount]),
LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[ConfounderSignFlipCount]) >=
10), IF({{HypothesisId}} = "H-causal-latent",
AND(LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[ColliderSelectionManifestCount])
= 0, LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[ColliderSelectionLatentOnlyCount])
>= 5), IF({{HypothesisId}} = "H-explained-
confounder", AND(LOOKUP("simpsons-paradox-
v1", ModelSummary[ModelSummaryId],
ModelSummary[ExplainedConfounderCount]) =
LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[ConfounderSignFlipCount]),
LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[ExplainedConfounderCount]) >=
10), IF({{HypothesisId}} = "H-unexplained-
nonconfounder", LOOKUP("simpsons-paradox-
v1", ModelSummary[ModelSummaryId],
ModelSummary[ContestedOrMediatorExplainedCount]
= 0, IF({{HypothesisId}} = "H-catalog-exact-
match", LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[TypePredictionMatchRate]) < 0.5,
IF({{HypothesisId}} = "H-catalog-flip-prediction",
LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[SignFlipPredictionMatchRate]) <
0.5, IF({{HypothesisId}} = "H-domain-flip-
geometry-controlled", AND(LOOKUP("simpsons-
paradox-v1", ModelSummary[ModelSummaryId],
ModelSummary[EconomicsHighImbalanceSignFlipCou
= 0, LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[EpidemiologyHighImbalanceSignFlipP
> 0.15), IF({{HypothesisId}} = "H-collider-no-
manifest-v2", LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[ColliderSelectionManifestCount])
= 0, IF({{HypothesisId}} = "H-cplus-magnitude",
AND(LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
collManifest=", LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[ColliderSelectionManifestCount])),
IF({{HypothesisId}} = "H-causal-latent",
CONCAT("collManifest=", LOOKUP("simpsons-
paradox-v1", ModelSummary[ModelSummaryId],
ModelSummary[ColliderSelectionManifestCount]),
" collLatent=", LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[ColliderSelectionLatentOnlyCount]),
" collN=", LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[ColliderSelectionCount])),
IF({{HypothesisId}} = "H-explained-confounder",
CONCAT("ExplainedConfounderCount=",
LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[ExplainedConfounderCount]), ",
ConfounderSignFlipCount=", LOOKUP("simpsons-
paradox-v1", ModelSummary[ModelSummaryId],
ModelSummary[ConfounderSignFlipCount])),
IF({{HypothesisId}} = "H-unexplained-
nonconfounder",
CONCAT("ContestedOrMediatorExplainedCount=",
LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[ContestedOrMediatorExplainedCount])),
IF({{HypothesisId}} = "H-catalog-exact-match",
CONCAT("exactRate=", LOOKUP("simpsons-
paradox-v1", ModelSummary[ModelSummaryId],
ModelSummary[TypePredictionMatchRate])),
IF({{HypothesisId}} = "H-catalog-flip-prediction",
CONCAT("flipPredRate=", LOOKUP("simpsons-
paradox-v1", ModelSummary[ModelSummaryId],
ModelSummary[SignFlipPredictionMatchRate])),
IF({{HypothesisId}} = "H-domain-flip-geometry-
controlled", CONCAT("econHighImbFlips=",
LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[EconomicsHighImbalanceSignFlipCount]),
"; epiRate=", LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[EpidemiologyHighImbalanceSignFlipRate])
"; legal=", LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[LegalHighImbalanceSignFlipRate]),
"; sports=", LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[SportsHighImbalanceSignFlipRate])),

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ModelSummary[CPlusAvgDistortion]) > LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[CMinusAvgDistortion]), LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[CPlusAvgDistortion]) > LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[TypeDAvgDistortion])), IF({{HypothesisId}} = "H-econ-encoding- selection", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[EconomicsExpectedAMismatchRate]) > 0.5, IF({{HypothesisId}} = "H-domain-profiles- stable", AND(LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[EducationLatentFraction]) > 0.5, LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[SportsLatentFraction]) > 0.5, LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[EconomicsSignFlipRate]) < 0.05), IF({{HypothesisId}} = "H-corrected-gap- invariant", AND(LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[MaxStudySweepCorrectedGapRange] < 0.0001, LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[CorrectedGapInvariantFailCount]) = 0), IF({{HypothesisId}} = "H-explained- bidirectional", AND(LOOKUP("simpsons-paradox- v1", ModelSummary[ModelSummaryId], ModelSummary[ExplainedConfounderCount]) = LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[ConfounderSignFlipCount]), LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[FalsePositiveExplainedCount]) = 0, LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[UnexplainedConfounderSignFlipCount] = 0), IF({{HypothesisId}} = "H-collider-no- manifest-theorem", AND(LOOKUP("simpsons- paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[ColliderSelectionManifestCount]) = 0, LOOKUP("simpsons-paradox-v1",	IF({{HypothesisId}} = "H-collider-no-manifest-v2", CONCAT("collManifest=", LOOKUP("simpsons- paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[ColliderSelectionManifestCount])), IF({{HypothesisId}} = "H-cplus-magnitude", CONCAT("C+=", LOOKUP("simpsons-paradox- v1", ModelSummary[ModelSummaryId], ModelSummary[CPlusAvgDistortion]), " C-=", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[CMinusAvgDistortion]), " D=", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[TypeDAvgDistortion])), IF({{HypothesisId}} = "H-econ-encoding- selection", CONCAT("EconExpectedAMismatchRate=", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[EconomicsExpectedAMismatchRate])), IF({{HypothesisId}} = "H-domain-profiles-stable", CONCAT("eduLatent=", LOOKUP("simpsons- paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[EducationLatentFraction]), " " sportsLatent=", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[SportsLatentFraction]), " " econFlipRate=", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[EconomicsSignFlipRate])), IF({{HypothesisId}} = "H-corrected-gap- invariant", CONCAT("maxRange=", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[MaxStudySweepCorrectedGapRange]), " fails=", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[CorrectedGapInvariantFailCount])), IF({{HypothesisId}} = "H-explained-bidirectional", CONCAT("explained=", LOOKUP("simpsons- paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[ExplainedConfounderCount]), " " confFlip=", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[ConfounderSignFlipCount]), " " falsePos=", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId], ModelSummary[FalsePositiveExplainedCount]), " " unexplained=", LOOKUP("simpsons-paradox-v1",
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ModelSummary[ModelSummaryId],
 ModelSummary[ColliderSelectionCount]) >= 10),
 IF({{HypothesisId}} = "H-theorem-portfolio",
 LOOKUP("simpsons-paradox-v1",
 ModelSummary[ModelSummaryId],
 ModelSummary[TheoremCount]) >= 4,
 IF({{HypothesisId}} = "H-age-identity-flip-rate",
 AND(LOOKUP("simpsons-paradox-v1",
 ModelSummary[ModelSummaryId],
 ModelSummary[AgeIdentityManifestFlipRate]) >= 0.40, LOOKUP("simpsons-paradox-v1",
 ModelSummary[ModelSummaryId],
 ModelSummary[AgeIdentityStudyCount]) >= 10),
 IF({{HypothesisId}} = "H-identity-map-coverage",
 LOOKUP("simpsons-paradox-v1",
 ModelSummary[ModelSummaryId],
 ModelSummary[IdentityMapCoverageRate]) >= 0.95, IF({{HypothesisId}} = "H-severity-medicine-manifest", LOOKUP("simpsons-paradox-v1",
 ModelSummary[ModelSummaryId],
 ModelSummary[SeverityMedicineManifestFlipRate]) >= 0.45, IF({{HypothesisId}} = "H-severity-epi-latent-only", LOOKUP("simpsons-paradox-v1",
 ModelSummary[ModelSummaryId],
 ModelSummary[SeverityEpiManifestFlipCount]) = 0, IF({{HypothesisId}} = "H-mechanism-other-drain", LOOKUP("cluster-id-mechanism-other",
 IdentityClusterSummaries[IdentityClusterId],
 IdentityClusterSummaries[StudyCount]) <= 5, IF({{HypothesisId}} = "H-selection-frailty-zero-manifest", AND(LOOKUP("cluster-id-selection-frailty",
 IdentityClusterSummaries[IdentityClusterId],
 IdentityClusterSummaries[ManifestFlipCount]) = 0, LOOKUP("cluster-id-selection-frailty",
 IdentityClusterSummaries[IdentityClusterId],
 IdentityClusterSummaries[StudyCount]) >= 3), IF({{HypothesisId}} = "H-collider-identity-low-manifest", AND(LOOKUP("simpsons-paradox-v1",
 ModelSummary[ModelSummaryId],
 ModelSummary[ColliderIdentityManifestFlipRate]) <= 0.15, LOOKUP("cluster-id-collider-proxy",
 IdentityClusterSummaries[IdentityClusterId],
 IdentityClusterSummaries[StudyCount]) >= 5), IF({{HypothesisId}} = "H-geographic-stable-type-d", AND(LOOKUP("simpsons-paradox-v1",
 ModelSummary[ModelSummaryId],
 ModelSummary[GeographicTypeDFraction]) >= 0.25, LOOKUP("cluster-id-geographic-

ModelSummary[ModelSummaryId],
 ModelSummary[UnexplainedConfounderSignFlipCount])),
 IF({{HypothesisId}} = "H-collider-no-manifest-theorem", CONCAT("collN=",
 LOOKUP("simpsons-paradox-v1",
 ModelSummary[ModelSummaryId],
 ModelSummary[ColliderSelectionCount]), "collManifest=", LOOKUP("simpsons-paradox-v1",
 ModelSummary[ModelSummaryId],
 ModelSummary[ColliderSelectionManifestCount])),
 IF({{HypothesisId}} = "H-theorem-portfolio", CONCAT("theoremCount=", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId],
 ModelSummary[TheoremCount])),
 IF({{HypothesisId}} = "H-age-identity-flip-rate", CONCAT("rate=", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId],
 ModelSummary[AgeIdentityManifestFlipRate]), ", n=", LOOKUP("simpsons-paradox-v1",
 ModelSummary[ModelSummaryId],
 ModelSummary[AgeIdentityStudyCount])),
 IF({{HypothesisId}} = "H-identity-map-coverage", CONCAT("coverage=", LOOKUP("simpsons-paradox-v1", ModelSummary[ModelSummaryId],
 ModelSummary[IdentityMapCoverageRate])),
 IF({{HypothesisId}} = "H-severity-medicine-manifest", CONCAT("medicine_flip_rate=", LOOKUP("simpsons-paradox-v1",
 ModelSummary[ModelSummaryId],
 ModelSummary[SeverityMedicineManifestFlipRate])),
 IF({{HypothesisId}} = "H-severity-epi-latent-only", CONCAT("epi_manifest_count=", LOOKUP("simpsons-paradox-v1",
 ModelSummary[ModelSummaryId],
 ModelSummary[SeverityEpiManifestFlipCount])),
 IF({{HypothesisId}} = "H-mechanism-other-drain", CONCAT("other_count=", LOOKUP("cluster-id-mechanism-other",
 IdentityClusterSummaries[IdentityClusterId],
 IdentityClusterSummaries[StudyCount])),
 IF({{HypothesisId}} = "H-selection-frailty-zero-manifest", CONCAT("manifest=", LOOKUP("cluster-id-selection-frailty",
 IdentityClusterSummaries[IdentityClusterId],
 IdentityClusterSummaries[ManifestFlipCount]), ", n=", LOOKUP("cluster-id-selection-frailty",
 IdentityClusterSummaries[IdentityClusterId],
 IdentityClusterSummaries[StudyCount])),
 IF({{HypothesisId}} = "H-collider-identity-low-


```
composition",
IdentityClusterSummaries[IdentityClusterId],
IdentityClusterSummaries[StudyCount]) >= 3),
IF({{HypothesisId}} = "H-unexplained-flips-only-
nonconfounders", LOOKUP("simpsons-paradox-
v1", ModelSummary[ModelSummaryId],
ModelSummary[UnexplainedConfounderSignFlipCount]
= 0, IF({{HypothesisId}} = "H-confounder-drift-
type-d", AND(OR(LOOKUP("simpsons-paradox-
v1", ModelSummary[ModelSummaryId],
ModelSummary[SeverityDriftDirection]) = "rising",
LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[SeverityDriftDirection]) = "single-
decade"), OR(LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[GeographicDriftDirection]) =
"rising", LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[GeographicDriftDirection]) =
"single-decade"), OR(LOOKUP("simpsons-
paradox-v1", ModelSummary[ModelSummaryId],
ModelSummary[SituationalDriftDirection]) =
"rising", LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[SituationalDriftDirection]) =
"single-decade"), LOOKUP("simpsons-paradox-
v1", ModelSummary[ModelSummaryId],
ModelSummary[AgeDriftDirection]) <> "rising",
LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[InstitutionalDriftDirection]) <>
"rising", LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[SocioeconomicDriftDirection]) <>
"rising"), FALSE((((((((((((((((((((((((((((((((((((
)))))))))))))))))))))))))))))))))))))))))))))))))))).
```

```
manifest", CONCAT("rate=", LOOKUP("simpsons-
paradox-v1", ModelSummary[ModelSummaryId],
ModelSummary[ColliderIdentityManifestFlipRate])),
IF({{HypothesisId}} = "H-geographic-stable-type-
d", CONCAT("type_d_frac=",
LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[GeographicTypeDFraction])),
IF({{HypothesisId}} = "H-unexplained-flips-only-
nonconfounders",
CONCAT("unexplained_confounder_flips=",
LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[UnexplainedConfounderSignFlipCount])),
IF({{HypothesisId}} = "H-confounder-drift-type-
d", CONCAT("severity=", LOOKUP("simpsons-
paradox-v1", ModelSummary[ModelSummaryId],
ModelSummary[SeverityDriftDirection]), "
geography=", LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[GeographicDriftDirection]), "
situational=", LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[SituationalDriftDirection]), "
age=", LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[AgeDriftDirection]), "
institutional=", LOOKUP("simpsons-paradox-v1",
ModelSummary[ModelSummaryId],
ModelSummary[InstitutionalDriftDirection]), "
socioeconomic=", LOOKUP("simpsons-paradox-
v1", ModelSummary[ModelSummaryId],
ModelSummary[SocioeconomicDriftDirection])),
"")))))))))))))))))))))))))))))))))))))))))))))))).
```

► **Ground truth at the leaves** **1** the raw values this ultimately depends on

Pass/fail witness string for UI and conclusions.

Witnessed in Loop Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

LoopId where this finding was witnessed.

Corpus Domain

A corpus domain is identified by its [name](#).

f **Name** Calculated 1 hop from ground truth string DR-384

IN RULESPEAK®

A corpus domain's [name](#) is the same as its domain ID.

▼ **Inputs** 1 fields this rule reads

● **Domain ID** Corpus Domain
Unique slug (matches
CandidateStudyCatalog.Domain for expansion
wave).

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

| *Display label.*

● **Display Name** Ground truth string

Ground truth. Written directly to the database (datatype string); not computed from anything else. Everything derived ultimately traces back to values like this one.

| *Human-readable domain name.*

● **Expansion Target** Ground truth integer

Ground truth. Written directly to the database (datatype integer); not computed from anything else. Everything derived ultimately traces back to values like this one.

| *Target study count from corpus-expansion-plan.md.*

● **Legacy Studies Domain** Ground truth string

Ground truth. Written directly to the database (datatype string); not computed from anything else. Everything derived ultimately traces back to values like this one.

| *Studies.Domain alias for counting already-imported corpus rows.*

● **Plan Source Section** Ground truth string

Ground truth. Written directly to the database (datatype string); not computed from anything else. Everything derived ultimately traces back to values like this one.

Confounder Identity

A confounder identity is identified by its name.

f **Name** Calculated 1 hop from ground truth string DR-385

IN RULESPEAK®

A confounder identity's name is the same as its confounder identity ID.

▼ **Inputs** 1 fields this rule reads

- **Confounder Identity ID** Confounder Identity
Unique slug (e.g. id-age-composition).

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Display label.

Display Name Ground truth string

Ground truth. Written directly to the database (datatype string); not computed from anything else. Everything derived ultimately traces back to values like this one.

Human-readable archetype name.

Mechanism Class Ground truth string

Ground truth. Written directly to the database (datatype string); not computed from anything else. Everything derived ultimately traces back to values like this one.

Mechanism taxonomy token.

Description Ground truth string

Ground truth. Written directly to the database (datatype string); not computed from anything else. Everything derived ultimately traces back to values like this one.

Narrative description of confounding mechanism.

Policy Default Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Default analytic policy for this confounding archetype.

Σ Study Count Aggregation 1 hop from ground truth integer DR-386

IN RULESPEAK®

A confounder identity's study count is the number of the confounder identity's studies.

▼ **Inputs** 2 fields this rule reads

● **Map ID** Stratum Variable Identity Map
Unique slug.

● **Confounder Identity ID** Confounder Identity
Unique slug (e.g. id-age-composition).

► **Ground truth at the leaves** 2 the raw values this ultimately depends on

Studies with a StratumVariable mapped to this identity.

Stratum Variable Identity Map

A stratum variable identity map is identified by its name and is related to a stratum variable and a confounder identity.

f Name Calculated 1 hop from ground truth string DR-387

IN RULESPEAK®

A stratum variable identity map's name is the same as its map ID.

▼ **Inputs** 1 fields this rule reads

● **Map ID** Stratum Variable Identity Map
Unique slug.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Display label.

... Stratum Variable Relationship string

A pointer to Stratum Variable. It is not derived from anything else — it is the link the rest of the DAG hangs off.

FK → StratumVariables.



Confounder Identity

Relationship

string

A pointer to [Confounder Identity](#). It is not derived from anything else — it is the link the rest of the DAG hangs off.

FK → ConfounderIdentities.



Variable Name Normalized

Ground truth

string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Lowercase normalized variable token for dedup audit.

Identity Cluster Summary

An identity cluster summary is identified by its [name](#) and is related to a [confounder identity](#).



Name

Calculated

1 hop from ground truth

string

DR-388

IN RULESPEAK®

An identity cluster summary's [name](#) is the same as its identity cluster ID.

▼ **Inputs** 1 fields this rule reads



Identity Cluster ID

Identity Cluster Summary

Unique slug; one row per ConfounderIdentity cluster.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Display label.



Confounder Identity

Relationship

string

A pointer to [Confounder Identity](#). It is not derived from anything else — it is the link the rest of the DAG hangs off.

FK → ConfounderIdentities.



Display Name

Lookup

1 hop from ground truth

string

DR-389

IN RULESPEAK®

An identity cluster summary's [display name](#) is the value of LOOKUP(ConfounderIdentity, ConfounderIdentities[ConfounderIdentityId], ConfounderIdentities[DisplayName]).

▼ **Inputs** 1 fields this rule reads

...→ **Confounder Identity** Identity Cluster Summary
FK → ConfounderIdentities.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

| *Lookup archetype display name.*

Study Count Ground truth integer

Ground truth. Written directly to the database (datatype integer); not computed from anything else. Everything derived ultimately traces back to values like this one.

| *Real studies in cluster (customize SQL).*

Manifest Flip Count Ground truth integer

Ground truth. Written directly to the database (datatype integer); not computed from anything else. Everything derived ultimately traces back to values like this one.

| *IsSignFlip=TRUE count in cluster.*

Latent Flip Count Ground truth integer

Ground truth. Written directly to the database (datatype integer); not computed from anything else. Everything derived ultimately traces back to values like this one.

| *LatentFlipPotential=TRUE without manifest flip.*

Type D Count Ground truth integer

Ground truth. Written directly to the database (datatype integer); not computed from anything else. Everything derived ultimately traces back to values like this one.

| *Type-D studies in cluster.*

Avg Allocation Distortion Ground truth number

Ground truth. Written directly to the database (datatype number); not computed from anything else. Everything derived ultimately traces back to values like this one.

| *Mean allocation distortion in cluster.*



Latent Fraction Among Type D

Ground truth

number

Ground truth. Written directly to the database (datatype `number`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Fraction of Type-D rows with LatentFlipPotential.



Domain Count

Ground truth

integer

Ground truth. Written directly to the database (datatype `integer`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Distinct Studies.Domain values in cluster.



Manifest Flip Rate

Calculated

1 hop from ground truth

number

DR-390

IN RULESPEAK®

The identity cluster summary's manifest flip rate is determined by the following priority:

1. an empty string if the study count is 0
2. the manifest flip count divided by the study count in all other cases

▼ Inputs 2 fields this rule reads

● Study Count

Identity Cluster Summary

Real studies in cluster (customize SQL).

● Manifest Flip Count

Identity Cluster Summary

IsSignFlip=TRUE count in cluster.

► Ground truth at the leaves 2 the raw values this ultimately depends on

ManifestFlipCount / StudyCount.

Identity Domain Cell

An identity domain cell is identified by its name and is related to a confounder identity.



Name

Calculated

1 hop from ground truth

string

DR-391

IN RULESPEAK®

An identity domain cell's name is the same as its cell ID.

▼ Inputs 1 fields this rule reads

● Cell ID

Identity Domain Cell

Unique slug: identity-id + '-x-' + domain slug.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

| *Display label.*

...► **Confounder Identity** Relationship string

A pointer to [Confounder Identity](#). It is not derived from anything else — it is the link the rest of the DAG hangs off.

| *FK → ConfounderIdentities.*

● **Domain** Ground truth string

Ground truth. Written directly to the database (datatype `string`); not computed from anything else. Everything derived ultimately traces back to values like this one.

| *Studies.Domain value (e.g. medicine, epidemiology).*

● **Study Count** Ground truth integer

Ground truth. Written directly to the database (datatype `integer`); not computed from anything else. Everything derived ultimately traces back to values like this one.

| *Studies with this identity in this domain (customize SQL).*

● **Manifest Flip Count** Ground truth integer

Ground truth. Written directly to the database (datatype `integer`); not computed from anything else. Everything derived ultimately traces back to values like this one.

| *IsSignFlip=TRUE count in cell.*

● **Latent Flip Count** Ground truth integer

Ground truth. Written directly to the database (datatype `integer`); not computed from anything else. Everything derived ultimately traces back to values like this one.

| *LatentFlipPotential=TRUE without manifest flip.*

● **Type D Count** Ground truth integer

Ground truth. Written directly to the database (datatype `integer`); not computed from anything else. Everything derived ultimately traces back to values like this one.

| *Type-D studies in cell.*



Latent Fraction Among Type D

Ground truth

number

Ground truth. Written directly to the database (datatype `number`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Fraction of Type-D rows with LatentFlipPotential in cell.



Manifest Flip Rate

Calculated

1 hop from ground truth

number

DR-392

IN RULESPEAK®

The identity domain cell's manifest flip rate is determined by the following priority:

1. an empty string if the study count is 0
2. the manifest flip count divided by the study count in all other cases

▼ **Inputs** 2 fields this rule reads

● Study Count

Identity Domain Cell

Studies with this identity in this domain
(customize SQL).

● Manifest Flip Count

Identity Domain Cell

IsSignFlip=TRUE count in cell.

► **Ground truth at the leaves** 2 the raw values this ultimately depends on

ManifestFlipCount / StudyCount.



Identity Display Name

Lookup

1 hop from ground truth

string

DR-393

IN RULESPEAK®

An identity domain cell's identity display name is the value of LOOKUP(ConfounderIdentity, ConfounderIdentities[ConfounderIdentityId], ConfounderIdentities[DisplayName]).

▼ **Inputs** 1 fields this rule reads

↗ Confounder Identity

Identity Domain Cell

FK → ConfounderIdentities.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup archetype display name.

Confounder Distortion Timeline

A confounder distortion timeline is identified by its name and is related to a confounder identity.

f Name Calculated 1 hop from ground truth string DR-394

IN RULESPEAK®

A confounder distortion timeline's name is the same as its cell ID.

▼ **Inputs** 1 fields this rule reads

● **Cell ID** Confounder Distortion Timeline

Unique slug: confounder-identity-id + '-x-' + valid-time decade (e.g. id-disease-severity-x-1980).

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

| *Display label.*

...→ **Confounder Identity** Relationship string

A pointer to [Confounder Identity](#). It is not derived from anything else — it is the link the rest of the DAG hangs off.

| *FK -> ConfounderIdentities.*

● **Valid Time Decade** Ground truth integer

Ground truth. Written directly to the database (datatype `integer`); not computed from anything else. Everything derived ultimately traces back to values like this one.

| *Studies.ValidTimeDecade bucket this cell aggregates over (e.g. 1980 covers valid-time midpoints 1980-1989).*

● **Study Count** Ground truth integer

Ground truth. Written directly to the database (datatype `integer`); not computed from anything else. Everything derived ultimately traces back to values like this one.

| *Studies carrying this ConfounderIdentity whose ValidTimeDecade falls in this bucket (refresh_confounder_distortion_timeline()).*

● **Type D Count** Ground truth integer

Ground truth. Written directly to the database (datatype `integer`); not computed from anything else. Everything derived ultimately traces back to values like this one.

| *Of those, count with TreatmentRankings.DistortionType = 'D' (clean, non-reversing).*



Sign Flip Count

Ground truth

integer

Ground truth. Written directly to the database (datatype `integer`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Of those, count with `TreatmentRankings.IsSignFlip = TRUE`.



Sum Signal Purity

Ground truth

number

Ground truth. Written directly to the database (datatype `number`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Sum of `TreatmentRankings.SignalPurity` across cell studies (numerator for `MeanSignalPurity`; summed rather than pre-averaged so the aggregate survives refresh re-runs without drift).



Sum Allocation Distortion

Ground truth

number

Ground truth. Written directly to the database (datatype `number`); not computed from anything else. Everything derived ultimately traces back to values like this one.

Sum of `TreatmentRankings.AllocationDistortion` across cell studies (numerator for `MeanAllocationDistortion`).



Type D Fraction

Calculated

1 hop from ground truth

number

DR-395

IN RULESPEAK®

The confounder distortion timeline's [type d fraction](#) is determined by the following priority:

1. [an empty string](#) if any of these hold:
 - the [study count](#) is blank
 - the [study count](#) is 0
2. [the type d count divided by the study count](#) in all other cases

▼ Inputs 2 fields this rule reads



Study Count

Confounder Distortion Timeline

Studies carrying this `ConfounderIdentity` whose `ValidTimeDecade` falls in this bucket (`refresh_confounder_distortion_timeline()`).



Type D Count

Confounder Distortion Timeline

Of those, count with `TreatmentRankings.DistortionType = 'D'` (clean, non-reversing).

► Ground truth at the leaves 2 the raw values this ultimately depends on

`TypeDCount / StudyCount`.



Sign Flip Fraction

Calculated

1 hop from ground truth

number

DR-396

The confounder distortion timeline's sign flip fraction is determined by the following priority:

1. **an empty string** if any of these hold:
 - the study count is blank
 - the study count is 0
2. **the sign flip count divided by the study count** in all other cases

▼ **Inputs** 2 fields this rule reads

● **Study Count** Confounder Distortion Timeline

Studies carrying this ConfounderIdentity whose ValidTimeDecade falls in this bucket (refresh_confounder_distortion_timeline()).

● **Sign Flip Count** Confounder Distortion Timeline

Of those, count with TreatmentRankings.IsSignFlip = TRUE.

► **Ground truth at the leaves** 2 the raw values this ultimately depends on

SignFlipCount / StudyCount.



Mean Signal Purity

Calculated

1 hop from ground truth

number

DR-397

The confounder distortion timeline's mean signal purity is determined by the following priority:

1. **an empty string** if any of these hold:
 - the study count is blank
 - the study count is 0
2. **the sum signal purity divided by the study count** in all other cases

▼ **Inputs** 2 fields this rule reads

● **Study Count** Confounder Distortion Timeline

Studies carrying this ConfounderIdentity whose ValidTimeDecade falls in this bucket (refresh_confounder_distortion_timeline()).

● **Sum Signal Purity** Confounder Distortion Timeline

Sum of TreatmentRankings.SignalPurity across cell studies (numerator for MeanSignalPurity; summed rather than pre-averaged so the aggregate survives refresh re-runs without drift).

► **Ground truth at the leaves** 2 the raw values this ultimately depends on

SumSignalPurity / StudyCount.



Mean Allocation Distortion

Calculated

1 hop from ground truth

number

DR-398

The confounder distortion timeline's [mean allocation distortion](#) is determined by the following priority:

1. **an empty string** if any of these hold:
 - the [study count](#) is blank
 - the [study count](#) is 0
2. [the sum allocation distortion divided by the study count](#) in all other cases

▼ **Inputs** 2 fields this rule reads

● **Study Count** Confounder Distortion Timeline
Studies carrying this ConfounderIdentity whose ValidTimeDecade falls in this bucket (refresh_confounder_distortion_timeline()).

● **Sum Allocation Distortion** Confounder Distortion Timeline
Sum of TreatmentRankings.AllocationDistortion across cell studies (numerator for MeanAllocationDistortion).

► **Ground truth at the leaves** 2 the raw values this ultimately depends on

| *SumAllocationDistortion / StudyCount.*

↗ **Identity Display Name** **Lookup** 1 hop from ground truth string DR-399

IN RULESPEAK®

A confounder distortion timeline's [identity display name](#) is the value of LOOKUP(ConfounderIdentity, ConfounderIdentities[ConfounderIdentityId], ConfounderIdentities[DisplayName]).

▼ **Inputs** 1 fields this rule reads

...→ **Confounder Identity** Confounder Distortion Timeline
FK -> ConfounderIdentities.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

| *Lookup: ConfounderIdentities.DisplayName via ConfounderIdentity.*

↗ **Mechanism Class** **Lookup** 1 hop from ground truth string DR-400

IN RULESPEAK®

A confounder distortion timeline's [mechanism class](#) is the value of LOOKUP(ConfounderIdentity, ConfounderIdentities[ConfounderIdentityId], ConfounderIdentities[MechanismClass]).

▼ **Inputs** 1 fields this rule reads

Confounder Identity

Confounder Distortion
Timeline

FK -> ConfounderIdentities.

- **Ground truth at the leaves** 1 the raw values this ultimately depends on

Lookup: *ConfounderIdentities.MechanismClass* via *ConfounderIdentity* — the archetype group used to test whether Type-D drift is confounder-specific.



Identity Earliest Decade

Aggregation

1 hop from ground truth

integer

DR-401

IN RULESPEAK®

A confounder distortion timeline's identity earliest decade is the smallest valid time decade across the confounder distortion timeline related to the confounder distortion timeline.

- ▼ **Inputs** 2 fields this rule reads

Valid Time Decade

Confounder Distortion
Timeline

Studies.ValidTimeDecade bucket this cell aggregates over (e.g. 1980 covers valid-time midpoints 1980-1989).

Confounder Identity

Confounder Distortion
Timeline

FK -> ConfounderIdentities.

- **Ground truth at the leaves** 2 the raw values this ultimately depends on

Earliest ValidTimeDecade present in this table for this ConfounderIdentity.



Identity Latest Decade

Aggregation

1 hop from ground truth

integer

DR-402

IN RULESPEAK®

A confounder distortion timeline's identity latest decade is the largest valid time decade across the confounder distortion timeline related to the confounder distortion timeline.

- ▼ **Inputs** 2 fields this rule reads

Valid Time Decade

Confounder Distortion
Timeline

Studies.ValidTimeDecade bucket this cell aggregates over (e.g. 1980 covers valid-time midpoints 1980-1989).

Confounder Identity

Confounder Distortion
Timeline

FK -> ConfounderIdentities.

- **Ground truth at the leaves** 2 the raw values this ultimately depends on

Latest ValidTimeDecade present in this table for this ConfounderIdentity.

Identity Earliest Type D Fraction Lookup 2 hops from ground truth number DR-403

IN RULESPEAK®

A confounder distortion timeline's identity earliest type d fraction is the value of `LOOKUP(CONCAT(ConfounderIdentity, "-x-", IdentityEarliestDecade), ConfounderDistortionTimeline[CellId], ConfounderDistortionTimeline[TypeDFraction])`.

▼ Inputs 2 fields this rule reads

...→ **Confounder Identity** Confounder Distortion Timeline
FK -> ConfounderIdentities.

Σ **Identity Earliest Decade** Confounder Distortion Timeline
Smallest valid time decade across all confounder distortion timeline whose confounder identity is confounder identity.

► Ground truth at the leaves 2 the raw values this ultimately depends on

TypeDFraction of this ConfounderIdentity's earliest decade cell -- the start point for the drift comparison.

Identity Latest Type D Fraction Lookup 2 hops from ground truth number DR-404

IN RULESPEAK®

A confounder distortion timeline's identity latest type d fraction is the value of `LOOKUP(CONCAT(ConfounderIdentity, "-x-", IdentityLatestDecade), ConfounderDistortionTimeline[CellId], ConfounderDistortionTimeline[TypeDFraction])`.

▼ Inputs 2 fields this rule reads

...→ **Confounder Identity** Confounder Distortion Timeline
FK -> ConfounderIdentities.

Σ **Identity Latest Decade** Confounder Distortion Timeline
Largest valid time decade across all confounder distortion timeline whose confounder identity is confounder identity.

► Ground truth at the leaves 2 the raw values this ultimately depends on

TypeDFraction of this ConfounderIdentity's latest decade cell -- the end point for the drift comparison.

Drift Direction Calculated 3 hops from ground truth string DR-405

IN RULESPEAK®

The confounder distortion timeline's drift direction is determined by the following priority:

1. **"single-decade"** if the identity earliest decade is the identity latest decade

2. **an empty string** if any of these hold:
 - the identity earliest type d fraction is blank
 - the identity latest type d fraction is blank
3. **"rising"** if the identity latest type d fraction is greater than the identity earliest type d fraction plus 0.05
4. **"falling"** if the identity latest type d fraction is less than the identity earliest type d fraction minus 0.05
5. **"flat"** in all other cases

▼ **Inputs** 4 fields this rule reads

Σ **Identity Earliest Decade**

Confounder Distortion Timeline

Smallest valid time decade across all confounder distortion timeline whose confounder identity is confounder identity.

Σ **Identity Latest Decade**

Confounder Distortion Timeline

Largest valid time decade across all confounder distortion timeline whose confounder identity is confounder identity.

↗ **Identity Earliest Type D Fraction**

Confounder Distortion Timeline

LOOKUP(CONCAT({{ConfounderIdentity}}, "-x-", {{IdentityEarliestDecade}}),
ConfounderDistortionTimeline[CellId],
ConfounderDistortionTimeline[TypeDFraction]).

↗ **Identity Latest Type D Fraction**

Confounder Distortion Timeline

LOOKUP(CONCAT({{ConfounderIdentity}}, "-x-", {{IdentityLatestDecade}}),
ConfounderDistortionTimeline[CellId],
ConfounderDistortionTimeline[TypeDFraction]).

► **Ground truth at the leaves** 2 the raw values this ultimately depends on

rising | falling | flat | single-decade -- compares this ConfounderIdentity's earliest-decade TypeDFraction to its latest-decade TypeDFraction across the whole table (same value on every row for a given ConfounderIdentity by construction).

Corpus Balance

A corpus balance is identified by its name.



Name

Calculated

1 hop from ground truth

string

DR-406

IN RULESPEAK®

A corpus balance's name is the same as its balance ID.

▼ **Inputs** 1 fields this rule reads

● **Balance ID**

Corpus Balance

Primary key (singleton: 'corpus-balance')

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

| =BalanceId

f **Total Study Count** Calculated 2 hops from ground truth integer DR-407

IN RULESPEAK®

A corpus balance's total study count is computed as the simpson study count plus the control study count.

▼ **Inputs** 2 fields this rule reads

Σ **Simpson Study Count** Corpus Balance
Count of studies whose is control study is false().

Σ **Control Study Count** Corpus Balance
Count of studies whose is control study is true().

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Total studies in corpus (Simpson + control). Fixed loop-90: was authored with COUNTA/bareword refs and an uppercase Formula key, which the transpiler cannot parse. COUNTIFS(...,"<>") also mistranslates (known transpiler bug: literal '<>' string instead of IS NOT NULL), so this is expressed as SimpsonStudyCount + ControlStudyCount instead.

Σ **Simpson Study Count** Aggregation 1 hop from ground truth integer DR-408

IN RULESPEAK®

A corpus balance's simpson study count is the number of studies related to the corpus balance.

▼ **Inputs** 1 fields this rule reads

● **Is Control Study** Study
TRUE if this study is a control (no Simpson's paradox expected/found); FALSE for the main Simpson's Paradox corpus.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

| *Studies where IsControlStudy=FALSE. Fixed loop-90 (see TotalStudyCount).*

Σ **Control Study Count** Aggregation 1 hop from ground truth integer DR-409

IN RULESPEAK®

A corpus balance's control study count is the number of studies related to the corpus balance.

▼ **Inputs** 1 fields this rule reads

● Is Control Study

Study

TRUE if this study is a control (no Simpson's paradox expected/found); FALSE for the main Simpson's Paradox corpus.

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

Studies where `IsControlStudy=TRUE`. Fixed loop-90 (see `TotalStudyCount`).

f

Control Fraction

Calculated

3 hops from ground truth

number

DR-410

IN RULESPEAK®

The corpus balance's control fraction is determined by the following priority:

1. 0 if the total study count is 0
2. the control study count divided by the total study count in all other cases

▼ **Inputs** 2 fields this rule reads

f Total Study Count

Corpus Balance

Simpson study count plus control study count.

Σ Control Study Count

Corpus Balance

Count of studies whose is control study is true().

► **Ground truth at the leaves** 1 the raw values this ultimately depends on

`ControlStudyCount / TotalStudyCount`. Fixed loop-90 (see `TotalStudyCount`); returns 0 for the degenerate empty-corpus edge instead of "" (see `TreatmentRankings.DistortionRatio` for why `IF(x=0,"",...)` is unsafe on a numeric-typed field).

Rendered from the rulebook by the shared **RuleSpeak®** engine (`rulebook-rulespeak-core`). The same engine narrates the React Explainer DAG, so every narration mode reads identically across both.

● Ground truth

f Calculated

Σ Aggregation

↗ Lookup

↔ Relationship